An Upper Bound on the Compactification Scale of $SU(7)$ Grand Gauge-Higgs Unification, and the Dijet Angular Distribution That Tests It

Once the Higgs mass is pinned, the electroweak hierarchy stops being the output of a minimisation and becomes an algebraic function of two moments of the bulk content — both of them quantised. Maximising it over the lattice of admissible contents is then an integer program whose relaxation has a two-variable dual, which can be solved exactly in rationals; and once the vacuum is required to be the true one — a fourth condition, and linear like the others — the bound is not only certified but attained*: on the three-rung algebraic surface, no bulk content puts $1/R_5$ above 9.22 TeV. The maximum sits on the smallest rung because the per-rung ceiling provably falls, from 10.03 TeV there — on the seed [2] print — to a Lambert-W asymptote of 2.678 TeV. And the measurement that pushes from the other side is not a resonance search: the Kaluza-Klein tower resums into a closed-form propagator whose spacelike part deforms the dijet* angular *distribution, which CMS publishes unfolded — so it can be confronted with a calculation, and it is, here.*

Part VII — a continuation, inside their own construction, of the study of the model Y. Komori and N. Maru set out in arXiv:2503.04090. Parts I–V run the inverse direction and Part VI runs the direct one as far as a *sign*; this paper carries it to a *number*, and the number is an upper bound, which is the useful direction: the model cannot retreat to higher scales by adding matter. The parity-split structure the closed form rests on is not ours: it is published, one rung below, and is credited where it is used

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Abstract

In gauge-Higgs unification the compactification scale is tied to the weak scale by $\alpha_{\min} = 2m_W R_5$, so the Wilson-line vacuum *is* the electroweak hierarchy. Part VI computed the one-loop curvature of that vacuum for the $SU(7)$ model of [2] and reached a sign. Here we compute the vacuum itself. Expanding the one-loop effective potential — a sum of polylogarithms of the Wilson-line phase — about the symmetric point produces one ladder, indexed by $p = 5 - 2k$, whose k-th rung pairs the $2k$-th moment of the bulk content against $\zeta(p)$; the rungs are $\zeta(5)$, $\zeta(3)$ and the pole, and the pole is where the $\alpha^4 \log \alpha$ branch point comes from. Solving the stationarity condition gives $\alpha_{\min}$ in a Lambert-W closed form in *three* moments (D, A_4, G), $x^2 = -d/W(-d\,e^{-b})$ — not an algebraic root, since it carries $\ln x$, but solved rather than iterated, and the two branches W_{-1} and W_0 are the minimum and the maximum with the fold $W = -1$ at $m_h^2 = 0$ — accurate to a median $0.13\,\%$ over 272 contents and $0.209\,\%$ on the five published rows against our own numerical minimisation of the same potential; the same expansion gives the Higgs mass, so *both* columns of their Table 1 become functions of the moments with no minimisation anywhere — *our* values of those two columns, since §2 records

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that our $\alpha_{\min}$ does not reproduce the one they publish. Pinning m_h then eliminates the logarithm entirely and leaves two identities exact *within that truncation*, verified to 6×10^{-14} — and *that* is where $\alpha_{\min}$ becomes genuinely algebraic, in the two quantised moments (D, A_4) alone, which is precisely the role the measured Higgs mass plays. The moments are quantised: $8D \equiv 2A_4 + 3 \pmod 6$, because $c^4 \equiv c^2 \pmod 3$. What holds whatever the gauge sector contributes is the *matter* half of that — every bulk multiplet moves $8D - 2A_4$ by a multiple of six — while the residue is the gauge base point's own; both seeds considered here give the same 9, so the congruence itself holds on either branch. Whether it also makes $8D$ *odd* does not: for that A_4 must be integral, which is what the coefficients printed in [2]'s eq. (68) give. **We flag that step because those coefficients are the one input this paper does not derive from the product-orbifold geometry — though §13 now derives its leading parity-resolved determinant, and gets the other split.** Read at face value they correspond to five gauge degrees of freedom, where the general $-(D-2)$ gauge-plus-ghost rule stated in [17], evaluated in six dimensions and realised explicitly in [48] and [3], gives four. The oddness turns on them, and so do the quantum $D = 1/8$ and the compactification scales below; the congruence does not. So maximising $1/R_5$ is an integer program over a rational cone; its relaxation has a two-variable dual whose vertices we enumerate exactly, giving $1/R_5 \leq 10.03$ TeV for arbitrary multiplicities of [2]'s eight (representation, parity) bulk types — arbitrary in the multiplicities throughout, never in the list — at the quantum $D = 1/8$; writing G in the basis $\{1, \ln 2, \ln 3\}$ lifts the lattice to four dimensions, where $\mathbb{Z}^4/L \cong \mathbb{Z}_{18} \times \mathbb{Z}_{648}$ instead of the projection's $\mathbb{Z}_6$, and the relaxation's vertex turns out to be empty, sharpening that to 10.01 TeV *attained*. Put on the exact potential, however, the content attaining it is a *false vacuum*: it is deeper at the other symmetric point. Requiring the electroweak point to be the true minimum is the orbifold vacuum-stability criterion of [8], whose arithmetic is theirs; summed over a bulk content it is a *fourth* linear functional, $F(1) - F(0) = \frac{31}{16}\zeta(5)W$, and what is new is only that a linear functional can be carried inside the integer program. That functional is also what makes the fork cheap to state: the candidate seed's ceiling is lower still and imposing the same condition can only lower it further, so between the two seeds considered here 9.22 is the conservative number and the gauge-seed ambiguity moves the explanation rather than the bound. With it imposed the bound becomes $1/R_5 \leq 9.22$ TeV on that surface, attained by a content whose exact global minimiser gives 9157 GeV. Untruncated, and until the true-vacuum condition is carried through the interval machinery too, that is deliberately not written as an interval: the exact-potential witness gives 9157 GeV, the untruncated potential is bounded above by 10.037 TeV *over the first seventy odd rungs*, and a global untruncated bound over all rungs remains open. It is not a bound over a screened family: on the rung where the bound is decided the condition is *necessary* as well, because the well is shallower than the quantised gap by a factor of 9755, so a content that could beat 9.22 and fail it does not exist. Writing the true-vacuum requirement out with no screen at all turns it into a semi-infinite integer program, which on that rung terminates having generated exactly one cut — the screen itself. Two further numbers belong beside it. The window $[125, 127]$ is [2]'s and its top sits 16.4 standard deviations above the measured Higgs mass; at the measured mass the bound is $1/R_5 \leq 9.09$ TeV, which is the physical-mass benchmark to quote — subject to the anchor residual and, for finite-rung quantities whose affine base point matters, to the gauge-seed condition; the asymptote (27) carries neither. And keeping the discarded tail of the expansion rather than bounding the answer with it gives 10.037 TeV, untruncated, over the first seventy odd rungs — past which the remainder bound loosens faster than the ceiling falls and says nothing, while the truncated ceiling is proved on every rung — so three rungs cost two and a half GeV rather than two and a half TeV. Along the way the five coordinates $(A_4, 8D, 2U, V, 2W)$ turn out to be *complete*: two bulk contents have the same one-loop potential, as a function of α, if and only if they agree on all five, and $\mathbb{Z}^5/L \cong \mathbb{Z}_2 \times \mathbb{Z}_{72} \times \mathbb{Z}_{2592}$. That also explains a gap we had left open, between a certificate of 10.03 and an enumeration that stopped at 9.16: the enumeration had been screening false vacua silently and the certificate was screening nothing. And if the model must also afford Part VI's escape from proton decay, the same certificate gives 3.97 TeV. The previously quoted 2.18 TeV was a bound on the *size* of the content, not on the model: eight multiplets already reach 9.14 TeV. Since the Kaluza-Klein tower is tied to $1/R_5$ the class is, *on the three-rung algebraic surface*, bounded above and therefore finitely testable — extending that to the full reduced Li_5 potential is the uniform remainder bound §13 asks for — and §10 says by which state, from the orbifold parities alone. The parity that breaks colour is the one acting on the small circle, so *every* colour-changing gauge boson is odd under it: its tower starts at $1/R_6$, which [2] is free to put at the Planck scale, and its wavefunction *vanishes* at the fixed point where that same paper localises the quarks — which answers, for the gauge channel, the rapid proton decay [2] flags in its conclusions and leaves to future work. What is left at the brane is two colour-diagonal towers, so the lightest one, whose *unshifted* tower begins at $1/(2R_5)$ — $\langle W \rangle$ moves it with α — is a colour *singlet* and not what a dijet search sees; the one that is has $M_n = n/R_5$, coupling $\sqrt{2}g_s$ and $\Gamma(G_n \to q\bar{q})/M_n = 2\alpha_s$, none of the three ours. And what it moves is the dijet *angular* distribution rather than the mass spectrum, through a colour-octet operator, so that the published contact-interaction limits, being stated for a colour-singlet one, *do not directly apply* to it — the data behind them do constrain the octet, which is what the recast below does with them; it is their quoted numbers that cannot be carried across, and not by a colour factor either, because which operator a dataset constrains more tightly turns on the flavour composition of the sample: the translation is a *calculation*, not a rescaling, and it is done rather than estimated. Carrying that operator through [39]'s own unfolded distribution, 77×77 covariance and NNLO baseline — in the spacelike channels alone, so that neither the width nor $\mathrm{BR}(G_n \to jj)$ enters — gives a nominal recast sensitivity of $1/R_5 > 10.86$ TeV, and 13.75 TeV on the mass range they set their own contact-interaction limit on. *Both lie numerically above the ceiling*, and that the two calculations cross is why the two halves of the title are one paper. It is deliberately not called an exclusion: a Gaussian χ^2 over unfolded data is not [39]'s detector-level CL_s, and §13 says what closing that gap needs. The closure tests measure it rather than assert it — the hadron-level layer reproduces CIJET's leading-order octet interference to 0.2 % bin by bin, their own NNLO through this χ^2 returns $\chi^2/\mathrm{ndf} = 0.81$, aimed at their published colour-singlet limit on their own selection it returns 17.5 against 17 TeV and 29.5 against 37, and on their *expected* limits it returns closely similar ratios — 1.08 and 0.85 against the observed 1.03 and 0.80 — so the gap to their CL_s is the method and not a fluctuation of these data; and changing the closure template from LL to the vector-like VV the tower actually is moves the ratio by two to three points and no more, so it is not a chirality artefact either. **And because $8D$ is quantised, the question can be sharpened past a continuous limit**: the per-rung ceilings form a comb of their own — one tooth per quantised $8D$, and each tooth the most a rung can reach rather than the whole of what it can reach — and reading the same likelihood at those teeth rather than at an arbitrary mass gives $\Delta\chi^2 \geq 12.0$ *everywhere on it* — read at the relaxation ceiling of the least constrained rung on the seed [2] print, 10.03 TeV at $8D = 1$, against a threshold of 3.84, with the next tooth at $\Delta\chi^2 = 244$. Two independent published computations are used as controls, and one of them — [7], whose five-dimensional single-parity model is exactly the $B = 0$ corner of our formula — reproduces our criterion, its three critical flavour numbers and one of its two published minima exactly, which narrows Part VI's unresolved anchor residual to the model-specific dictionary or to the antiperiodic sector this anchor does not reach.

Most of the machinery is not specific to this model. The ladder, the closed form and the two identities hold for any one-loop Wilson-line effective potential built from Li_5 of a phase on an orbifold — which is why §11 can rebuild [7]'s five-dimensional $SU(2)$ and $SU(3)$ results from them. *The completeness of the five coordinates is narrower and does not travel with them*: what bounds the function space is the *charge alphabet*, here $c = 2|q| \leq 3$, and the three is a box count in [2]'s largest representation rather than anything about $SU(7)$. This model's own are the model-specific arithmetic, and it separates: a mod-six congruence common to both gauge seeds considered here, and an odd-eighth corollary that depends on the gauge sector inherited from [2].

Scope, and it is stated throughout: one loop, flat space, a free Kaluza-Klein tower, a single Wilson phase — which for this model is the whole Wilson-line moduli space and not a restriction, since A_6 has no zero mode at all and A_5 has exactly the Higgs doublet — and every theory-side scale inherits Part VI's open question about the published $\alpha_{\min}$ column.

The first job is to say what a bound on the compactification scale would even mean here, and why a one-loop potential is entitled to produce one.

1 Introduction

In gauge-Higgs unification the Higgs is the extra-dimensional component of a higher-dimensional gauge field [4, 5]. Its potential is forbidden at tree level by the higher dimensional gauge symmetry and generated at one loop, where the contribution of the five-dimensional tower is finite — the second circle is another matter, and (53) quotes its logarithmic sensitivity to the cut-off rather than denying it; the Higgs mass is protected by a symmetry rather than by a cancellation. The price of that protection is the content of this paper. The potential is a function of the Wilson-line phase α, the vacuum sits at some $\alpha_{\min}$, and the compactification scale follows from it: in the model considered here, by its eq. (82),

$$\frac{1}{R_5} = \frac{2m_W}{\alpha_{\min}}, \tag{1}$$

so $\alpha_{\min}$ *is* the electroweak hierarchy. A small phase is a large scale, and there is no freedom to decouple: fixing the vacuum fixes the scale.

That makes the following question well posed, and it is the question of this paper. *How large can $1/R_5$ be, over arbitrary multiplicities of the representations [2] put in the bulk, subject to the Higgs mass coming out right?* A model class that can answer it is a model class that can be excluded.

Where this sits in the series. Parts I–III classify: which fermion completions are anomaly- and tadpole-compatible [28], which $SU(4)$ representations contain a Standard-Model generation [29], and a centre-charge selection rule on the Wilson-line potential [30]. Part IV gives the closed form that makes such a potential computable [31] and Part V implements it [32]. All five run the *inverse* direction — given an observation, what is excluded. Part VI [33] runs the direct one for the $SU(7)$ model of Komori and Maru [2]: given a content, does electroweak symmetry break? It reaches a sign, the sign of a quantity D built from the content, and stops there. **This paper is the next step and the natural one: the direct map stops being a sign and becomes a closed form, and then a number.**

And what it stands on, which is worth saying at the start rather than the end. A paper like this one is mostly other people's equations. Fifty-nine of them are quoted here, from eleven sources, and the load is not spread evenly. Twenty-five come from [2], the model under study. Fourteen come from [8], on the vacuum stability of orbifold gauge breaking. Seven come from Haba, Hosotani and Kawamura [11] and five from Haba and Yamashita [17], the two papers that made Wilson-line potentials on orbifolds something one computes rather than models. The rest arrive one and two at a time. That list is not decoration: it is kept by a gate which assigns every quoted equation to a source, in the order the text cites them, and fails if the two drift apart — a paper standing on eleven others should be able to say exactly where it is standing.

And here that turns out to be more than a courtesy. The chain from a bulk content to a number passes through one coefficient this paper does not compute: it is read off [2]'s eq. (68). §13 shows that the total gauge-plus-ghost degree-of-freedom count needed to check that coefficient — though not the parity-resolved split it would take to *derive* it — was written down twice in 2004, by [17] and by [48], and that neither paper is in the citation chain this model inherits. Which raises the second question of this paper, and it is not the one we set out to ask: *what happens to a bound when the number holding it up is one that nobody downstream ever re-derived?* The answer is in §13, and it is neither "nothing" nor "everything".

Part IV's dichotomy returns, one derivative lower down, and it is worth naming at the outset because it organises everything below. There, a graded dimension $\Sigma_\lambda(t) = s_\lambda(1, 1, t, 1/t)$ that cannot cancel was set against an index $D_\lambda(t) = s_\lambda(1, -1, t, 1/t)$ that can. Here the same pair appears as the two moments of the bulk content — one blind to the orbifold parity, positive definite; one weighted by it, and therefore an index. Everything this paper computes is a statement about how a universal kernel, the polylogarithm, pairs those two.

What is new here. Seven things, in the order they are derived.

1. A structural reason the answer cannot be a polynomial (§3): a compactified gauge theory sits at an *odd* polylogarithm index, where the finite-temperature theory sits at an even one and has Gross-Pisarski-Yaffe's exact quartic. Our ladder never lands on an even index.

2. $\alpha_{\min}$ in closed form (§5), and m_h from the same two moments (§6), so both columns of their Table 1 follow without minimising — our values of them, the $\alpha_{\min}$ one being the column §2 records as unreproduced. The same section settles what kind of expansion it is: the ladder continues past its own pole into rational territory, so there is *exactly one* logarithm in it and what is discarded is the tail of a convergent series, not an asymptotic estimate.

3. The arithmetic of the content (§7): $8D$ is odd *on the gauge seed [2] print* — conditionally, and §13 says why the condition is worth printing — so *the symmetric vacuum* is never marginal; and only that one, since D is its curvature; the broken branch has its own marginal point, the Lambert fold $W = -1$ of Figure 4, which this argument says nothing about. Also $8D \equiv 2A_4 + 3 \pmod 6$, which holds on *either* gauge seed; and the same argument at the top rung bounds the potential's *value* away from zero, which does not — it reads the parity character (63) of the gauge base point exactly as the odd eighths do. A second independent parity is what supplies the extra affine character that *can* make a rung odd — not the antiperiodic channel itself, which five dimensions already have: the same computation in the five-dimensional class [17] covers is even at all three rungs, and reaches zero with three multiplets — but which coset the offset then lands in is the separate question §13 asks. The whole ladder of the gauge sector collapses to one number, $2^pO_k = -39 + 3 \cdot 2^{p-1}$ on that seed, and that formula is also why the obstruction dies at the fourth moment there: the antiperiodic half enters with $2-2^p$, which is zero at $p = 1$, leaving the periodic weight alone to decide — and on the candidate seed that same vanishing makes the fourth moment the one *odd* rung of the three.

4. The ceiling (§8), which is the first half of the title: an integer program over a rational cone, closed by an exact two-variable dual, giving $1/R_5 \leq 10.03$ TeV for arbitrary content on the gauge seed [2] print — and the correction of a number we ourselves quoted. Asked to also afford Part VI's escape, the same certificate returns 3.97 TeV: the content with the largest hierarchy is exactly the one that cannot pay for it.

5. And the certificate is not the last word, twice over (§8, Figure 9). First because the content lives on a lattice with two more axes than the paper had been using: writing G in the basis $\{1, \ln 2, \ln 3\}$ — which §5 shows is the *complete* span of the potential's transcendentals — lifts it to $(A_4, 8D, 2U, V)$, where $\mathbb{Z}^4/L \cong \mathbb{Z}_{18} \times \mathbb{Z}_{648}$ against the projection's $\mathbb{Z}_6$, so Theorem 2 is one of *two* independent congruences and not the only one; there the condition deciding the ceiling stops being transcendental, becomes a finite integer maximisation, and shows the relaxation's vertex *empty*, sharpening the bound to 10.01 TeV attained. And second because that attaining content, evaluated on the exact potential, turns out to be a **false vacuum**. The test it fails is [8]'s orbifold-stability criterion, and the criterion, its arithmetic and its sign rule are all theirs; summed over a content it becomes a fourth linear functional, $F(1) - F(0) = \frac{31}{16}\zeta(5)W$, and *that* is what is new — not the criterion but the observation that it fits inside the same integer program. With it the ceiling is $1/R_5 \leq 9.22$ TeV, and its witness on the exact potential gives 9157 GeV with $m_h = 126.00$ GeV. It also explains the one gap this paper had been unable to account for.

6. **Which state a collider would actually bound** (§10), read off [2]'s three parity matrices and nothing else. Colour-changing gauge bosons are odd under the small-circle parity,

hence at the $1/R_6$ scale and identically zero on the quark brane, which settles for the gauge channel the proton-decay question that paper leaves open; the surviving coloured tower is the gluon's, and because the quarks sit at a point it acts through the dijet angular distribution and a colour-octet operator that the published contact-interaction limits, being stated for a colour-singlet one, do not *directly* constrain — the data behind them do, which is why what follows is a recast and not a rescaling. So it is computed here rather than borrowed: the tower's spacelike form factor, folded with parton densities and confronted with [39]'s unfolded distribution and published covariance, gives $1/R_5 > 10.86$ TeV — numerically above the ceiling of item 5, so the two calculations already cross; promoting the second to an experimental lower bound requires the detector-level likelihood, which is why the word *exclusion* is withheld for a reason §10 states and measures rather than waves at. The dictionary itself is assembled from results of others and credited as such; what is ours is that a ceiling stands over it, and a lower bound alone could never have closed the class.

7. A second, independent anchor (§11), which validates the instrument against a published one-loop computation in another group, another dimension and another decade.

Figure 1 is the whole of it on one page. What makes it a chain rather than a list is that its links are *identities*: the value and the curvature are the same rung formula at two values of k, and the ceiling sits on the quantum because the quantisation and the monotonicity are the same fact. The diagram draws the chain in the two moments the paper works in; item 5 adds a link it does not draw, and the two facts that make that link are already inside the diagram — the $\ln 2$ and $\ln 3$ of the charges, and the H_4 of the pole.

Two things that read as scope turn out to be results once measured, and both are stated where the scope is (§12). The Wilson-line moduli space of this model is *exactly* the Higgs doublet — fourteen zero modes in A_μ, four in A_5 and none at all in A_6 — so minimising in one phase is minimising over all of it, not a restriction we chose. And every α used here sits inside the radius of convergence by a factor of at least 2.7.

Before any of it: which equations are inherited, which are ours, and the one caveat that every theory-side scale in this paper carries.

2 What is inherited, and what is computed

The model of [2] is six-dimensional $SU(7)$ on $S^1/\mathbb{Z}_2 \times S^1/\mathbb{Z}_2$. Its five-dimensional ancestor is [12]; its six-dimensional sibling is the $SU(4)$ construction of [3] on $T^2/\mathbb{Z}_2$, which predicts $\sin^2\theta_W = 1/4$ at the compactification scale and is the control §13 uses for the degree-of-freedom count. [2]'s orbifold parities are their eqs. (11)–(13). Their one-loop potential, their eqs. (63)–(68), is a sum over the bulk content of polylogarithms of the Wilson-line phase; in the form we use it,

$$F(\alpha) = \sum_{\text{terms}} m \,\mathrm{Re}\,\mathrm{Li}_5\big(s\, e^{ic\pi\alpha}\big)\,, \qquad c \in \mathbb{Z}, \quad s = \pm 1, \tag{2}$$

And (2) is already an approximation, inherited rather than made here, which has to be said before anything is called exact. [2]'s potential before their eqs. (68) and (71) carries $\coth(\pi k n)$ and $\mathrm{csch}(\pi k n)$ with $k = R_5/R_6$, and the n^{-5} form is what those become as $k \to \infty$; their model takes $R_5 \gg R_6$ throughout, with $1/R_6$ a free parameter they suggest may be the Planck scale. **So everywhere below, "exact" means exact with respect to** (2), the potential of [2] already reduced in that limit, and never with respect to the full six-dimensional one. The correction is suppressed by $e^{-2\pi k}$ and is numerically nothing for any $1/R_6$ the model wants; what it changes is the *status* of several statements, and they are stated with that scope. In particular Theorem 3's five coordinates are complete for the Li_5 space of (2), not for the full

the Higgs *is* the Wilson-line phase

so $\alpha_{\min} = 2m_W R_5$, and the electroweak hierarchy is a *minimum of a potential*, not a ratio of scales

one identity splits that potential in two: $\sum_n s^n/n^p = 2^{-p}\zeta(p) + s\,(1-2^{-p})\zeta(p)$

a parity-blind half S and a parity-sensitive half Δ, and nothing else of the content is seen

§4 — the split is one $\mathbb{Z}_2$ under four names: even/odd winding, S against Δ, the two orbifold parities $\eta\eta'$, and the identity/coset multiplicities. It is not four structures.

three rungs of that *same* identity: $O_k = \dfrac{S_k + (2^p-1)\Delta_k}{2^p}$, $p = 5-2k$

$D = O_1$ and $F(0) = \zeta(5)O_0$ — the curvature and the value are one formula read twice

§4, §7 — the 7 and the 31 are 2^p-1 at $p=3$ and $p=5$. At $p=1$ the periodic pole survives uncancelled: that is the logarithm, and why no polynomial exists.

and its moments are **quantised**: $8D = S + 7\Delta$ odd, $8D \equiv 2A_4 + 3 \pmod 6$

so $D \neq 0$: the symmetric point is never marginal, and $|D| \geq \frac{1}{8}$ (KM seed)

§7 — the odd eighths are absent in the five-dimensional class tested, and here need the seed [2] print: the second parity enables them, and on the candidate seed of §13 they fail anyway. The mod-six law follows from $c^4 \equiv c^2 \pmod 3$, which is Fermat, and holds on either seed.

pinning m_h kills the logarithm: $x^2 = \dfrac{12\zeta(3)D}{6\mu + A_4}$

content in, scale out — with no minimisation anywhere

§8 — two logarithm-free identities, verified to 6×10^{-14} on the five published rows.

the ceiling is monotone in D, so its maximum sits *on* the quantum

$1/R_5 \leq 10.01$ TeV; 9.22 with the true vacuum; 3.97 with the escape

all three in our normalisation, and all three sit outside the range the five published rows anchor

§8, §9 — an integer programme over a rational cone, closed by a two-variable dual whose vertices are enumerated exactly. The Kaluza-Klein gauge bosons sit at $1/R_5$, so on the three-rung algebraic surface the class is bounded above and finitely testable in one machine.

and there the chain stops, on *two* inherited inputs

the odd-$8D$ branch is conditional on [2]'s gauge seed (§13); on the candidate seed $8D$ is even, the quantum $\frac{2}{8}$, the ceiling 7.38 TeV — and every theory-side scale carries the anchor residual, finite-rung ones g_4 too

Figure 1: The chain, and it is one chain. Every link is an identity rather than a step. The Higgs being a phase makes the hierarchy a vacuum; one Bernoulli-free identity splits that vacuum's potential into a parity-blind and a parity-sensitive moment and sees nothing else of the content; the value, the curvature and the fourth moment are that one identity at $p = 5, 3, 1$; the resulting invariant is quantised, and *oddly* so on the seed [2] print, which is what keeps it from zero; pinning the measured Higgs mass then removes the logarithm and leaves the scale as an algebraic function of two moments; and because the ceiling is monotone in D, the quantisation is what puts its maximum on the smallest allowed D — Theorem 1 doing physical work rather than arithmetic. The red note is the scope that travels with both numbers; the dashed box at the foot is not a link but the edge of the chain.

winding weight, which mixes powers and does not close on five — §8 says so where the function space is counted, and it is the same caveat.

And "bulk content" means one thing throughout, so it is worth fixing here. It is a vector of non-negative multiplicities over the *eight* (representation, parity) pairs of [2]'s Table 1 — **7**, **28**, **48**, **84** at two parities each. Arbitrary below always means arbitrary in the multiplicities, never in the list. That distinction earns a sentence because §8 removes a cap on the *size* of the content and this one, on its *type*, stays: the Wilson line sees a weight through $c = 2|q|$ with q set by the number of index-7 boxes, so $c \leq 3$ here *because their largest representation has three boxes* — checked against the term tables themselves in `content_scope.py` — and nothing in the model, the orbifold or the potential caps the box count. A bulk field in a four-box representation of $SU(7)$ would bring $c = 4$, and it would move two things: A_4 collects mc^4, so a charge-four state is worth 256 against a charge-three state's 81 and the certificate of §8 would run over a different cone; and the function space of §5 would grow, because the duplication chain that stops at $c = 3$ would continue, taking eight functions to six dimensions instead of six to five. Theorem 3 is therefore a statement about *this* charge alphabet. None of that is a defect of what follows; all of it is scope, and scope has to be printed.

Each bulk multiplet contributes a table of terms (m, s, c): an integer charge c under the Wilson-line direction, a sign s recording whether that state's Kaluza-Klein tower is periodic or antiperiodic, and a multiplicity m. The gauge sector contributes $(-1, +1, 2)$, $(-2, +1, 1)$, $(-\frac{7}{2}, -1, 1)$, and those three multiplicities are *reconstructed* from their eqs. (63)–(67) rather than assumed — inferred backwards from their coefficients, never derived forwards from the geometry (§13): Part VI reconstructs the three coefficients of their eq. (68) from two unknowns. Three equations for two unknowns is overdetermined, so it could have failed — but only in one place. Two of the three rows contain no $P_6 = -1$ pair at all, so the spare equation tests the *periodic* weight and nothing else. The antiperiodic weight is fixed once and checked never. Both weights carry below, and it takes both: what decides the parity is $2n_+ + \frac{1}{2}n_-$ over the row that mixes them, which needs the periodic part even *and* the antiperiodic part odd.

The moments. Writing $A_2 = \sum_{s=+1} mc^2$, $B_2 = \sum_{s=-1} mc^2$, $A_4 = \sum_{s=+1} mc^4$, $B_4 = \sum_{s=-1} mc^4$ and $A_4^L = \sum_{s=+1} mc^4 \ln c$, the three combinations that this paper is about are

$$D = A_2 - \tfrac{3}{4}B_2\,, \qquad A_4\,, \qquad G = A_4 H_4 - A_4^L - \ln 2\, B_4\,, \qquad H_4 = \tfrac{25}{12}\,. \tag{3}$$

D is Part VI's curvature at the symmetric point, $V''(0) = -\pi^2\zeta(3)D$; the factor $\frac{3}{4}$ is $\eta(3)/\zeta(3)$ and comes from their eq. (67), not from a choice of ours.

The caveat, stated once and carried to the end. Part VI §7 records an unresolved discrepancy: our assembly of their potential does not reproduce their published $\alpha_{\min}$ column. The row-wise ratio $\alpha_{\min}^{\text{ours}}/\alpha_{\min}^{\text{theirs}}$ runs from 1.03 to 2.08; grouped, it is 1.94 on the two rows containing a **48** and 1.20 on the other three. That question is not settled here, and §11 does not settle it either — but it does move it: the second anchor validates the potential and the minimiser in the *periodic* $B = 0$ sector against a different published computation, which narrows what remains to be explained to the model-specific dictionary *or* to the antiperiodic sector, which no published check anchors and §13 says why. **Every scale below that is inferred through the Higgs-potential dictionary carries that residual**, which is every absolute number on the theory side of this paper. It is *not* the collider recast of §10: (56)'s 10.86 TeV and the 13.75 beside it are read off [39]'s data through the tower's form factor and never pass through $\alpha_{\min}$ at all. The comb they are read *at*, (57), is theory-side and does carry it. What also does not carry it is the *matter* sublattice of §7: the eight generators are group theory, and the anchor residual cannot touch them. The affine offset those generators are added to is a separate matter, and it is §13's — *the lattice is a property of the content; the coset it sits in is a property of the gauge seed*, and the distinction is not pedantic, because moving only the base point is exactly what turns $8D$ from odd to even.

And the residual is not a *band*, which changes how far it may be carried. A band is unexplained scatter and propagates as a multiplier. This does not scatter: sorted by D the five ratios fall 2.076, 1.803, 1.286, 1.286, 1.025, monotonically and with the tie in D carrying a tie in the ratio — an ordering shared by only 2 of the 120 pairings of these five ratios to these five rows, so $p = 0.017$. It is a trend in the content, and it rises as D falls. Two consequences, and both bear on §8. First, *which* moment drives it is not decided here: A_4 orders the same five rows exactly as well as D does, the two being nearly collinear over them, and five points cannot separate them. Second, and this is the sharp one, the ratio may only be carried to contents the five rows actually surround. They anchor $D \in [1.125, 3.625]$ and $A_4 \in [189, 271]$; the ceiling of §8 is certified at $(A_4, 8D) = (215, 1)$, whose D is nine times below the smallest anchored one, and the escape ceiling at $(727, 11)$, whose A_4 is 2.7 times the largest. **Each ceiling is anchored in one moment and unanchored in the other**, and with the two moments unseparated neither can inherit the range as an uncertainty. Where a rescaled number appears below it is therefore labelled an extrapolation, not a bound. Measured in `anchor_trend.py`.

And now the structural question, which decides what kind of answer is available at all: why is there no exact formula for this vacuum, when the analogous thermal problem has had one since 1981?

3 Why there is no polynomial: Bernoulli against Clausen

A Wilson-line potential is a *holonomy* potential. It is the same object Gross, Pisarski and Yaffe wrote for a gauge theory at finite temperature [6]: a sum over windings of the compact direction, weighted by the holonomy eigenvalues. The two problems differ in one index, and the index decides whether the answer is a polynomial.

The classical fact is a Fourier series. For $0 \le x \le 1$,

$$\sum_{n\geq 1} \frac{\cos 2\pi n x}{n^{2k}} = \frac{(-1)^{k+1}(2\pi)^{2k}}{2\,(2k)!}\, B_{2k}(x)\,, \tag{4}$$

a polynomial with rational coefficients; while the same series with an *odd* exponent is a Clausen function, for which no such expression is known. The tell is the value at $x = 0$: for even p, $\zeta(p)/\pi^p \in \mathbb{Q}$ is a theorem; for odd p it is not known either way, and an integer-relation search at 40 digits finds no relation for $\zeta(3)$ or $\zeta(5)$ while returning 1/6, 1/90 and 1/945 for $\zeta(2)$, $\zeta(4)$ and $\zeta(6)$. **That asymmetry is evidence and motivation, not proof**, and it is not what the section rests on: a 40-digit null is a statement about a search, and while $\zeta(3)$ itself is irrational [26], the irrationality of $\zeta(3)/\pi^3$ does not follow and is open. What is rigorous is one line further down — the $p = 5$ potential carries an $x^4 \ln x$ branch point, established in §4 from the expansion itself, and *no polynomial has a branch point.* The arithmetic of $\zeta(3)$ and $\zeta(5)$ says why one should have expected it; the branch point is why there cannot be one.

A gauge theory in d dimensions with one compact direction contributes $\sum_n \cos(2\pi n x)/n^d$. Hence:

theory	index	parity	closed form of the holonomy potential
four-dimensional, finite temperature	4	even	$B_4(x) = x^2(1-x)^2 - \frac{1}{30}$ — **exact**
five-dimensional on S^1 [7, 17]	5	odd	Clausen; $\zeta(5)$, $\zeta(3)$; no polynomial
six-dimensional on $S^1 \times S^1$ [2]	5	odd	the same

and the Gross-Pisarski-Yaffe quartic is recovered from (4) to 10^{-41} by the ancillary script.

The ladder of this paper is $p = 5 - 2k$, so it runs 5, 3, 1: odd, odd, odd. Stepping by two from an odd start cannot reach the solvable case. There is no rung at which this potential becomes a polynomial, and the last rung is the pole of ζ — which is exactly where the $\alpha^4 \log \alpha$ branch point of §5 comes from.

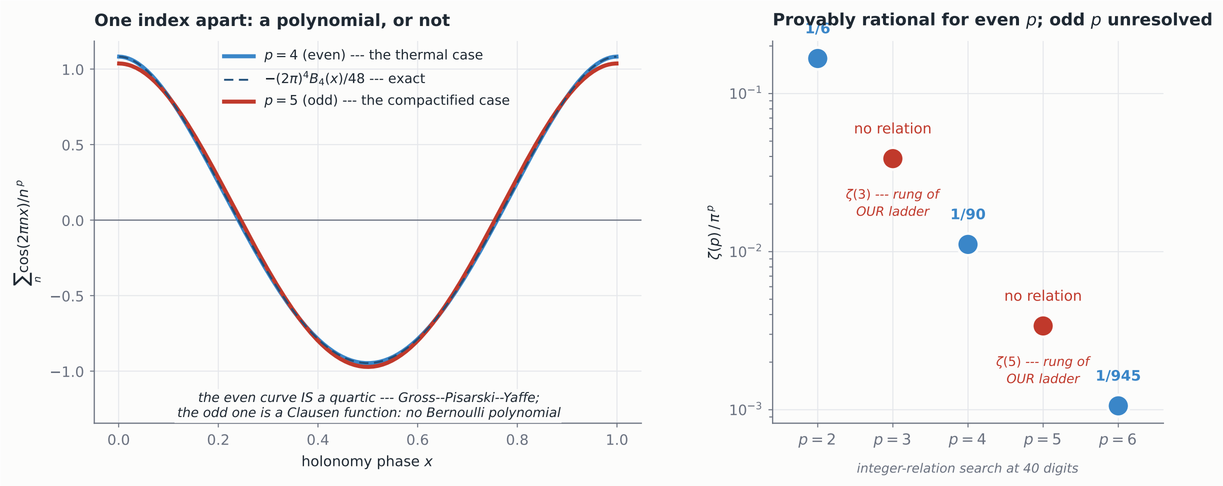


Figure 2: One index apart. The same Fourier sum at an even and an odd exponent. At $p = 4$ it *is* a quartic — the dashed curve is $-(2\pi)^4 B_4(x)/48$, the Gross-Pisarski-Yaffe potential, and the two coincide to 10^{-41}. At $p = 5$ no such expression exists. Right: the arithmetic tell. $\zeta(p)/\pi^p$ is rational for even p, a theorem; for odd p it is open, and an integer-relation search at 40 digits returns $1/6$, $1/90$ and $1/945$ and finds nothing for $\zeta(3)$ or $\zeta(5)$ — which are the two rungs of our ladder. Evidence, not proof: what rules out a polynomial rigorously is the $x^4 \ln x$ branch point.

This is worth stating plainly because it explains a literature gap rather than merely noting one. The finite-temperature community never needed an expansion of the holonomy potential: it has the polynomial, and every question about the vacuum is algebra. The extra-dimensional community works at an odd index, has no polynomial, and computes numerically model by model. A closed form for the vacuum is the object that is missing, and the reason it is missing is arithmetic rather than technical. Figure 2 carries both halves of that: the quartic that exists at $p = 4$, and the integer-relation search that finds nothing at $p = 5$.

With the shape of the answer settled, the expansion itself: one series, read rung by rung, each pairing a moment of the content against a zeta value. Three rungs carry the physics, and where the fourth would go there is a pole — which turns out to be the last place a new transcendental can enter.

4 One expansion, three rungs

Expanding (2) in the phase is a single operation:

$$\operatorname{Re}\operatorname{Li}_5\!\left(s\, e^{ic\pi\alpha}\right) \;=\; \sum_{k\geq 0} \frac{(-1)^k (c\pi\alpha)^{2k}}{(2k)!} \sum_{n\geq 1} \frac{s^n}{n^{5-2k}}\,, \tag{5}$$

so the $2k$-th derivative in α is governed by $p = 5 - 2k$ *and* by the $2k$-th moment of the content. The inner sum splits once and for all, because $s^n = +1$ on even windings whatever s is:

$$\sum_{n\geq 1} \frac{s^n}{n^p} \;=\; 2^{-p}\zeta(p) \;+\; s\,(1 - 2^{-p})\,\zeta(p)\,. \tag{6}$$

Summed over the content, the two brackets collect the two moments at once — one blind to the parity, one weighted by it — and the rung-k coefficient is, with the last column of the table below read on the gauge seed [2] print,

$$\frac{S_k + (2^p - 1)\,\Delta_k}{2^p}\,, \qquad S_k = \sum m\, c^{2k}\,, \qquad \Delta_k = \sum m\, c^{2k} s\,. \tag{7}$$

k	p	object	coefficient	gauge sector
0	5	the *value* $F(0)$	$(S_0+31\Delta_0)/32$	9
1	3	the *curvature* $V''(0)=-\pi^2\zeta(3)D$	$(S_1+7\Delta_1)/8$	−27
2	1	the fourth moment	$(S_2+\Delta_2)/2=\mathcal{A}_2=A_4$ — periodic only	−36

The eighths of Part VI are 2^3 and its seven is 2^3-1; the denominators $32, 8, 2$ are 2^p. At $p=1$ both halves of (6) diverge, and what becomes of them depends on the parity: at $s=-1$ they cancel and leave a *finite* $-\ln 2$, while at $s=+1$ they add and the pole survives. **So the $x^4 \ln x$ is the *periodic* sector's uncancelled pole**, and the antiperiodic sector contributes instead the $\ln 2$ that (3) carries on B_4. Li_5 has exactly one logarithm in x, and it is periodic-only — which is the $p=1$ row of the table above, read as an analytic statement rather than a bookkeeping one.

The ladder is geometric, and charge two is its fixed point. Since $2^p = 32/4^k$, a single term (m, s, c) walks the ladder as a geometric sequence:

$$s=+1:\quad 32\,m\,\sigma^k \ \text{ with } \ \sigma=\tfrac{c^2}{4}; \qquad s=-1:\quad 2\,m\,(c^2)^k - 32\,m\,\sigma^k \quad \text{— two modes.} \tag{8}$$

So $O_k = \sum_\sigma A_\sigma\, \sigma^k$, with $\sigma \in \{\frac{1}{4}, 1, \frac{9}{4}\}$ from $c=1,2,3$ and, for antiperiodic states only, additionally $\sigma \in \{1, 4, 9\}$.

A *charge-two* periodic state has $\sigma=1$: it weighs exactly the same on the value, on the curvature and on the fourth moment. Charge one decays by four per rung; charge three grows. The gauge sector reaches only $\{\frac{1}{4}, 1\}$, so its ladder has two parameters rather than three, and

$$O(5)-O(3) \;=\; 4\left[\,O(3)-O(1)\,\right]$$

identically — zero counterexamples over all 4096 admissible parity assignments. Its two amplitudes are computed in §7, (19), where the identity becomes a line of algebra rather than a census.

The geometric relation survives any matter content that opens no third mode, and dies as soon as one appears. Predicted before the table was computed and confirmed on all eight multiplets: it holds for $\mathbf{7}^{(\pm)}$, $\mathbf{28}^{(+,+)}$ and $\mathbf{48}^{(+,+)}$, and breaks for $\mathbf{28}^{(+,-)}$ and $\mathbf{48}^{(+,-)}$ — which differ from their partners only by whether their charge-two state is periodic or antiperiodic — and for both **84**s, the only multiplets carrying a charge three. Every row of their Table 1 contains an **84**, so for every published row the value, the curvature and the fourth moment are three independent numbers; without one, they are two.

And that last difference is sharper than it looks: the charge-two parity flip is the only log-free one. Move one unit of multiplicity at charge c from the periodic tower to the antiperiodic one. The three moments shift by

$$\Delta D = -\tfrac{7}{4}c^2, \qquad \Delta A_4 = -c^4, \qquad \Delta G = -c^4\Big(H_4 - \ln\tfrac{c}{2}\Big),$$

and ΔG is *rational* exactly when $c=2$, because $\ln c$ is then the same $\ln 2$ that the antiperiodic tower carries universally in (3). So at charge two, and at no other charge, the parity flip is log-free, with the exact signature

$$(\Delta D,\ \Delta A_4,\ \Delta G) \;=\; \big(-7,\ -16,\ -\tfrac{100}{3}\big) \text{ per unit,} \tag{9}$$

and, equivalently, $J \equiv G - H_4 A_4 = -A_4^L - \ln 2\, B_4$ is completely blind to the parity of a charge-two state. That this is the same charge that sits at the fixed point of (8) is not a consequence of it: $\sigma = 1$ is about $c^2/4$ and this is about $\ln(c/2)$, and the two coincide only because both ask c to be 2.

It suggested a test of Part VI's anchor residual, since the rows that fail are the ones containing a **48** and the parity of a charge-two state is what distinguishes $\mathbf{28}^{(+,+)}$ from $\mathbf{28}^{(+,-)}$ and $\mathbf{48}^{(+,+)}$ from $\mathbf{48}^{(+,-)}$: if the residual were a misread charge-two parity, then their published $(\alpha_{\min}, m_h)$ would sit at our moments shifted by an integer multiple of (9). Solving that multiple from (II) and then checking (I), which the solve never saw, gives non-integer multiples between 0.03 and 0.15 and residuals in (I) of order 10^{-1} to 1; two decoy vectors, the log-carrying $c = 1$ and $c = 3$ flips, do no worse. **The test is negative and the reading is withdrawn**: whatever the anchor residual is, it is not a charge-two parity. It is recorded because the vector is exact and cheap to re-test against any future candidate.

But the log-free property itself outlives the test it was invented for, and it is not an accident of $c = 2$ being small. §8 writes G in the basis $\{1, \ln 2, \ln 3\}$ and finds the content sitting on a *four*-dimensional lattice whose logarithmic coordinates are the $\ln 2$ and $\ln 3$ weights U and V. In that lattice the charge-two flip is the move with $\Delta U = \Delta V = 0$ — the unique generator direction the third axis cannot see — and *that* is why its ΔG came out rational: with no logarithm moving, ΔG can only be $\frac{25}{12}\Delta A_4$, which is $-\frac{100}{3}$. Measured in `crossed_equations.py` and `semigroup.py`.

The top rung is not ours, and neither is more of it than we first took. Cacciapaglia *et al.* [8] decide orbifold stability by comparing the potential at $a = 0$ and $a = 1/2$, and their eqs. (2.11)–(2.13) give $F_-(0) = -\frac{15}{16}F_+(0)$, with $\frac{15}{16} = \eta(5)/\zeta(5)$, the sign deciding whether the orbifold is stable — a weight already in print in [14] eq. (13) in 2001, as $V^{-,\text{scalar}} = -\frac{15}{16}V^{+,\text{scalar}}$. The *split* it comes from is in six dimensions too, in [15] eq. (3.18), where the two orbifold parities enter as $\mathrm{Li}_n(\pm q)$ — but not the weight: that paper never evaluates the ratio, and the strings 15/16, 31/32 and $\zeta(5)$ do not occur anywhere in its forty pages. **The parity-split architecture is theirs and is not claimed here.** And their eq. (2.11) says more than that value: $F_-(a) = -F_+(a) + \frac{1}{16}F_+(2a)$ is an identity in a, the duplication formula [13] rather than (6) at a point, and §5 uses it as such. Read only at $a = 0$ it is the top rung; read as a function it is why the potential spans five. What their method does not reach is the curvature: over 45 pages the words *curvature* and *second derivative* do not occur, nor *Dynkin* nor *trace*, and their single second derivative (their eq. (3.15)) is taken in an example where the potential has already collapsed to one term, so no parity split appears in it. Their own text says why they do not go there: small vacuum expectation values, they write, are "usually obtained via some form of fine tuning" — and the small-α interior minimum they set aside is precisely the regime of [2] and of this paper. §12 states the boundary in full.

Their stability criterion is theirs too, and it returns in §8: comparing the potential at the two symmetric points is what decides whether the small-α minimum this paper computes is the vacuum or merely a vacuum, and [8] state it, prove the identities it needs and draw its sign rule. (34) is that statement summed over a bulk content, in our notation and with nothing added. The one thing that is ours is the use: summed, it is *linear* in the multiplicities, so it can go inside the integer program that produces the ceiling — and the ceiling is computed twice for exactly that reason, the two answers differing by 0.8 TeV. Figure 3 is the rung structure the curvature adds, with the gauge sector's own three rungs beside it.

The curvature rung is where the vacuum is decided, so that is the rung to solve.

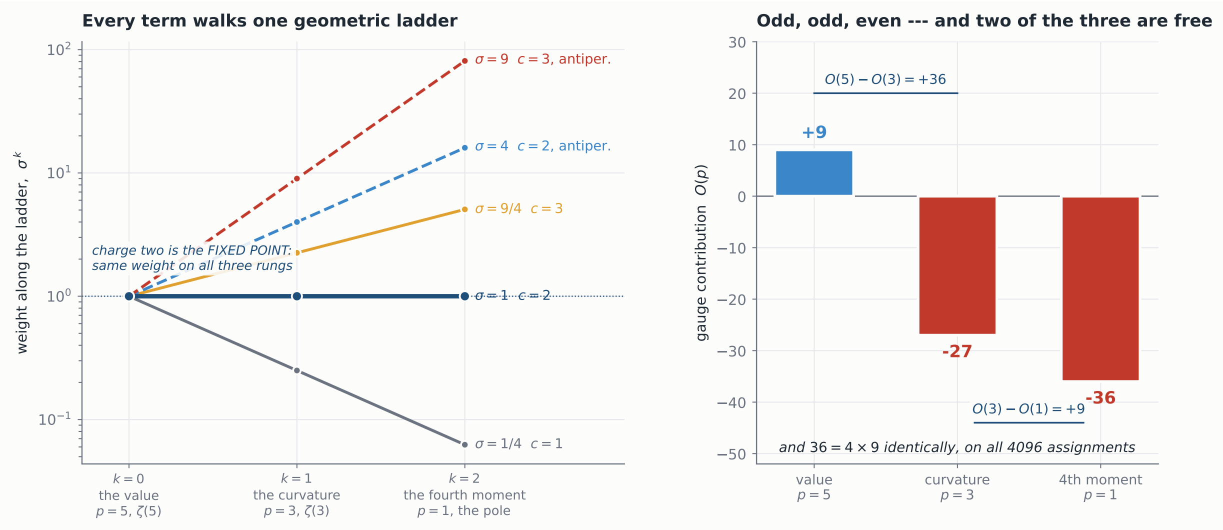


Figure 3: The ladder is geometric. Left: every term of the potential walks the three rungs as σ^k with $\sigma = c^2/4$, and antiperiodic terms carry a second mode $\sigma = c^2$. A periodic charge-two state has $\sigma = 1$: it weighs the same on the value, the curvature and the fourth moment, and is the flat spine everything else fans away from. Right: the gauge sector's own three rungs on the seed [2] print, odd, odd and even. It reaches only $\sigma \in \{1/4, 1\}$, so two of the three numbers determine the third — on every one of the 4096 admissible parity assignments.

5 The hierarchy in closed form

Keeping the three rungs of §4 and writing $x = \pi\alpha$,

$$F(x) \;=\; \text{const} \;-\; \frac{\zeta(3)D}{2}\,x^2 \;+\; \frac{x^4}{24}\big[\,G - A_4 \ln x\,\big] \;+\; O(x^6)\,, \tag{10}$$

the logarithm being the $p = 1$ rung. The symmetric point is therefore *stationary but not analytic*: $x^4 \ln x$ and its first two derivatives vanish at the origin, so $F'(0) = 0$ and $F''(0) = -\zeta(3)D$ are perfectly well defined — which is what entitles Theorem 1 to speak of the curvature there at all — while the branch point sits two derivatives higher up and no ordinary Taylor series exists. And $\mathrm{d}F/\mathrm{d}x = 0$ gives, not a root of a polynomial, but

$$\boxed{x^2 \;=\; \frac{24\,\zeta(3)\,D}{4G - A_4\,(4\ln x + 1)}\,, \qquad x = \pi\alpha_{\min}.} \tag{11}$$

The hierarchy is a ratio of two moments of the content, corrected by a third.

And it is a closed form, which we had been calling a fixed point. Earlier versions of this paper solved (11) by iteration and described it as a fixed-point equation rather than a solvable one. It is solvable. Putting $y = x^2$, so that $\ln x = \frac{1}{2}\ln y$, it is $y\,(b - \ln y) = d$ with $b = (4G - A_4)/(2A_4)$ and $d = 12\zeta(3)D/A_4$; setting $u = \ln y - b$ turns that into $u\,e^u = -d\,e^{-b}$, which is the definition of the Lambert W function. Hence

$$\boxed{x^2 \;=\; \frac{-d}{W(-d\,e^{-b})}\,, \qquad b = \frac{4G - A_4}{2A_4}\,, \qquad d = \frac{12\,\zeta(3)\,D}{A_4}\,.} \tag{12}$$

A negative argument is necessary for two real branches and not sufficient: W is real only for $z \geq -1/e$, which here reads $b - 1 - \ln d \geq 0$. That condition holds on every content used here, and it is not an extra hypothesis. **Which branch is the minimum is not a labelling but a theorem**, and one line of algebra: eliminating x^2 between (12) and identity (II) of (22) gives

$$\mu \;=\; -\frac{A_4}{6}\,(W+1)\,, \qquad \text{equivalently} \qquad W_{-1} = -1 - \frac{6\mu}{A_4}\,, \tag{13}$$

with $\mu = m_h^2/(K\pi^2)^2$. Since $A_4 > 0$, the branch with $W \le -1$ carries $\mu \ge 0$ and the branch with $-1 < W < 0$ carries $\mu < 0$: *the sign of the Higgs mass squared is the sign of* $-(W+1)$. So W_{-1} is the minimum and W_0 a maximum, proved rather than assigned; on the five published rows μ runs 81 to 115 on the first and -30 to -38 on the second. It also says the measured Higgs mass *selects a point of the Lambert sheet*, and that $m_h \to 0$ is $W \to -1$ — the branch point, which is the fold Theorem 1 is careful not to claim. Verified in `lambert_criticality.py`. **Neither is one of the two symmetric points $\alpha = 0, 1$ that §8 compares**: those are endpoints of the domain, their difference (34) is evaluated exactly and not through this expansion, and the outer root plays no part in the ceiling. Checked against the iteration on the five published rows, agreeing to 1.8×10^{-14}, in `lambert_amin.py`. We owe a piece of our own narrative with it: §3 rules out a *polynomial*, which the branch point does rigorously, and that is all it rules out. It never ruled out a closed form, and the same W was already in this paper one section later, at (27), before the equation it belongs to was recognised.

And in the right coordinates the whole family is one picture. Put $L = b - 1 - \ln d$, the discriminant itself — two real branches exist exactly when $L \ge 0$. Then $b = L + 1 + \ln d$ and the argument of the Lambert function loses its second parameter outright, $z = -d\,e^{-b} = -e^{-(1+L)}$, so (12) separates:

$$x^2 \;=\; d\, f(L), \qquad f(L) = \frac{-1}{W\big(-e^{-(1+L)}\big)}\,, \tag{14}$$

one factor per coordinate. The two branches are then two sheets of one surface over (L, d), and they meet where $W = -1$, that is $f = 1$, that is the *straight* edge $L = 0$ — along the whole of which $\mu = 0$ by (13). **The closed form is the upper sheet of a fold, and the Higgs mass is the height above its edge.** Figure 4 is that surface, with their five published rows and the vertex where §8's ceiling is attained standing on it.

Verification, against our own minimisation and not against their column. What is tested here is the expansion, not the anchor: a disagreement would be a defect of (11) and of nothing else. Over 272 synthetic contents spanning twelve values of D and $\alpha_{\min} \in [0.018, 0.119]$ — the range of their Table 1 — the median error is 0.13 %, and 0.004 % at the small-α end where the truncated tail is smallest. On the five published rows:

row	D	A_4	numerical $\alpha_{\min}$	closed form	error
(1)	15/8	258	0.055304	0.055423	0.215 %
(2)	29/8	271	0.083062	0.083579	0.623 %
(3)	9/8	189	0.043593	0.043621	0.065 %
(4)	11/8	223	0.046879	0.046929	0.107 %
(5)	15/8	255	0.055277	0.055393	0.209 %

Median 0.209 %, maximum 0.623 % — and the maximum falls on the row with the largest α, which is where the truncated tail is largest. Both are well inside the two significant figures to which that table is printed.

Three controls run alongside, and the last is the one that could have failed. The D column comes out in exact eighths. The published ratio pattern of Part VI §7 is reproduced: 1.939 on the two rows with a **48** against 1.199 on the other three. And *dropping the logarithm* — using $x^2 = 24\zeta(3)D/4G$, which is what one would write without §3 — fails by +104 to +174 %. The branch point is not a decoration.

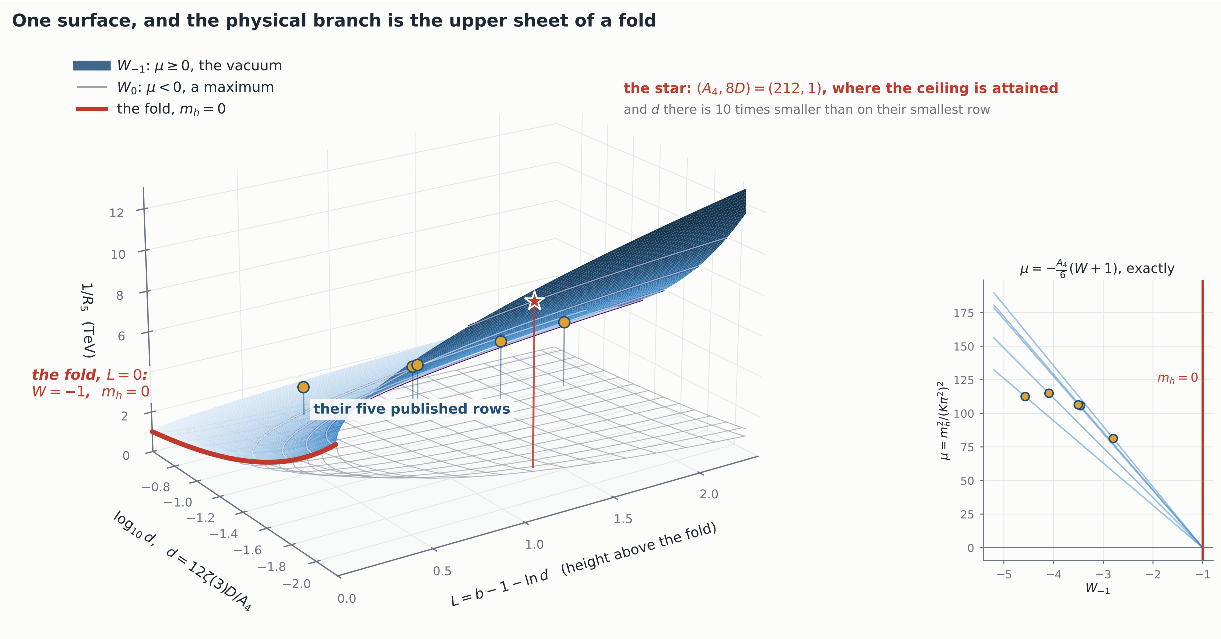


Figure 4: The closed form is one sheet of a fold. $1/R_5 = 2\pi m_W/x$ over the two coordinates of (14): $L = b - 1 - \ln d$, the discriminant, and $d = 12\zeta(3)D/A_4$, on a logarithmic axis because the certificate's vertex and their published rows differ by a factor of ten in it. The blue sheet is W_{-1}, where $\mu \geq 0$ and the stationary point is the vacuum; the open grey mesh below it is W_0, where $\mu < 0$ and the same point is a maximum. They meet along the red edge $L = 0$, where $W = -1$ and therefore $m_h = 0$: this is the *Lambert* branch point, where the electroweak minimum and the outer root coalesce, and not the $\alpha^4 \ln \alpha$ one at the origin that §3 turns on — it is the fold Theorem 1 is careful not to claim, drawn rather than described. Amber discs are [2]'s five rows; the star is $(A_4, 8D) = (212, 1)$, the vertex the ceiling is attained at. **The star is not at** 10.03 **TeV and should not be**: the surface returns each content's *own* stationary point, 9967 GeV there, while the certificate reads (22)(II) at the top of the Higgs window instead — a 0.7 % difference, and the whole of it is which m_h is being asked for. *Inset:* (13) as the straight line it is, one line per row, all five meeting at $W = -1$, $\mu = 0$. Drawn by `make_fig_lambert.py`, which refuses to draw unless the surface returns every plotted $\alpha_{\min}$ — it does, to 2×10^{-14}.

And the truncation is a tail, not an asymptotic estimate — which is a stronger statement than it looks, and it is what §8 will need. The ladder does not stop at $p = 1$. Its index continues to $p = -1, -3, -5, \ldots$, where ζ is *rational* by Bernoulli, so past its own pole the expansion generates no further transcendentals: the rung at order x^{2k} is $\zeta(5 - 2k)\,[\,A_{2k} + (2^{2k-4} - 1)B_{2k}\,]$, which is (18) read at negative p, and the antiperiodic weight runs $3, 15, 63, 255, \dots = 2^{2k-4} - 1$. **There is exactly one logarithm in the whole expansion**, at order x^4, and every term beyond it is rational in the moments; the transcendental content of the potential is $\zeta(5)$, $\zeta(3)$, $\ln x$, $\ln 2$ and $\ln 3$, and nothing else. This is the classical expansion of $\mathrm{Li}_n(e^{\mu})$ about $\mu = 0$ [25], valid for $|\mu| < 2\pi$, and its $H_{n-1} - \ln(-\mu)$ term at $n = 5$ is the $H_4 = \frac{25}{12}$ already carried by G in (3) — the same constant, arriving by a second route. Once that single logarithm is extracted the remainder is therefore the tail of a *convergent* series, with radius fixed by the nearest singularity of $\mathrm{Li}_5(s\,e^{ic\pi\alpha})$, at $s\,e^{ic\pi\alpha} = 1$: $\alpha = 2n/c$ for the periodic tower and $\alpha = (2n+1)/c$ for the antiperiodic one, so $|\alpha| < 1/c_{\max} = 1/3$ covers both. Measured term by term, the asymptotic ratio is $(c\alpha)^2$ antiperiodic and $(c\alpha/2)^2$ periodic, with controls on both sides of both radii — the series is convergent at $\alpha = 0.32$ and divergent at $\alpha = 0.40$ for the antiperiodic tower, and at $\alpha = 0.40$ and $\alpha = 0.70$ for the periodic one. Every α this paper uses is inside $1/3$ by a factor of at least 2.7. **So the error committed here is a convergent remainder and admits a rigorous bound rather than an estimate**, which is what the companion bound of `remainder_theorem.py` supplies. Measured in `crossed_equations.py`.

That the transcendental content is exactly $\{\zeta(5), \zeta(3), \ln x, \ln 2, \ln 3\}$ is not a curiosity in passing; it is a coordinate system, and §8 spends it. Because the only logarithms of *numbers* are $\ln 2$ and $\ln 3$ — charges run to three and no further — the moment G decomposes exactly in the basis $\{1, \ln 2, \ln 3\}$, which turns the transcendental condition that decides the ceiling into an integer program. The two statements are one, read at different rungs of the same ladder.

The same expansion has a second output, and it is the column one does not expect for free.

6 The Higgs mass, from the same two moments

Differentiating (10) twice and using the stationarity condition to eliminate $G - A_4 \ln x$ produces a cancellation worth isolating:

$$F''(\alpha_{\min}) \;=\; \pi^2 \Big[2\zeta(3) D - \tfrac{1}{6} A_4 x^2 \Big]. \tag{15}$$

The logarithm and G cancel identically at the stationary point.

With their eq. (80), $m_h = K\sqrt{F''(\alpha_{\min})}/\alpha_{\min}$ and $K = (\sqrt{3}/2\pi^3) m_W g_4$, both columns of their Table 1 are functions of (D, A_4) alone, with no minimisation anywhere — and they return *our* values of the two, since by §2 the $\alpha_{\min}$ one does not reproduce theirs. Against a numerical second derivative of the same potential the closed form agrees to 0.13–1.22 % across the five rows, and the resulting m_h to 0.07–0.61 %.

A control that must fire, and does: their eq. (80) returns no real m_h where $F'' < 0$, and at *their* published α on row (3) our assembly gives $F'' = -2.45$. That is a symptom of the anchor discrepancy of §2, recorded in Part VI and not resolved here.

Both observables now come from two integers in disguise. The next question is what those integers are allowed to be.

7 The arithmetic of the content

In the sum/difference basis $S = A_2 + B_2$, $\Delta = A_2 - B_2$, the curvature rung reads $8D = S + 7\Delta$, and the seven is $2^3 - 1$ from the zeta ratio, not from $SU(7)$.

And that is not a coincidence of notation: D *is* a rung. Writing the rung of §4 as $O_k = [S_k + (2^p - 1)\Delta_k]/2^p$ with $p = 5 - 2k$ and reading it at the two values of k that this paper uses,

$$k=1: \quad O_1 = \frac{S + 7\Delta}{8} = D, \qquad\qquad k=0: \quad O_0 = \frac{S_0 + 31\Delta_0}{32}, \quad F(0) = \zeta(5)\, O_0.$$

So the curvature invariant of this section and the stability criterion of [8] are the same object evaluated at two rungs of one ladder, and the 7 and the 31 are $2^p - 1$ in each. The odd eighths of Theorem 1 and the bound $|F(0)| \geq \zeta(5)/32$ are then one statement read twice, which is why the gauge sector's numerators 9, -27, -36 at $k = 0, 1, 2$ are odd, odd, even *on the seed [2] print*: the obstruction is alive at the value and the curvature and dies at the fourth moment, where $p = 1$, the antiperiodic half cancels to a finite $\ln 2$ while the periodic pole survives as the $x^4 \ln x$. On the candidate seed of §13 the same three numerators are 18, -18, -27, and the pattern *permutes* to even, even, odd. What vanishes at $p = 1$ is the antiperiodic half, not the obstruction: it leaves the *periodic* weight alone to carry the parity, and that weight is exactly the one the fork of §13 turns on. **Read across the three rungs at once, the ladder is a *tomography* of the gauge seed**: at $p = 5$ and $p = 3$ the parity character reads the two P_6 sectors in combination, and at $p = 1$, where $2 - 2^p$ vanishes, it reads the periodic sector alone. So moving the seed does not switch the protection off — it moves it to the rung that reads the half the seed changed.

Theorem 1 (the odd eighths, conditional)**.** If *the gauge sector's contribution to* $8D$ *is odd, then* $8D$ *is an odd integer for every bulk content. Hence* $D \neq 0$ *and* $|D| \geq \frac{1}{8}$*:* the symmetric point *is never marginal at one loop, and by a quantised amount.*

$$\boxed{8D_{\text{gauge}}\ \text{odd} \quad \Longrightarrow \quad 8D \in 2\mathbb{Z}+1 \quad \Longrightarrow \quad |D| \ \geq \ \tfrac{1}{8}} \tag{16}$$

Corollary 1 (for the seed printed in [2])**.** *The gauge coefficients of [2]'s eq. (68) give* $8D_{\text{gauge}} = -27$*, which is odd. The hypothesis of Theorem 1 is therefore satisfied, and every consequence drawn from it below holds in this model.*

Why the hypothesis is separated out. It is not a formality. The proof needs two things: that matter contributes an even amount, and that the gauge sector contributes an odd one. The first is arithmetic and cannot fail. The second is not computed here — it is read off someone else's equation, and §13 shows that reading is under active question. Separating it costs nothing, since Corollary 1 restores every consequence verbatim for the published seed. What it buys is the distinction that matters: *if the seed moves, the theorem stays true and its hypothesis stops being met*, which is a different situation entirely from a theorem that fails. A conditional statement also says something the unconditional one cannot — exactly which downstream number is hostage to which upstream input.

And only the symmetric point. D is the coefficient of α^2 at $\alpha = 0$, so what is bounded away from zero is the curvature there. Whether the *broken* vacuum can be marginal — $F''(\alpha_\star) = 0$ at $\alpha_\star \neq 0$ — is a different condition, and $D \neq 0$ does not touch it. In the closed form of §5 it is the Lambert fold $W = -1$, equivalently $m_h = 0$; it sits at α several times larger than any used here, where the expansion's error scales up by the square of that ratio and, a fold being a coalescence of roots, the position is in any case not controlled by a truncation. This paper therefore neither asserts nor excludes it. Measured in `lambert_criticality.py`.

Proof. For matter, c and m are integers, so $S + \Delta = 2A_2$ is even and $S + 7\Delta = (S + \Delta) + 6\Delta$ is even — for *every* content, with no reference to the gauge sector. By hypothesis the gauge sector contributes an odd amount. Even plus odd is odd. For Corollary 1, the coefficients of [2]'s eq. (68) give $S + 7\Delta = -27$. □

The obstruction sits in the one row of the gauge table that mixes the two P_6 sectors — the $-\frac{7}{2}$ of the $P_6 = -1$ entry, reconstructed from their eqs. (63)–(67) rather than derived from the geometry (§13). **And what carries the parity is the whole coefficient of that row, not either weight inside it.** Written over its $n_+ = 2$ periodic and $n_- = 6$ antiperiodic pairs the row is $2a + 6b$, which is $2 \times 2 + 6 \times \frac{1}{2} = 7$ on the seed [2] print. The per-pair $\frac{1}{2}$ does not do it alone: $6 \times \frac{1}{2} = 3$ is an *integer*. What is odd is the *sum*, and that needs the periodic part even just as much as it needs the antiperiodic part odd — which is the parity character (63) of the affine gauge base point, and §13 is where that becomes the object the whole fork turns on. A census over all 4096 admissible diagonal parity assignments gives the same statement in group-theoretic form: D can never vanish precisely when the block left unbroken by both projections has odd dimension, and in their assignment that block is $SU(3)_C$. *On the seed [2] print,* $D \neq 0$ *because the number of colours is odd* — and that reading is a property of the weights $(2, \frac{1}{2})$ and not of the group: at $(\frac{3}{2}, \frac{1}{2})$ the same three colours are still there, the census still returns the same count, and $8D$ is even all the same (§13). Within the five-dimensional $SU(N)$, $S^1/\mathbb{Z}_2$, single-Wilson-phase class [17]'s formula covers, no odd rung occurs; the second independent parity of this product orbifold is what supplies the extra affine character that *can* make one odd. Which parity the base point then lands on is (63).

The theorem has a dimension, and the obvious reason is not the right one. In the single-parity corner of §11 every tower is integer-moded, $B = 0$, and $8D = 8A_2$ is even; it would be easy to stop there. But a five-dimensional model on $S^1/\mathbb{Z}_2$ need not sit in that corner — with an antiperiodic sector $B \neq 0$ — and $8D$ is even there too, so $B = 0$ is not what is doing the work. Running Haba and Yamashita's general five-dimensional formula [17] over every boundary condition of $SU(N)$ on S^1/Z_2 carrying a single Wilson phase, up to $N = 12$, returns no odd $8D$ at all. The mechanism is the counting of degrees of freedom: a Dirac fermion enters with 4 and a complex scalar with -2, both even, and the one odd coefficient — the -3 of the gauge and ghost sector — multiplies only the adjoint, whose weight reaches $8D$ as $-24(2+\delta_1)+18\delta_2$ with δ_1, δ_2 the block leftovers of their eq. (5.9). Here the gauge sector instead delivers the half-integer $-\frac{7}{2}$ of their eq. (68), and $6 \times \frac{7}{2} = 21$ is odd. On Haba and Yamashita's own $SU(3)$ example three multiplets suffice to lose the protection: one adjoint Dirac fermion with $\eta\eta' = +$ and two fundamental ones with $\eta\eta' = -$ give $A_2 = 3$, $B_2 = 4$ and $8D = 0$ exactly. Marginality is not a fine-tuned corner in five dimensions; it is ordinary. What forbids it here is the sixth dimension *together with the gauge seed* [2] print: the candidate seed of §13 is equally six-dimensional, has the same three colours, and reaches $8D = 0$ on the lattice. The second parity supplies the extra affine character; which coset the offset then lands in is a separate question, and it is the open one.

Theorem 2 (a congruence mod six)**.** *For every bulk content,* $8D \equiv 2A_4 + 3 \pmod 6$.

$$8D \;\equiv\; 2A_4 + 3 \pmod 6 \qquad \text{for every bulk content, and on } \textit{either} \text{ gauge seed} \tag{17}$$

Proof. Work with the single combination $8D - 2A_4$, and separate the content from the gauge sector, which enters only as a fixed base point. *Matter:* a state with $s = -1$ contributes $-6mc^2$ to $8D$ and nothing to A_4, so $8D - 2A_4$ receives $-6mc^2 \equiv 0 \pmod 6$; a state with $s = +1$ contributes $8mc^2$ and mc^4, so it receives $8mc^2 - 2mc^4 = 2mc^2(4 - c^2)$, which is even, and divisible by three because $c^4 \equiv c^2 \pmod 3$ for every integer c. Hence every bulk multiplet has $8D - 2A_4 \equiv 0 \pmod 6$, and the eight generators confirm it one by one. *Gauge:* the coefficients [2] print give $8D - 2A_4 = -27 - 2(-18) = 9 \equiv 3 \pmod 6$. Adding the two, $8D - 2A_4 \equiv 3 \pmod 6$ for every content. □

Note that the proof never mentions a parity, and that is not a stylistic preference. An earlier version closed with "modulo two the statement is Theorem 1", which is true for the published seed but makes a congruence about the lattice depend on a theorem about one gauge offset. The argument above does not: the matter sublattice is what it is, and the gauge sector enters only through the residue of $8D - 2A_4$ at its base point. That matters because the candidate seed of §13 gives $-18 - 2(-\frac{27}{2}) = 9$ — *the same* 9 — so Theorem 2 holds unchanged on either branch, **while the hypothesis of Theorem 1 is not satisfied on the candidate branch** — the theorem itself is as true there as here. *The mod-six law is the general statement; the odd-eighth conclusion of Corollary 1 followed from it under "A_4 integral", and it is A_4 turning half-integral that flips $2A_4 + 3$ from odd to even.* Which of the two is the deeper result is not a matter of taste: one survives the change of hypothesis and the other does not.
The mod-three half removes two thirds of the $(A_4, 8D)$ plane and is what makes the certificate of §8 a statement about *contents* rather than about a relaxation. Their five published rows give $8D + A_4 = 273, 300, 198, 234, 270$, all divisible by three. That divisibility is not a coincidence of the five and not the end of the statement: it is (64) read modulo three, and the same congruence holds modulo *nine* once $2U$ is carried along.

And the same argument one rung up — in fact at every rung at once. Split the table into its periodic and antiperiodic halves. **The subscript here is the *rung*, not the power,**

so write them calligraphically to keep them apart from the moments of (3): $\mathcal{A}_k = \sum_{s=+} mc^{2k}$ and $\mathcal{B}_k = \sum_{s=-} mc^{2k}$, so that $\mathcal{A}_1 = A_2$ and $\mathcal{A}_2 = A_4$. **And $\mathcal{B}_k$ is not the per-pair weight b:** b weighs one $P_6 = -1$ pair, while $\mathcal{B}_k$ is the whole $s = -1$ sector summed over *both* P_6 classes, so the periodic weight a sits inside it. That is why $\mathcal{B}_k$ moves from $-\frac{7}{2}$ to -3 between the two seeds of §13 while $b = \frac{1}{2}$ does not move at all; the two letters are close enough to be worth separating once, here. Then $S_k = \mathcal{A}_k + \mathcal{B}_k$, $\Delta_k = \mathcal{A}_k - \mathcal{B}_k$, and the whole ladder collapses to

$$2^p O_k \;=\; 2^p \mathcal{A}_k \;+\; \big(2 - 2^p\big)\mathcal{B}_k, \qquad p = 5 - 2k. \tag{18}$$

For the gauge sector on the seed [2] print, $\mathcal{A}_k = -4^k - 2$ and $\mathcal{B}_k = -\frac{7}{2}$, and since $2^p 4^k = 2^5$ the k cancels, leaving one number for the whole ladder:

$$2^p O_k^{\text{gauge}} \;=\; -39 + 3 \cdot 2^{p-1} \qquad \longrightarrow \qquad 9,\ -27,\ -36 \quad \text{at } k = 0, 1, 2. \tag{19}$$

Equation (19) is §4's two-parameter ladder written out: since $2^{p-1} = 16 \cdot 4^{-k}$, it reads $-39 \cdot 1^k + 48 \cdot (\frac{1}{4})^k$, which are exactly the two modes $\sigma \in \{1, \frac{1}{4}\}$ the gauge sector was found to reach — now with their amplitudes, -39 and 48. The identity $O(5) - O(3) = 4[O(3) - O(1)]$ is then not a coincidence to be checked but an algebraic consequence: for any $O_k = A + B(\frac{1}{4})^k$ the two sides are $3B/4$ and $3B/16$.

Matter has integer m and c, and at the three rungs of integer normalisation, $p = 5, 3, 1$, both coefficients in (18) are even — 32 and -30, 8 and -6, 2 and 0 — so matter contributes an even integer to $2^p O_k$ on each. Past the pole, at $p = -1, -3, \ldots$, those coefficients are fractional and the *arithmetic* ladder has no rung to stand on, though the analytic one continues. On the three: **whatever parity a rung has is the gauge sector's alone**, and what decides it is the parity character (63) of the complete gauge coefficient — on the seed [2] print, $\mathcal{B}_k$ half-integral is what makes that character odd, and §13 is where the character itself becomes the object. That is a statement about the two parity-sensitive rungs and no others: at $p = 1$ the antiperiodic weight $2 - 2^p$ vanishes identically, $\mathcal{B}_k$ drops out, and the fourth-moment rung is left to the *periodic* gauge weight alone — which is the weight §13 puts in question. It is even on the seed [2] print, where the obstruction dies at that rung, and *odd* on the candidate seed, where that rung is the one the obstruction survives on. At $k = 0$ that gives $32F(0)/\zeta(5)$ always odd and

$$|F(0)| \;\geq\; \frac{\zeta(5)}{32} \qquad \text{for every bulk content, and attained — on the seed [2] print} \tag{20}$$

The zeta value is inherited, not derived here. [10] evaluate the Wilson-line potential at these special points and get $f_D(0) = \zeta_R(D)$ and $F = \sum(2n-1)^{-5} = (1 - 2^{-5})\zeta_R(5)$: that $\frac{31}{32}$ is ours only in the sense that we read it from them, and the $\zeta(3)$ one rung up is the same inheritance. The claim is the coefficient in front. It is quantised, so the value cannot be made small by any choice of content. Like the odd eighths it rests on $\mathcal{B}_k$ being a half-integer — which is the *same* statement as the parity character of the whole mixed row and not a rival to it, a distinction worth making because the row is where the two readings of §13 differ. $\mathcal{B}_k$ is $-(2a+6b)/2$, so the *periodic* weight a sits inside it: it moves from $-\frac{7}{2}$ to -3 on the candidate seed, while the per-pair $b = \frac{1}{2}$ does not move at all. What is not the reason is b on its own.

In the language of [8], whose stability criterion *is* this rung, the comparison of the symmetric point against the maximal one, applied to a model they did not consider, is never a tie. And (18) says why the obstruction dies at $k = 2$ on the seed [2] print, which is otherwise only an observation about the number -36: the antiperiodic half enters with $2 - 2^p$, and at $p = 1$ that is *exactly zero*. The fourth moment does not see the antiperiodic sector at all — $2O_2 = 2\mathcal{A}_2 = 2A_4$, and the half-integer never enters, which is why the parity left at that rung is the periodic weight's and travels with the seed. That is also why A_4, the entire non-analyticity, is the one thing matter *can* cancel: at that rung there is only a periodic sum to cancel. Seventeen contents of at most six multiplets have $A_4 = 0$ exactly, five of them with $D > 0$.

And cancelling it is not free: the analytic branch is charged twice over. $A_4 = 0$ is the one place where the vacuum becomes analytic at the symmetric point, so it is worth asking what such a content can do. Theorem 2 answers before any dynamics does: $8D \equiv 2A_4 + 3 \pmod 6$ forces $8D \equiv 3 \pmod 6$ when $A_4 = 0$, so $8D \in \{3, 9, 15, \dots\}$ and $8D = 1$ **is arithmetically forbidden** — and $8D = 1$ is exactly the quantum on which the ceiling of §8 sits. Maximal hierarchy and maximal analyticity are incompatible, and what makes them so is a congruence rather than a mechanism. The smallest $8D$ actually realised with $A_4 = 0$ and $D > 0$ is 15, not 3.

There is a second charge behind the first, and unlike the congruence it can be settled *exhaustively*, because the branch is finite in the only direction that matters. Exactly one multiplet has $A_4 = 0$, the $\mathbf{7}^{(+,+)}$, so it may be added freely without leaving the branch while every other multiplet has $A_4 > 0$ and must therefore make up the gauge sector's -18 exactly — which admits eleven solutions and no more. The free multiplet is then the only knob, and it moves the two functionals that matter at a fixed rate: $8D \to 8D - 6n$ and $W \to W + n$, six units of curvature bought for one of stability. Electroweak breaking asks $n < 8D_{\text{base}}/6$ and a true vacuum asks $n > -W_{\text{base}}$, so both together need an integer in an interval, and over the eleven base solutions there is none:

$$A_4 = 0 \implies \text{no content has both } D > 0 \text{ and } W > 0. \tag{21}$$

Cancelling the branch point, breaking the electroweak symmetry and landing in the true vacuum cannot be done at once. And it is close: the widest window is $18 \times \mathbf{7}^{(+,-)}$, with $8D = 117$ and $W = -\frac{39}{2}$, where $n = 19$ gives $8D = 3 > 0$ but $W = -\frac{1}{2} < 0$, and $n = 20$ gives $W = \frac{1}{2} > 0$ but $8D = -3 < 0$. One step of the only knob there is. The statement is exact arithmetic on two integers — no expansion enters, which matters here because the closed form has no business on this branch at all: its champion sits at $\alpha = 2m_W/181 \simeq 0.89$, a factor 2.7 *outside* the radius 1/3 that §5 proves. **An earlier version of this paper quoted $m_h \le 29.6$ GeV here; that number was computed outside the radius and is withdrawn.** The exact potential agrees with (21): every base family with $D > 0$ has no global interior minimum at all. Measured in `analytic_branch_nogo.py` and `analytic_branch_exact.py`; the enumeration of the seventeen is `moment_ladder.py`.

All of §7 is arithmetic of the content, and Figure 5 is what it looks like from the outside: the same 4319 contents plotted against their own size, against the two observables they produce, and against the precision a measurement would need to tell them apart.

Two quantised moments, one universal kernel, and a hierarchy that is their ratio. What is the largest hierarchy the lattice allows?

8 The ceiling

First, two identities with no logarithm in them. Write $\mu = m_h^2/(K\pi^2)^2$ — the Higgs mass in the units the potential works in. Eliminating the logarithm between (15) and the fixed point (11) leaves

$$\text{(I)} \quad \ln x = \frac{G - 3\mu}{A_4} - \frac{3}{4}, \qquad \text{(II)} \quad x^2 = \frac{12\,\zeta(3)\,D}{6\mu + A_4}. \tag{22}$$

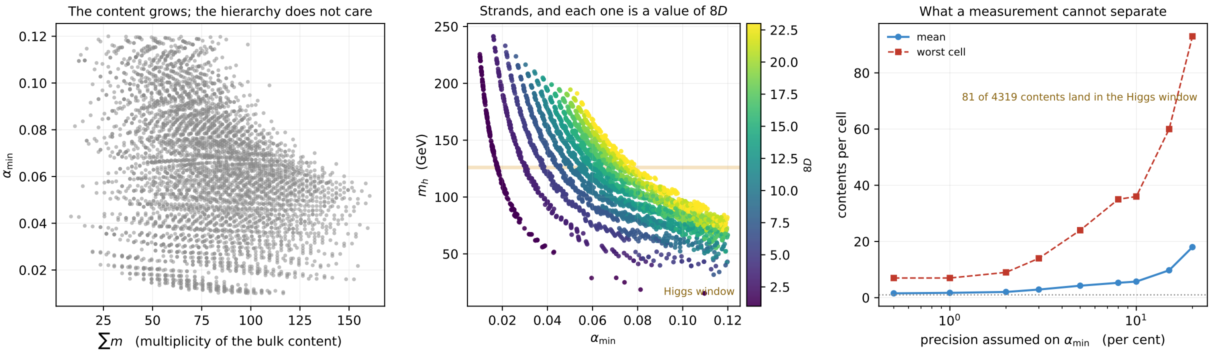


Figure 5: The arithmetic, seen from the observable side. Every content built from at most four copies of each of the eight multiplets, kept when $0 < D \leq 3$ and $\alpha_{\min} \in [0.010, 0.120]$: 4319 of them. *Left:* $\alpha_{\min}$ against the total multiplicity of the bulk content. There is no relation — the hierarchy does not care how much matter there is, only how it is distributed over charge and parity. *Centre:* the same contents in the plane of the two observables they predict. The cloud separates into strands, and each strand is one value of $8D$; the Higgs window crosses them, so which contents are viable is a question about the second moment rather than about the size of the model. Coloured by multiplicity instead, the same panel shows nothing. *Right:* how many contents share one cell of the observable plane, against the precision assumed on $\alpha_{\min}$ — with the m_h side of the cell relaxed alongside it, from 1 to 5 GeV, because a measurement that loosens one loosens the other. At the finest cell drawn the mean is 1.55 and the worst cell holds 7; at the loosest, 20 % in $\alpha_{\min}$ and 5 GeV in m_h, the mean is 18.0 and the worst cell holds 93. The two observables therefore determine the content only to the extent that they are measured well, and the quantisation of §7 is what sets the scale of that trade. 81 of the 4319 land inside 125–127 GeV.

Both are verified to 6×10^{-14} on the five published rows — and that number should be read for what it is. (I) and (II) are obtained *algebraically* from (15) and (11), so 6×10^{-14} tests the implementation and the algebra and nothing else: an identity derived from a truncated expansion verifies to machine precision while inheriting every bit of the truncation error of its parents. The physical validation is a different measurement and it is the one in §5: 0.13 % median and 0.62 % worst against numerical minimisation of the untruncated potential. Internal exactness and external accuracy are separate claims and are kept separate here. (II) is the useful one: *once the Higgs mass is pinned, $\alpha_{\min}$ is algebraic in the two moments* — no fixed point, no logarithm. Since $1/R_5 = 2\pi m_W/x$, maximising the compactification scale means making D small and A_4 large; and D cannot go below 1/8 by Theorem 1. So the ceiling is the question of how far A_4 can be pushed before (I) runs out of contents that satisfy it.

That question is an integer program. All three moments are *linear* in the multiplicity vector $n \in \mathbb{Z}^8_{\geq 0}$, and A_4 and $8D$ are integers. For a content at $(A_4, 8D) = (t, k)$ whose Higgs mass sits at the top of the window, (II) fixes x with no freedom left, and (I) then demands a specific value of G:

$$G^\star(t,k) \;=\; t\Big(\ln x + \tfrac{3}{4}\Big) + 3\mu\,, \qquad x = \sqrt{\frac{12\zeta(3)(k/8)}{6\mu + t}}\,, \tag{23}$$

so the pair (t, k) is realisable only if some content there has $G \leq G^\star(t, k)$. Relaxing the multiplicities to the reals turns "the smallest G at (t, k)" into a linear program with two equality constraints — whose *dual* therefore has two variables [27], so its vertices can be enumerated exactly in rational arithmetic, with ln 2 and ln 3 rounded in the direction that keeps the bound a bound. Admissible (t, k) are restricted by Theorem 2, and the restriction bites: the optimum of the relaxation falls at $(216, 1)$, where $216 + 1$ is not divisible by three, so no content sits there at all. Figure 6 is the lattice the program runs on, and its right-hand panel is where the one-in-six comes from.

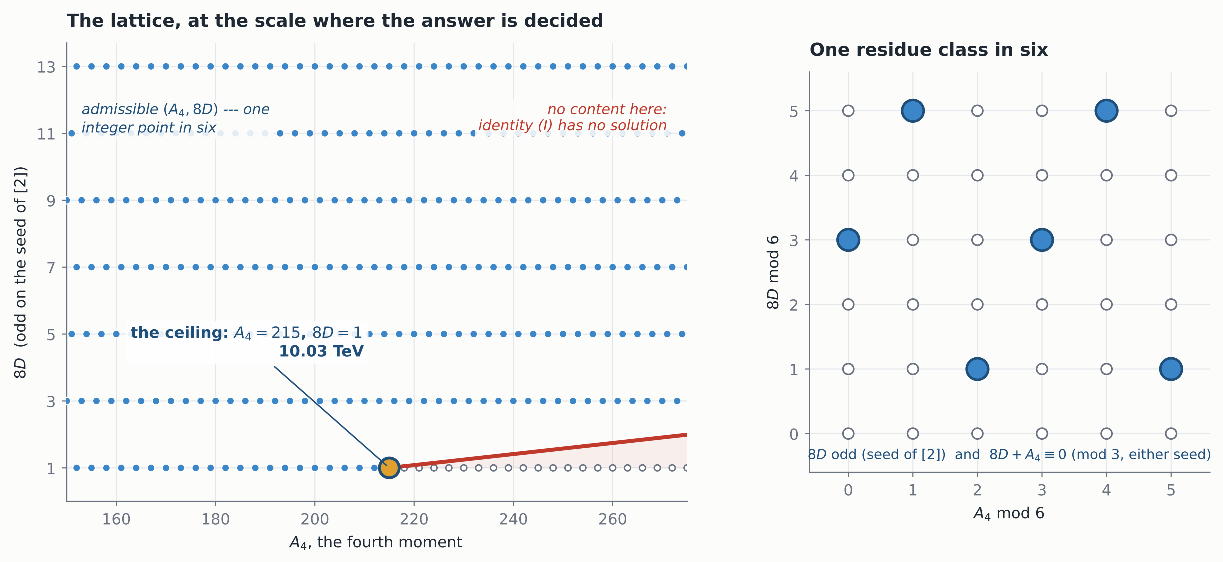


Figure 6: The lattice the integer program runs on. Left, at the scale where the answer is decided: the admissible $(A_4, 8D)$ are one integer point in six, the red curve is where identity (I) runs out of contents in the relaxation, and the hollow points beyond it are unrealisable. The relaxation's maximum sits on the last admissible point of the row $8D = 1$ — and Figure 9 shows that *that* point is empty once the third coordinate is counted, so the integer maximum is the one below it. Right: why one in six — $8D$ odd on the seed [2] print, which is Corollary 1, together with $8D + A_4 \equiv 0 \pmod 3$, which is Theorem 2 on either seed. The geometry is computed exactly in Sage: the cone has extreme rays $(0, -1)$ and $(1, 8)$, dual description $\{A_4 \geq 0,\ 8D \leq 8A_4\}$, and the eight multiplets generate a sublattice of index exactly 6 in $\mathbb{Z}^2$ — the congruence, restated as a lattice index. Its Hilbert basis is $\{(0, -1), (1, 8)\}$, and $(0, -1)$ is *not* reachable: the $\mathbf{7}^{(+,+)}$ steps down by six, never by one.

$$\boxed{\frac{1}{R_5} \leq 10.03\ \text{TeV}} \qquad \text{certified at } A_4 = 215,\ D = \tfrac{1}{8}. \tag{24}$$

And the quantifier deserves saying exactly. The dual certifies each rung it is run on, for arbitrary content *on that rung*. Extending that to a maximum over *all* rungs uses one more thing: that $t^\star(k)/k$ is non-increasing. That was measured to $k = 501$ for two weeks and is now proved, in the paragraph after next.

The per-D ceiling is monotone decreasing — 10.03, 6.27, 4.21, 3.14, 2.80 TeV at $8D = 1, 3, 9, 33, 129$ — flattening towards about 2.7 TeV. So the maximum sits *on the quantum* $D = 1/8$: Part VI observed that the eight contents of smallest $\alpha_{\min}$ all had $D = 1/8$; here that is not an observation about an enumeration but where the argument maximum is, and Theorem 1 is doing the physical work. That monotonicity is worth saying carefully, because the whole ceiling rests on it. From (II), $(1/R_5)^2 \propto (6\mu + t^\star(k))/k$ with $t^\star(k)$ the largest admissible A_4 on rung k, so a *sufficient* condition is that $t^\star(k)/k$ be non-increasing — and it is, falling from 215 at $k = 1$ through 70 at $k = 9$ to 50.8 at $k = 501$, monotonically at every step. Measured in `fissures.py`.

And it is a sign, not a scan. Since $1/R_5 = 2\pi m_W/x$ and (II) fixes x, the ceiling depends on the rung *only* through the slope $r_k = t^\star(k)/k$:

$$M(k)^2 \;=\; \frac{8\pi^2 m_W^2}{3\zeta(3)}\Big(r_k + \frac{6\mu}{k}\Big), \tag{25}$$

so the question is a derivative of one function of one variable. Two facts settle it. First, the dual of §8's program has **exactly two vertices**, and *one* of them is active at every rung — so the piecewise-affine argument has a single piece and there are no boundaries to check. Second, writing $t = rk$ makes $x^2 = (3\zeta(3)/2)/(r + 6\mu/k)$, and the active vertex's condition becomes $\Phi(r,k) \geq 0$ with $\Phi = r\big(A - \frac{1}{2}\ln(r + 6\mu/k)\big) - \nu - C/k$, $A = \frac{1}{2}\ln\frac{3\zeta(3)}{2} + \frac{3}{4} - \lambda$ and $C = G_{\text{gauge}} - \lambda A_{\text{gauge}} - \nu\,(8D)_{\text{gauge}} - 3\mu$. Then

$$k^2\,\frac{\partial\Phi}{\partial k} \;=\; \frac{3\mu\,r}{r + 6\mu/k} \;+\; C, \tag{26}$$

whose first term is positive and strictly below 3μ. So $\partial\Phi/\partial k < 0$ as soon as $C \leq -3\mu$, which is the statement $G_{\text{gauge}} - \lambda A_{\text{gauge}} - \nu(8D)_{\text{gauge}} \leq 0$ — **a condition on the gauge sector alone**, and it holds with room: $-37.43 \leq 0$. With $\partial\Phi/\partial r < 0$ at the upper root, $dr^\star/dk < 0$. Both terms of (25) therefore fall, $M(k)$ strictly decreases, and since the integer ceiling never exceeds the real relaxation, $M(k) \leq M_{\text{real}}(3) = 6268$ GeV for every $k \geq 3$, below the 10 034 GeV of $8D = 1$. **The maximum sits on the quantum because it must.**

And the flattening has a value. Φ's large-k limit drops k entirely and leaves $r(A - \frac{1}{2}\ln r) = \nu$, a Lambert equation whose two real branches are the ends of an interval of admissible slopes: $r = -2\nu/W(-2\nu e^{-2A})$. Intersecting the two vertices' intervals,

$$r_\infty = 50.6809086439, \qquad M_\infty = \sqrt{\tfrac{8\pi^2 m_W^2}{3\zeta(3)}\,r_\infty} = 2\,678 \text{ GeV}, \tag{27}$$

approached from above and never reached. So the 2.7 TeV the table flattens towards is not a numerical coincidence: it is the asymptotic ray of the feasible cone, and the ceiling runs from 10.03 TeV on the smallest rung — on the seed [2] print, since which rung is smallest is what the seed decides — to 2.678 TeV at infinity. The asymptote itself is not conditional: it carries no gauge quantity at all. Measured in `asymptotic_ray.py`.

A number of ours, corrected. An earlier note of this series quoted $1/R_5 \leq 2.18$ TeV under the same Higgs window. That is a bound on the *size* of the content — it is what an enumeration over contents of at most six multiplets returns, six being the size used in their Table 1 — and not a bound on the model, which does not cap the content at all. Opening the cap:

N	in window	best $1/R_5$	D	content
5	1	1924	29/8	$\mathbf{28}^{++} + 4 \times \mathbf{84}^{++}$
6	3	2182	21/8	$\mathbf{28}^{++} + \mathbf{28}^{+-} + \mathbf{48}^{++} + 3 \times \mathbf{84}^{++}$
7	15	3604	7/8	$3 \times \mathbf{7}^{++} + 2 \times \mathbf{28}^{++} + 2 \times \mathbf{84}^{++}$
8	30	**9138**	**1/8**	$3 \times \mathbf{7}^{++} + \mathbf{7}^{+-} + 2 \times \mathbf{48}^{++} + \mathbf{48}^{+-} + \mathbf{84}^{++}$
14	1097	9156	1/8	$4 \times \mathbf{7}^{++} + 2 \times \mathbf{28}^{++} + 5 \times \mathbf{28}^{+-} + \mathbf{84}^{++}$

The jump is between seven and eight multiplets and is a factor of 2.5: eight is where the lattice first manages to sit on $D = 1/8$ *inside* the Higgs window. Two more multiplets than their Table 1 uses, not seventy. Figure 7 plots both halves of the claim: the jump against the cap on the content, and the ceiling's monotonicity in D.

And that table is where the certificate stops being the answer. The relaxation's optimum sits at $(A_4, 8D) = (215, 1)$ and the enumeration climbs only to 9.16 TeV. The dual bounds the whole cone, and that is a theorem; whether an integer content *sits* on the certifying vertex is a further question, and it is decidable rather than open — because in the right coordinates the obstruction stops being transcendental.

Write G exactly. The charges are $c \in \{1, 2, 3\}$, so A_4^L and B_4 contribute only $\ln 2$ and $\ln 3$, and

$$G \;=\; \tfrac{25}{12}A_4 \;-\; U\ln 2 \;-\; V\ln 3\,, \tag{28}$$

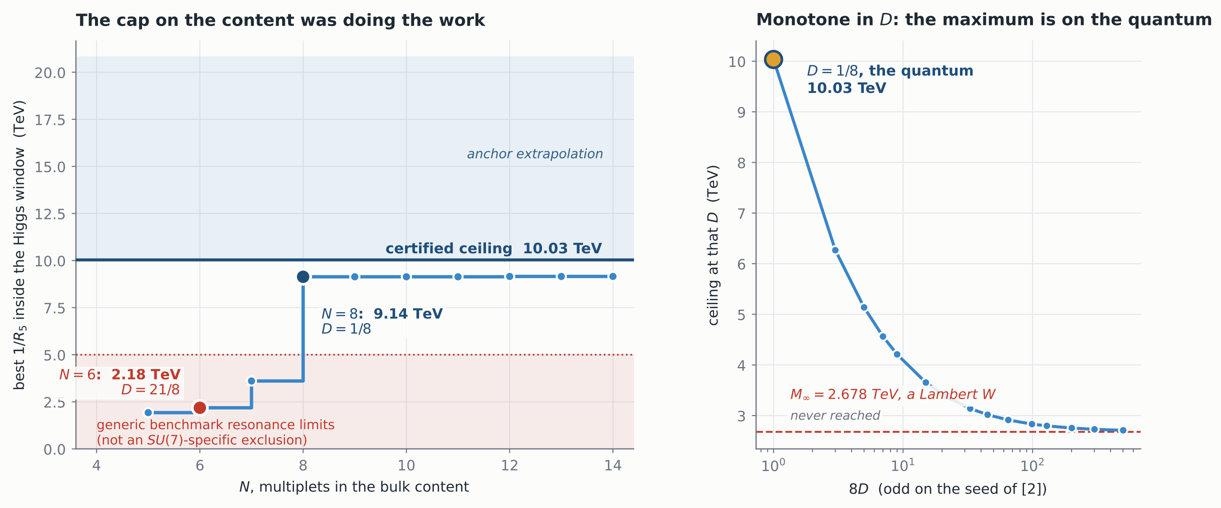


Figure 7: The cap on the content was doing the work. Left: the best $1/R_5$ inside the Higgs window against the number of multiplets allowed. The 2.18 TeV quoted earlier in this series is the $N = 6$ point; at $N = 8$ the lattice first reaches the quantum $D = 1/8$ of the seed [2] print and the scale jumps by a factor 2.5. The horizontal line is the certified ceiling, the band above it the anchor *extrapolation* of §2 — which §2 is careful to say is a trend and not an uncertainty, and the figure labels the same way — and the shaded region the reach of generic benchmark resonance searches — **not an** $SU(7)$**-specific exclusion**: §10 shows those limits are set on a *narrow* resonance and cannot be carried to an object with $\Gamma/M \simeq 0.16$ and an unfixed $\mathrm{BR}(G_1 \to jj)$. Right: the ceiling as a function of D — monotone, so the maximum sits on the quantum, which is Theorem 1 doing physical work.

with U half-integer and V integer, both linear in the multiplicities. Since $\ln 2/\ln 3$ is irrational — $2^q = 3^p$ has no solution — the pair (U, V) is not lost inside G, and the content lives on a *four*-dimensional lattice $(A_4, 8D, 2U, V)$ rather than the two-dimensional one of Figure 6. It is worth being careful about the fourth coordinate, because there is a tempting false step: $\ln 3$ enters through the single generator $\mathbf{84}^{(+,-)}$, so $V = 81\, n_{\mathbf{84}^{(+,-)}}$ — and that makes V cheap to describe but *does not* make it dependent on the other three. The seven generators with $V = 0$ already reach rank three in $(A_4, 8D, 2U)$, so the eighth can only add a direction, and the rank is four.

The right object is then the finite group $\mathbb{Z}^r/L$, whose invariant factors are the congruences. A Smith normal form gives

$$\mathbb{Z}^2/L \cong \mathbb{Z}_6\,, \qquad \mathbb{Z}^3/L \cong \mathbb{Z}_2 \times \mathbb{Z}_{72}\,, \qquad \mathbb{Z}^4/L \cong \mathbb{Z}_{18} \times \mathbb{Z}_{648}\,, \tag{29}$$

of orders 6, 144 and 11664. So the projected law of Theorem 2 is the single congruence $\mathbb{Z}_6$, and the lifted content is **two** congruences and not one — an index is the order of a group, not a count of conditions, and the number of independent conditions is at most the rank. Two earlier sections turn out to have been describing this lattice without naming it. §5 is what licenses the basis: the transcendental content of the whole potential is $\{\zeta(5), \zeta(3), \ln x, \ln 2, \ln 3\}$ and nothing else, so $\{1, \ln 2, \ln 3\}$ is not a convenient truncation but the complete span. And the log-free charge-two flip of §4, (9), is exactly the direction with $\Delta U = \Delta V = 0$: the one generator move the third axis cannot see, which is why its ΔG was rational. Figure 9 is the two pictures side by side.

And lifting once more closes it, because the fifth coordinate is $2W$**.** Carrying W up as a coordinate rather than a screen — Figure 9 with one axis more — gives $(A_4, 8D, 2U, V, 2W)$, integral in every entry once U and W are doubled, of rank *five*, with

$$\mathbb{Z}^5/L \;\cong\; \mathbb{Z}_2 \times \mathbb{Z}_{72} \times \mathbb{Z}_{2592}, \qquad \text{index } 373\,248. \tag{30}$$

Five is also the dimension of the function space of §5, and that is not a coincidence:

Theorem 3 (complete invariants)**.** *Two bulk contents have the same one-loop potential, as a function of α, if and only if they agree on $(A_4, 8D, 2U, V, 2W)$.*

The proof is two kernels meeting. Content $\mapsto$ potential has a three-dimensional kernel, spanned by (42); content $\mapsto$ the five coordinates has a three-dimensional kernel because the rank is five; and the first is contained in the second because each coordinate is a linear functional of the potential. Equal dimensions and one inside the other make them equal. The two kernels are computed here by routes sharing no arithmetic — one from the Fourier vectors, one from the moment definitions — and come out generator for generator the same. Figure 8 is the statement and its reason side by side.

What of that is not new, and one instance of it is in print. That the potential is a *linear form in the bulk content*, with the gauge sector as a constant offset, is standard and explicit: [17] eq. (3.20) writes exactly that and closes with "this counting rule for the coefficients of the effective potential is applicable in general", and [8] eqs. (2.7) and (2.9) restate it. Degenerate contents therefore *exist* as a matter of counting, and one is printed: [8] eq. (3.26) has two SU(6) models whose bulk fermions differ — two **15**'s against $\mathbf{6} + \bar{\mathbf{6}} + \mathbf{20}$ — and whose gauge-scalar potentials agree identically, recorded as "accidentally the same". **That is an element of the kernel below, and it is not an accident.** A second sits unremarked in the equations of the five-dimensional ancestor of this model: [12] eqs. (3.10)–(3.12) decompose **21**, **28** and **48**, and $\mathbf{21} \oplus \mathbf{28}$ reproduces the decomposition of **48** term for term, the leftover $49 - 48 = 1$ being a singlet the potential cannot see.
Nor is the idea of reducing the content-dependence to a short list new. [11] eqs. (4.13)–(4.14) reduce a supersymmetric Scherk–Schwarz vacuum energy to *two* integer functionals a and b, and their $b = n_F^{(+)} - n_F^{(-)}$ is W in another notation: their $h(N) - h(0) = -Nb$ is our $F(1) - F(0)$, a signed count of matter by parity deciding which of two symmetric points wins.[1] What *they* compute, and say so four times including in their abstract, is the vacuum energy *at vanishing Wilson-line phase*: their $h(x)$ is a function of a boundary-condition integer, not of the phase.
So what is added here is three things, and none of them is the phenomenon. The basis is *irredundant* — the naive coefficient list overcounts, and it is [8]'s own eq. (2.11) that says by how much, a use they make of it only as an algebraic simplifier. The kernel is a *subspace with generators*, (42), rather than a list of accidents; no statement of the form "these invariants are complete" appears in any of [17, 11, 8], and none of them counts the dimension of the space of achievable potentials. And the invariants are of a particular kind: A_4, U and V come from the *small-α expansion*, so Theorem 3 says that local expansion data together with a single global comparison, W, pin the potential everywhere on $[0, 1]$. That last is only worth saying because the space is five-dimensional and not one: were it one, the theorem would be a restatement of linearity. **The lattice of this section and the function space of §5 are one object**, and these five numbers are coordinates on it.

Which makes all three parities inherited — and, it turns out, not inherited together. Every matter multiplet contributes an *even* amount to $8D$, to $2U$ and to $2W$. That much is robust and is a property of the content alone: all three parities are decided by the gauge sector and by nothing a bulk content can do. With the coefficients [2] print, the gauge sector contributes -27, -39 and -3 — odd in all three — so for every bulk content

$$8D \;\equiv\; 2U \;\equiv\; 2W \;\equiv\; 1 \pmod 2. \tag{31}$$

But that is a statement about this seed, not a structural identity, and we could not see the difference until there was a second seed to test it against. An earlier version of

[1] Their eq. (4.7) substitutes $N_{\mathrm{Ad}}^{(+-)} + N_{\mathrm{Ad}}^{(-+)} = (p+s)(q+r)$ where their own eq. (3.20) gives twice that; the factor is inert, since their eq. (4.13) uses the correct value, but eq. (4.7) is where the gauge offset is fixed and the coefficient is the one this paragraph turns on. Rebuilt from their eqs. (4.5)–(4.6) in `hhk_gauge_offset.py`.

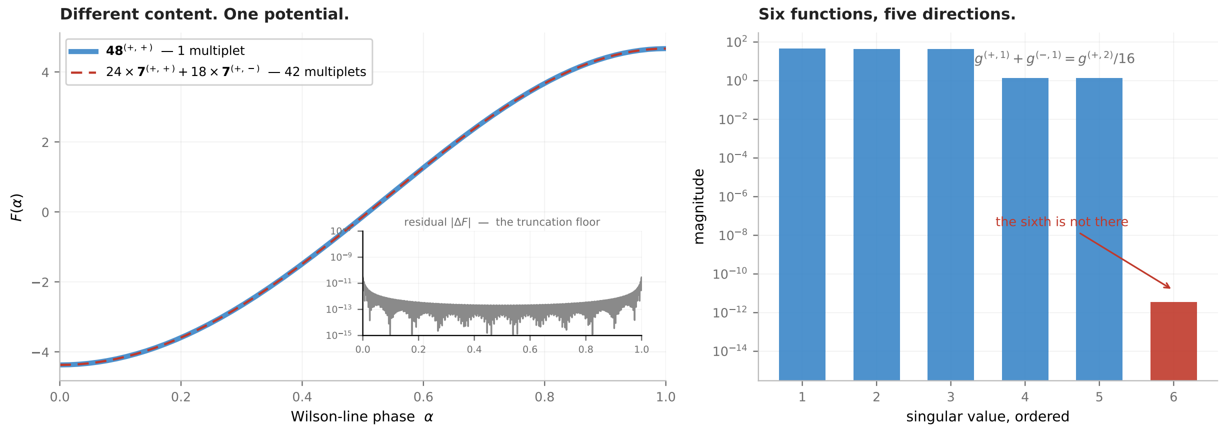


Figure 8: Theorem 3, in the two halves it is made of. *Left:* what it asserts. A single $\mathbf{48}^{(+,+)}$ and $24\times\mathbf{7}^{(+,+)}+18\times\mathbf{7}^{(+,-)}$ — one multiplet against forty-two, different representations, different dimensions — give *one curve*, on the exact polylogarithmic potential and across the whole fundamental domain. The inset is the residual: over 1600 points of $[0,1]$ its worst value is 2.9×10^{-11}, which is not the identity failing but the floor of this evaluation, the truncation of the winding sum at 600 modes. The relation is exact over $\mathbb{Q}$, proved in `dimension_five.sage`. *Right:* why it happens. The six functions $\mathrm{Re}\,\mathrm{Li}_5(s\,e^{ic\pi\alpha})$ look like six directions and are five: three singular values of order 45, two of order 1.4, and a sixth at 3.5×10^{-12} — eleven orders below the smallest of the five. The missing one is the duplication formula at $s=5$, which is [8]'s eq. (2.11). A content is invisible to the potential exactly when it is invisible to those five, which is the theorem. Drawn by `make_fig_kernel.py`.

this paragraph read (31) as one congruence on three coordinates, sourced to the single half-integer $-\frac{7}{2}$. Substitute the candidate gauge-plus-ghost split of §13 and the three come apart: $8D$ goes $-27\to-18$ and $2U$ goes $-39\to-30$, both losing their oddness, while $2W$ stays at -3. *The parity of* $2W$ *is robust. The parity of* $8D$ *is not.*

The reason is legible in the coordinates, and it is prettier than the claim it replaces. The candidate moves the two charge-one weights, periodic and antiperiodic, by the same $+\frac{1}{2}$, and the charge-two weight by $+\frac{1}{4}$. Collect those: $8D$ and $2U$ each shift by exactly 9 — odd, so both flip — while $2W$ depends on the *difference* $-u_{(+,1)}+u_{(-,1)}$, in which the two half-unit shifts cancel, and moves by 0. So there is a shared mechanism, but not the one we claimed: it is that the gauge sector supplies all three parities, and $2W$ is the one combination that a uniform shift of the charge-one weights cannot reach. **A theorem that survives a change of hypothesis is worth more than one that does not, and until we had the second seed we had no way to tell which of ours were which.** The third reading, $2U$ odd, is stated because the character says it; what it forbids a content from doing is not known here, and A_4 and V are not forced, so (31) is a statement about *which* three. It is not, however, an independent reading: (64) makes $2U$ odd and $2W$ odd *equivalent* on any content, by a congruence that never mentions the $-\frac{7}{2}$. Computed in `lattice_lift.py` and independently, with the transformation matrices and the three congruences written out, in `lattice_lift.sage`.

That is what makes the vertex decidable. At fixed $(A_4,8D)$ the condition $G\le G^\star$ reads $U\ln 2+V\ln 3\ \ge\ \frac{25}{12}A_4-G^\star$: maximising a linear form over the contents with two moments pinned. Every generator has $A_4\ge 0$ and only $\mathbf{7}^{(+,+)}$ has $A_4=0$, whose multiplicity is then forced by $8D$ — so the feasible set is *finite* and the maximum is a computation, not a search. It comes out

A_4 on $8D=1$	required	largest available	shortfall	
215	681.084	681.017	-0.067	no content, at any size
212	667.670	669.927	$+2.257$	attained

so the relaxation's vertex is *empty* and the one below it is not — which is what Figure 9 draws, rung by rung. The ceiling therefore has a witness, and the bound sharpens:

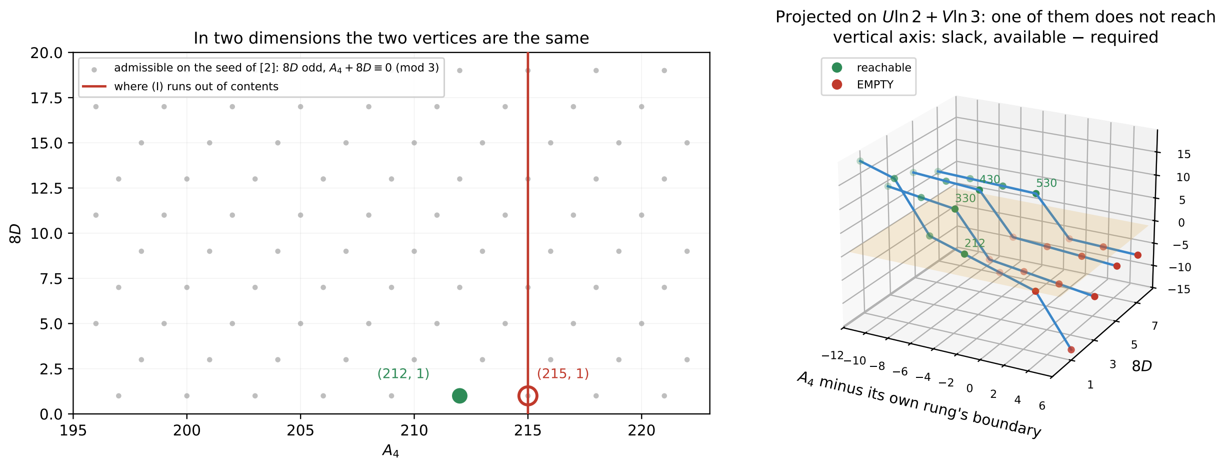


Figure 9: The axis the projection was hiding. *Left:* the plane the ceiling has been computed in. Both $(215, 1)$ and $(212, 1)$ are admissible lattice points — $8D$ odd on the seed [2] print, and $A_4 + 8D \equiv 0 \pmod 3$ — and both sit at or under the boundary where identity (I) runs out of contents. Nothing in this picture separates them. *Right:* the same points with the third coordinate of (28), plotted as the *slack* between the logarithmic budget a content can supply and the budget $G \le G^\star$ demands; the raw budget runs past 680 and the whole verdict is 0.067 wide, so the slack is what can be drawn. Each rung is shown against distance to *its own* boundary, which is what makes the four comparable. The amber plane is zero. Green is reachable, red is empty, and the green label is that rung's integer optimum. The slack falls monotonically, so each crossing is single and each optimum well defined — and the pattern is the same on every rung: **the relaxation's boundary is empty and the integer optimum sits one or two steps of three below it**, at 212, 330, 430, 530 for $8D = 1, 3, 5, 7$. Drawn by `make_fig_lift.py`; the arithmetic is `semigroup.py`.

$$\boxed{\frac{1}{R_5} \le 10.01\ \text{TeV}} \qquad \text{optimum at } A_4 = 212,\ D = \tfrac{1}{8}. \tag{32}$$

The witness is $17 \times \mathbf{7}^{(+,+)} + 2 \times \mathbf{7}^{(+,-)} + 57 \times \mathbf{28}^{(+,-)}$, seventy-six multiplets — large, and nothing in the model caps the content, which was the point of §8 to begin with. Three things keep that honest. The 215 verdict is checked at fifty digits and at both ends of the Higgs window — the smallest G available there is -233.100 against a $G^\star(127) = -233.168$, so it fails at the *most permissive* end and therefore everywhere. The witness is rebuilt through the ordinary `moments()` route, which knows nothing about U and V, and reproduces $A_4 = 212$, $8D = 1$, $G = -228.260$ inside the band. And the margin that separates 215 from 212 is one part in 10^4, an order below the closed form's own 0.13–0.62 %. Measured in `semigroup.py`.

And the ordering survives the truncation, which is not obvious and nearly failed. Everything so far lives on the surface (11), which keeps three rungs; the discarded remainder R is bounded at the end of this section, in (38), and what is needed here is only that it can be *evaluated* on the handful of contents in question. Redoing the elimination without discarding it gives the corrected pair

$$x^2(6\mu + A_4) = 12\zeta(3)D - \frac{18R'(x)}{x} + 6R''(x)\,, \qquad \ln x = \frac{G - 3\mu}{A_4} - \frac{3}{4} + \frac{3\,[xR''(x) - R'(x)]}{A_4 x^3}\,, \tag{33}$$

so the demanded $G^\star$ moves by $\Delta G^\star = -3[xR'' - R']/x^3$. Evaluated at the contents that actually sit at these vertices, $\Delta G^\star \simeq -0.45$ at $(215, 1)$ and -0.33 at $(212, 1)$ — **seven times the** 0.067 **that separates them**. Comparing the correction with that gap is therefore the wrong test, and it is worth saying why: $\Delta G^\star$ is *common mode*, since the contents at the two vertices differ by a

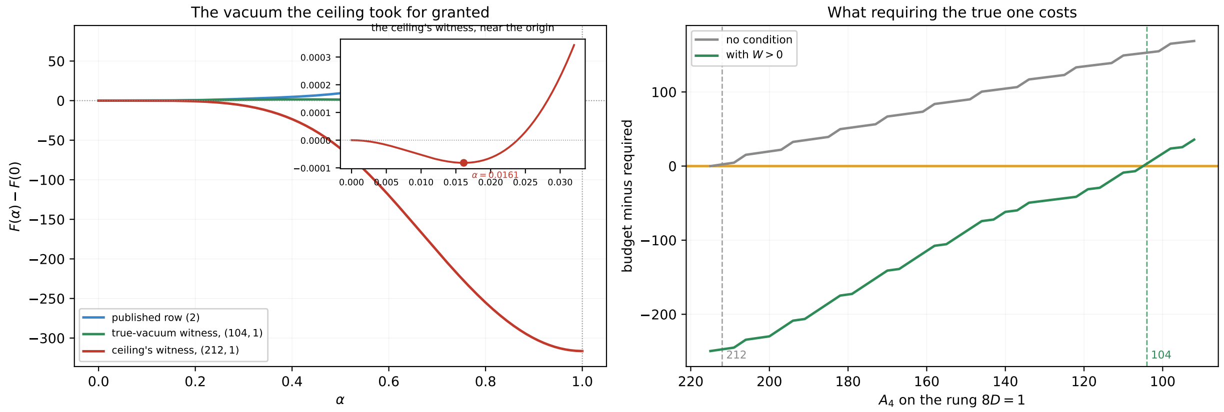


Figure 10: The vacuum the certificate took for granted. *Left:* the exact $F(\alpha)$ across the whole fundamental domain, each curve shifted so that $F(0) = 0$. A published row rises to the right: its electroweak minimum is the deepest point there is. The content that attains 10.01 TeV falls off a cliff instead — at the other symmetric point $\alpha = 1$ it is 316 below the origin, so the minimum the ceiling was built on is a false one. The third curve is the content the constrained ceiling (36) is attained by: it rises too, but only by 3.5, which is why it hugs the axis at this scale — passing the test is not the same as passing it comfortably. The inset is that same content near the origin, where its minimum is perfectly real and sits within 0.02 % of where (11) puts it: nothing is wrong with the closed form, only with the question it was asked. *Right:* what excluding such contents costs. At each vertex of the rung $8D = 1$, the largest logarithmic budget available minus the budget (23) demands, with and without the stability condition $W > 0$ of (34), which is [8]'s and not ours. The unconstrained curve crosses zero at $A_4 = 212$ and the constrained one not until $A_4 = 104$; the distance between those two crossings is 0.8 TeV of ceiling. Drawn by `make_fig_vacuum.py`.

few multiplets out of eighty. What it can do is eat into each vertex's own slack, and that is the test:

vertex	slack, truncated	$\Delta G^\star$	slack, corrected	
$(215, 1)$	-0.067	-0.452	-0.519	still fails
$(212, 1)$	$+2.257$	-0.332	$+1.925$	still holds

Both verdicts survive: $(215, 1)$ **fails by more once the truncation is carried, not less, and** $(212, 1)$ **keeps** 1.93 **of slack.** What is not claimed is uniformity: $\Delta G^\star$ here is evaluated on the finitely many named contents rather than bounded over all of them, which is legitimate because they are finitely many and is not the same statement.

There is also a route needing no correction at all. At $(215, 1)$ the smallest G available over all 26933 contents is -233.100, while the Higgs window runs $[-238.553, -233.168]$; since m_h is monotone in G at fixed moments — $dG/d\mu = 18\mu/(6\mu + A_4) > 0$ — every content there lies above the window. That is an evaluation over a finite set. Measured in `certify_212_215.py`.

And then the witness is put on the exact potential, and it fails a test the certificate never applied. A witness is a specific content, so it can be evaluated with no expansion at all: minimise the polylogarithm numerically and look. The electroweak minimum is there and it is right — $\alpha = 0.016129$ against the closed form's 0.016133, agreement to 0.02 %, better than on any published row, with $m_h = 126.25$ GeV inside the window. But the potential at the *other* symmetric point, $\alpha = 1$, is -204.99 against $+111.44$ at that minimum. **The content attaining the ceiling sits in a false vacuum.** The five published rows do not, which is the control. Figure 10 is the two curves.

The test it fails is not ours, and neither is the arithmetic of it. Cacciapaglia *et al.* [8] state the criterion and prove the identities it needs, in their §2.2. Their words: *"it suffices to evaluate the potential at $a = 0$ and $a = 1/2$ and check which configuration gives the smallest* *value"*; their eqs. (2.12)–(2.13) give $F_+(0) = \frac{3}{2}\zeta(5) > 0$ and $F_-(0) = F_+(1/2) = -\frac{15}{16}F_+(0)$, $F_-(1/2) = F_+(0)$; and they draw the conclusion themselves — components with parities $(\pm,\pm)$ stabilise the starting orbifold and $(\pm,\mp)$ destabilise it. **The criterion, its arithmetic and its sign rule are all theirs.** Summing their (2.13) over a bulk content in the bookkeeping of (2), where their half-integer shift is our $\alpha \to \alpha+1$ and only odd charges therefore move, gives

$$F(1) - F(0) \;=\; \tfrac{31}{16}\,\zeta(5)\,W\,, \qquad W = \sum_{c\text{ odd}} m\,(-s)\,, \tag{34}$$

with $\frac{31}{16} = 1 + \frac{15}{16}$, and $W > 0$ sufficient for the electroweak point to be the deeper one. That is their statement in our notation and nothing more; W is their stabilising count minus their destabilising one. What is new here is only what it is *used for*: W is linear in the multiplicities, like A_4, $8D$ and G, so it can be carried inside the same integer program — and [8] do not compute a compactification scale, so they have no program to carry it in. For this model the per-multiplet values are $\pm1, \pm3, \pm6, \pm9$ for $\mathbf{7}, \mathbf{28}, \mathbf{48}, \mathbf{84}$ and $-\frac{3}{2}$ for the gauge sector, the sign being the parity; the five published rows have $W = \frac{63}{2}, \frac{75}{2}, \frac{75}{2}, \frac{73}{2}, \frac{69}{2}$ and the vertex at $(212,1)$ has $W = -\frac{315}{2}$. Checked against the exact potential to machine precision on all six.

And W cannot vanish — for a sturdier reason than Theorem 1. Matter contributes integers to W. The gauge sector does not: its odd-charge part is $2 - \frac{7}{2} = -\frac{3}{2}$. Hence $2W$ is an odd integer for every bulk content, and with (34),

$$\boxed{\frac{32}{31}\,\frac{F(1)-F(0)}{\zeta(5)} \text{ is an odd integer, so } \;|F(1)-F(0)| \;\geq\; \tfrac{31}{32}\,\zeta(5)\,.} \tag{35}$$

In this model the two symmetric points are never degenerate — and the scope of that sentence is not decoration, because in other models of the same architecture they are: see below. **And this one holds on both branches, which the odd eighths do not.** It is tempting to file it with Theorem 1 as the same half-integer read twice, and an earlier version of this paper did. The candidate seed of §13 separates them. It moves the two charge-one weights by the same $+\frac{1}{2}$, and $2W$ depends on the *difference* $-u_{(+,1)} + u_{(-,1)}$, in which that shift cancels exactly: $2W$ stays -3. The gauge contribution to $8D$ does not cancel and moves by 9, losing its parity. *So $2W$ odd is seed-robust and $8D$ odd is not* — and we would not have known which was which with one seed to look at. Verified over all 3003 contents of at most six multiplets: none has $W = 0$ and none has $2W$ even.

That matters more than a curiosity, because it closes a question [8] raise themselves. In their models the degeneracy *does* occur, twice and exactly: their eq. (3.14) is a function of $2a$ alone, and their eq. (3.44) is invariant under the simultaneous shift $(a, b) \to (a + \frac{1}{2}, b + \frac{1}{2})$, so in both cases the two symmetric points are degenerate and they read it correctly — the two are then *"two equivalent descriptions of the same orbifold"*. Were that the case here, a potential deeper at $\alpha = 1$ would be a relabelling and not a false vacuum at all. But equivalent descriptions must give equal potentials, and (35) says the two values are never equal — so for this model the deeper configuration is a genuinely different orbifold, and the reading above stands.

And the whole difference between the two situations is one number. The architecture is shared: [11] likewise reduce their vacuum energy to integer functionals and likewise find that a signed parity count decides between two symmetric configurations. But their gauge sector contributes the *even integer* $+2$ to their a and nothing at all to their b, so both can vanish — they open their classification with "the case with $a = 0$" and list $n_F^{(+)} = n_F^{(-)} \Rightarrow$ "completely degenerate" as an ordinary case (their eqs. (4.14), (4.15)). Here the gauge sector contributes a *half*-integer to W — $2 - \frac{7}{2} = -\frac{3}{2}$ on the seed [2] print and $\frac{3}{2} - 3 = -\frac{3}{2}$ on the candidate one, the same number both times because W reads the difference — and a half-integer cannot be cancelled by integers. **The degenerate case they enumerate is, in this model, arithmetically**

unreachable. Measured in `W_is_half_odd.py` and `hhk_gauge_offset.py`. The criterion remains theirs; what is added is that here it can never tie.

The ceiling, with the vacuum required to be the true one. Imposing $W > 0$ alongside $G \leq G^\star$ and walking the rung down, no vertex survives until $A_4 = 104$:

$$\boxed{\frac{1}{R_5} \leq 9.22\ \text{TeV}} \qquad \text{optimum at } A_4 = 104,\ D = \tfrac{1}{8},\ \text{true vacuum required.} \tag{36}$$

The witness is $16 \times \mathbf{7}^{(+,+)} + \mathbf{28}^{(+,-)} + \mathbf{48}^{(+,+)} + 4 \times \mathbf{48}^{(+,-)} + \mathbf{84}^{(+,+)}$, twenty-three multiplets, $W = \frac{5}{2}$; and on the *exact* polylogarithmic potential its global minimiser is $\alpha = 0.017560$, with $m_h = 126.00$ GeV and $1/R_5 = 9157$ GeV. **That last number is a statement about the potential and not about any expansion.**

And there is a second number, because 127 is not a mass the Higgs has. Every ceiling here is evaluated at the top of [2]'s window $[125, 127]$, which is the right end for an upper bound: $\mu \propto m_h^2$, so by (II) the ceiling rises with m_h, measured and not assumed. But the measurement is $m_h = 125.20 \pm 0.11$ GeV, and 127 sits 16.4 standard deviations above it. Evaluated instead at the mass the Higgs actually has, the vertex drops a whole lattice step and so does the answer:

$$\boxed{\frac{1}{R_5} \leq 9.09\ \text{TeV}} \qquad \text{at } A_4 = 101,\ \text{measured Higgs mass.} \tag{37}$$

against 9.22 TeV at $A_4 = 104$ over the window — 1.4 %, and a step of three in A_4. The one-sigma band does *not* move the vertex, 125.09 and 125.31 both giving 101, so (37) is stable against the measurement and unstable only against the width of a window nobody is obliged to use. The same substitution takes the unconstrained ceiling from 10.01 to 9.87 TeV. Both statements are true and they are not the same statement: 9.22 TeV bounds any content whose Higgs mass lies *anywhere* in $[125, 127]$, and 9.09 TeV bounds any content whose Higgs mass is the measured one. **A reader wanting a number about the world should take the second.** The window is kept for the headline because it is [2]'s and because a bound quoted over a stated window is the safer object. Measured in `higgs_window.py`.

And it explains a gap this paper had left unexplained. The table above tops out at 9156 GeV while the certificate says 10034, and that difference was called the gap between a relaxation and its lattice. It is not. The enumeration used the *global* minimiser, so it had been screening false vacua all along without saying so; the certificate screened nothing. Both of its champions have $W > 0$ — $\frac{31}{2}$ at $N = 8$ and $\frac{5}{2}$ at $N = 14$ — and evaluated exactly they give 9139 and 9157 GeV, which are the two numbers already in the table. **The constrained optimum *is* the enumeration's champion**, and a gap that looked like insufficient search was a physical condition nobody had written down. Measured in `socratic_witness.py` and `vacuum_constraint.py`.

And the last approximation left *inside* (2) is the truncation, and it is bounded rather than estimated. Everything above except the two exact evaluations is computed on the surface (11), which keeps three rungs of the ladder. Write $F = F_0 + R$, with R the part from $k \geq 3$. Two classical facts make R boundable and neither is ours: $\zeta(5 - 2k) = -B_{2k-4}/(2k - 4)$ with

$|B_{2n}| \le (\pi^2/3)(2n)!/(2\pi)^{2n}$ [24], and the expansion itself [25]. Bounding the bracket of (18) by the fourth moment, $A_4 + B_4$, and summing the geometric series gives, for $u = c_{\max}\alpha < 1$,

$$|R(\alpha)| \;\le\; \frac{\pi^6 S_4\, c_{\max}^2\, \alpha^6}{2160\,(1-u^2)}\,, \qquad |R'(\alpha)| \;\le\; \frac{\pi^6 S_4\, c_{\max}^2\, \alpha^5}{360\,(1-u^2)}\,. \tag{38}$$

Put to a falsification attempt on 32 contents at four values of α each — the attempt being to find one violation, and the outcome that there are none, tightest slack 1.4×. Two controls, and both can fail: the bound *refuses* to return anything at $u \ge 1$, where the true remainder does blow up; and it is a bound and not an estimate — the sharp estimate is $(A_6 + 3B_6)x^6/8640$, which is not rigorous, and mixing the two would be the easiest mistake here. An earlier version of (38) was off by a factor, and the falsification is what caught it.

Using it naively is worthless, and that is the instructive part. Replacing each rung's $1/R_5$ by its rigorous upper value and re-maximising does *not* give a rigorous ceiling of 10 TeV: it gives 52.429 TeV, and the maximum migrates from $8D = 1$ to $8D = 139$. The margin grows with A_4 while the truncated value falls, so the rungs that lose on the truncated surface win on the bounded one. **A bound on the remainder is not automatically a bound on the optimum**, and the distinction is not formal: taking it seriously moves the answer by a factor of five.

What repairs it is not a better inequality but not needing one. The charges run over $c \in \{1, 2, 3\}$ and there are two parities, so the potential is spanned by six functions *however much matter there is* — five, in fact, and the reduction is *not ours*. The polylogarithm duplication formula $\mathrm{Li}_s(z) + \mathrm{Li}_s(-z) = 2^{1-s}\mathrm{Li}_s(z^2)$ [13] at $s = 5$ gives $g^{(+,1)} + g^{(-,1)} = g^{(+,2)}/16$, and [8] already writes exactly this in exactly this setting — their eq. (2.11), $F_-(a) = -F_+(a) + \frac{1}{16}F_+(2a)$, repeated in [9] eq. (3.3). There is no second relation of that kind here because $c = 4$ and $c = 6$ are not charges of this model, so the six span five. What is used below is only that the number is finite; and the statement is about (2), the n^{-5} part of [2]'s potential, not about their full winding weight, which mixes powers and does not close on five. Hence $R(\alpha) = R_{\text{gauge}}(\alpha) + \sum_j n_j R_j(\alpha)$ exactly — the same shape as A_4, $8D$, G and W. The margin is therefore itself a linear functional and goes inside the very dual that produces the certificate, with the $G \le G^\star$ constraint released so that the answer stays a bound and the dual stays two-variable. That move is ordinary linear programming [27] and is claimed as nothing more. Over 70 rungs:

$$\boxed{\frac{1}{R_5} \;\le\; 10.037\ \text{TeV}} \qquad \text{untruncated, over the first 70 odd rungs.} \tag{39}$$

The maximum stays on $8D = 1$ across those seventy; the margin over the truncated certificate is 2.5 GeV, or 0.025 %; and against the 52.429 TeV of the uniform route it is a factor of four thousand. So the truncation is worth two and a half GeV, not two and a half TeV, and §8 is not an artefact of keeping three rungs.

And seventy rungs is where the statement stops, which an earlier version of this paragraph did not say. It read "arbitrary content", and that is false: the remainder *bound* loosens with the rung even as the truncated ceiling falls. Its relative bite $\delta/\hat{x}$ runs from 2.5×10^{-4} at $8D = 1$ to 9.1×10^{-2} at $8D = 129$ and 9.0×10^{-1} at $8D = 1201$ — a factor of three thousand six hundred, and at that last rung the bound eats nine tenths of the phase it is meant to certify. The scan there returns 27.8 TeV — not because the ceiling rises there, but because the bound stops saying anything. **The truncated ceiling is untouched by this**: (25) and the sign of $\partial\Phi/\partial k$ prove $M(k) \le 6268$ GeV for every $k \ge 3$, all rungs, no scan. What is restricted to seventy is (39) alone, and closing it needs a remainder bound uniform in the rung, which §13 now asks for.

And what in that is still numerical care rather than proof. Three places, named because a reader will find them otherwise. R'_j is a central difference rather than an analytic derivative, which the term-by-term expansion would fix immediately. The step δ uses the curvature *at* $\hat{x}$ rather than its minimum over $[\hat{x}-\delta, \hat{x}]$; at $\delta/\alpha = 2.5 \times 10^{-4}$ nothing moves, but the clean statement needs the minimum. And the winding sums are truncated at 4000, where the tail is 4×10^{-15} against $|R'| \sim 10^{-6}$. None of the three is a gap in the argument; all three are places where interval arithmetic would replace care with proof. Finally, (39) is the *unconstrained* ceiling: the true-vacuum condition has not been carried through the same machinery, and since it only removes contents, what may be said about (36) untruncated is this, and it is deliberately three statements rather than an interval. *The exact-potential witness gives* 9157 *GeV* — which needs no bound at all, being an evaluation. *Over the first seventy odd rungs the untruncated potential is bounded above by* 10.037 *TeV. A global untruncated upper bound, over arbitrary multiplicities and hence over all rungs, remains open*, and §13 names the missing piece: a remainder bound uniform in the rung. Writing those three as a single interval [9157 GeV, 10.037 TeV] would assert the third, and the third is precisely what is not proved. Measured in `remainder_theorem.py`, `rigorous_ceiling.py` and `rigorous_ceiling2.py`.

And $W > 0$ is *necessary* exactly where the bound is decided, which makes (36) a statement about every content and not about a screened family. Proving necessity in general is not required: a content only matters if it *beats* 9.22 TeV, and such a content has $\alpha < 2m_W/9220 = 0.01744$, that is $x < 0.0548$. Two facts then collide. The gap at the other symmetric point is quantised and cannot be small — by (35), $W < 0$ forces $F(1) \le F(0) - \frac{31}{32}\zeta(5) = F(0) - 1.0045$. And the electroweak well is microscopic, which is visible once its depth is written out: eliminating G with stationarity and then A_4 with (II) collapses it to two terms,

$$F(x_\star) - F(0) \;=\; -\,\frac{\zeta(3) D\, x_\star^2}{8} \;-\; \frac{\mu\, x_\star^4}{16}\,, \tag{40}$$

which reproduces the exact well to between 0.00 and 0.16 % on the five published rows and on the witness — better than the closed form's own accuracy, because the error cancels between the two eliminations. On the rung $8D = 1$, and at $x < 0.0548$, that depth is at most 1.03×10^{-4}.

The argument is then three lines. Only the rung $8D = 1$ can exceed 9.22 TeV, since the certificate already bounds $8D = 3$ at 6.27 TeV and every rung above it lower. So $D = \frac{1}{8}$ and the well is shallower than the quantised gap by a factor of 9755. If $W < 0$, then $F(1) \le F(0) - 1.0045 < F(0) - 1.03 \times 10^{-4} \le F(x_\star)$, so the electroweak point is not the global minimum and the content was not a candidate after all. $W = 0$ is arithmetically impossible. Hence $W > 0$, and the integer program over $W > 0$ has already been run. $1/R_5 \le 9.22$ **TeV therefore holds for any bulk content whose electroweak point is its true vacuum** — the screen was never a restriction where the bound is decided. The reach of the argument is $8D < 17815$, far beyond any rung that could compete. Measured in `necessity_of_W.py`. What it does *not* remove is the other caveat: step one uses the per-rung certificate, whose globality rests on the monotonicity of $t^\star(k)/k$ — which (26) now proves rather than measures.

And it is *sufficient* there too, because the screen can be replaced by the thing it stands for. Necessity leaves the other half open: $W > 0$ compares the electroweak point with the *other symmetric point* only, and a third minimum somewhere in $(0, 1)$ would be seen by neither. That half is settled by not screening at all. If the electroweak point is the global minimum then for every y

$$F(y) - F(0) \;\ge\; F(\alpha_{\rm EW}) - F(0) \;=\; -\,\big|\text{well}\big| \;\ge\; -\Theta, \tag{41}$$

where Θ denotes the bound on the well depth supplied by (40) — a separate letter from the Δ of (3), which is a moment of the content and not a depth. The left-hand side is *linear* in the multiplicities — the same shape as A_4, $8D$, G and W — while Θ is a constant of the rung. So "the electroweak point is the global minimum" is a semi-infinite integer program: solve with the cuts in hand, minimise the incumbent over y, add the cut where it is violated, repeat. It must be anchored at the origin and not at $x_\star$. The exact condition *is* $F(y) \geq F(x_\star)$ for all y — that is what a global minimum means, and it is not stronger than the truth but the truth itself. What cannot be carried into a linear program is its right-hand side, which depends on the content through $x_\star$ and its depth. Replacing it by the uniform bound $F(0) - \Theta$ keeps the program linear and keeps the relaxation *weaker* than the truth, which is the direction an upper bound needs: a maximum over too small a feasible set is not an upper bound at all.

Three things make it terminate. The cut at $y = 1$ *is* the screen, since (41) there reads $\frac{31}{16}\zeta(5)W \geq -\Theta$ and $2W$ odd then forces $W \geq \frac{1}{2}$ as soon as $\Theta < 1.0045$ — at the deciding vertex $\Theta = 1.02 \times 10^{-3}$, which is deliberately ten times the depth bound above and not a restatement of it, three orders of room even so, so the screen is a corollary of the programme rather than an input to it. The function space is finite, as above. And the eight multiplets are only *five* directions in it: the kernel of content $\mapsto$ potential has dimension three, with three relations of non-negative coefficients. Boldface is the representation and the upright digit before $\times$ is its multiplicity, so the third line has one $\mathbf{7}^{(+,+)}$ and not seven of anything:

$$\begin{aligned} &\mathbf{28}^{(+,+)} = 20 \times \mathbf{7}^{(+,+)} + 17 \times \mathbf{7}^{(+,-)}, \qquad && \mathbf{48}^{(+,+)} = 24 \times \mathbf{7}^{(+,+)} + 18 \times \mathbf{7}^{(+,-)}, \\ &\mathbf{48}^{(+,-)} = 1 \times \mathbf{7}^{(+,+)} + 4 \times \mathbf{7}^{(+,-)} + 1 \times \mathbf{28}^{(+,-)}, && \end{aligned} \tag{42}$$

so the fibre over $(A_4, 8D)$ is small enough to enumerate exhaustively. **That such degeneracies exist is [8]'s**: their §3.3 exhibits two models with different matter and the same gauge-scalar potential and calls it accidental, and their eqs. (3.25)–(3.27) are their own eq. (2.11) producing one. What is added here is that the accident is a subspace and can be listed: the kernel has dimension three and (42) spans it. The third relation is also Part VI's, found there for the opposite purpose — the two sides have different $\sum C_2$, so the degeneracy dies at two loops; the other two need thirty-seven and forty-two multiplets and that search was capped at six. Verified in exact rational arithmetic in `dimension_five.sage` on a retyped matrix.

And the coefficients being non-negative is worth a corollary. A relation with a negative coefficient would rewrite a content as a formal *difference* of contents, which is not a content. All seven coefficients in (42) are non-negative integers, so the rewriting never leaves the physical cone: every bulk content has a one-loop-equivalent representative built from five types only,

$$\begin{aligned} N_{\mathbf{7}^{(+,+)}} &= n_{\mathbf{7}^{(+,+)}} + 20\, n_{\mathbf{28}^{(+,+)}} + 24\, n_{\mathbf{48}^{(+,+)}} + n_{\mathbf{48}^{(+,-)}}, \\ N_{\mathbf{7}^{(+,-)}} &= n_{\mathbf{7}^{(+,-)}} + 17\, n_{\mathbf{28}^{(+,+)}} + 18\, n_{\mathbf{48}^{(+,+)}} + 4\, n_{\mathbf{48}^{(+,-)}}, \\ N_{\mathbf{28}^{(+,-)}} &= n_{\mathbf{28}^{(+,-)}} + n_{\mathbf{48}^{(+,-)}}, \end{aligned} \tag{43}$$

the two $\mathbf{84}$'s carrying through unchanged. The five are independent and minimal — deleting any one drops the rank to four — so this is a change of basis and not a projection.

The consequence, at its real strength. It is tempting to record this as a normality theorem: the semigroup equals its cone intersected with its lattice, no holes. That is true, and it is not a theorem. Writing B for the five generators, (43) makes $S = B\mathbb{N}^5$, $L = B\mathbb{Z}^5$ and $C = B\mathbb{R}^5_{\geq 0}$, so $x \in C$ *is* $B^{-1}x \geq 0$ and $x \in L$ *is* $B^{-1}x$ integral; "no holes" restates the definitions. What should be said instead is the shorter and stronger thing: **the semigroup of one-loop potentials is free on five generators, $S \cong \mathbb{N}^5$.** Free, not merely normal — there is nowhere to put a hole. All the content is in (43) and none of it in the deduction. Inverting B then turns the five complete invariants into five canonical multiplicities, so the congruences of §8 are the condition that B^{-1} return integers and the cone inequalities are $N \geq 0$.

And this does not settle the question §13 asks. That question is whether the *lifted* semigroup is normal in the four coordinates $(A_4, 8D, 2U, V)$, and projection does not preserve normality: a point can lie in the projected cone and the projected lattice while every preimage of it falls outside the cone. What the corollary changes is what a hole would *mean*. No potential is missing. A four-coordinate hole would record a $(A_4, 8D, 2U, V)$ whose required $2W$ no content supplies — a gap in the coordinate that was dropped, not in the space of potentials. Checked, with the controls that can fail separated from the ones that cannot, in `canonical_five.py`.

On the rung $8D = 1$ the programme stops at $A_4 = 104$ having generated exactly one cut, the one at $y = 1$ — and it is generated at $A_4 = 212$ by the content of Figure 10 itself, which the programme identifies as a false vacuum without being told. So where the bound is decided, $W > 0$ is not merely necessary: it is the *whole* of the true-vacuum condition, and (36) is a bound over every bulk content whose electroweak point is its vacuum. **The worry is not empty, though — it is real and it does not reach.** On $8D = 3$ the oracle demands a second cut, forced by $6 \times \mathbf{7}^{(+,+)} + \mathbf{7}^{(+,-)} + 5 \times \mathbf{28}^{(+,-)} + \mathbf{48}^{(+,-)} + 2 \times \mathbf{84}^{(+,+)}$, which has $W = \frac{1}{2}$ and so *passes* the screen, yet sits 4.82 below the origin there — far deeper than its own well. The oracle returns that cut as a number, $y = 0.6809$, and it need not be one. Since $\mathrm{Li}_5(z)+\mathrm{Li}_5(-z) = 2^{-4}\mathrm{Li}_5(z^2)$ puts every term of the potential in the span of $P_q(y) = \mathrm{Re}\,\mathrm{Li}_5(e^{iq\pi y})$ with $q \in \{1, 2, 3, 4, 6\}$ — Theorem 3 read off the phases rather than off the moments — and since every $P_q(2/3)$ is an exact rational multiple of $\zeta(5)$, the depth at the rational holonomy $y = 2/3$ is a closed-form linear functional of the content, in the same sense that $F(1) - F(0) = \frac{31}{16}\zeta(5)W$ is: writing $F = \sum_q C_q P_q$,

$$F(\tfrac{2}{3}) - F(0) \;=\; -\tfrac{121}{81}\,\zeta(5)\left[\,C_1 + C_2 + C_4\,\right],$$

which on this content is $-\frac{1331}{288}\zeta(5) = -4.7922$, some 579 times the allowance at the vertex that forced it, $\Theta(147, 3) = 8.28 \times 10^{-3}$ — and Θ is per vertex, eight times the 1.02×10^{-3} of the deciding one, so it is the rung's own allowance that is used here. So the second cut can be taken exactly, and the disqualification never rests on a numerical minimum. A third minimum invisible to both symmetric points exists and binds; it simply binds on a rung the ceiling does not live on. *What does not follow* is that the cuts are rational in general: over all 6560 contents of multiplicity at most two the global minimiser is strictly interior in 2956 cases, those sit a mere 1.05 times closer to $\{1/3, 1/2, 2/3\}$ than a uniform draw on the same range, and *not one* of the three skeleton bins is so much as a local maximum of where they land. The one excess that is there — 32.5 % of them in $[0.20, 0.30)$ against a uniform 12.7 % — sits against the electroweak end and not at any rational. The charge alphabet supplies a skeleton; it does not generate the cuts. Both measured, in `rational_holonomies.py` and `holonomy_skeleton.py`. Finally the winning witness is taken off the grid altogether: with the winding tails bounded by $\sum_{n>N} n^{k-5} \le N^{k-4}/(4-k)$ and $|F^{(k+1)}| \le \sum |m|(c\pi)^{k+1}\zeta(4-k)$ closing the gaps between grid points, $F'' < 0$ from the origin with $F'(0) = 0$ exactly, then $F' < 0$, then a convex window containing a sign change of F', then a certified margin of 0.42 on the rest of $[0, 1]$: **its electroweak point is the global minimum of F on** $[0, 1]$, not merely the better of two symmetric points. Measured in `semi_infinite.py`.

So there are two ceilings and they answer different questions. 10.01 TeV bounds any content whose electroweak point is stationary, which is what an integer program over the moments can see; 9.22 TeV bounds any content whose electroweak point is also the *vacuum*, which is what the model has to mean. The second is the physical one. The first is kept because the argument that produces it is what makes the second computable, and because the distance between them is the price of a question the certificate cannot ask.

Controls. Every champion of that curve was re-minimised numerically on the original potential rather than trusted to the closed form: the worst error is +0.62 %, on the row of largest α, and every champion's m_h is still inside 125–127 GeV at the numerical α. The certified ceiling dominates all 1097 window contents found up to $N = 14$. And the control that must fail does: ignoring Theorem 2 places the optimum on the phantom point $(216, 1)$ and inflates the ceiling by 7 GeV.

And Part VI's escape moves it. Part VI [33] obstructs the one-generation version of this model and classifies the minimal escapes; the surviving one is a family-dependent charge that must be hosted by an $\mathbf{84}^{(+,+)}$ donated to the brane, which removes it from the Higgs potential. The lattice of §7 prices that donation without a new computation: the host carries $8D = +10$, so paying for the escape costs exactly 10/8 of D. Electroweak breaking needs $D > 0$, and $8D$ is odd by Theorem 1 on the seed [2] print, so

$$\text{a content can afford the escape} \quad \Longleftrightarrow \quad 8D \geq 11, \tag{44}$$

an integer condition with no slack in it. The per-D ceiling is monotone decreasing, so (44) does not merely forbid contents — it forbids the rungs the ceiling lives on. Running the same exact rational dual over every odd rung from 11 upwards:

$$\boxed{\frac{1}{R_5} \leq 3.97 \text{ TeV}} \quad \text{for any content that can afford the escape.} \tag{45}$$

Against 10.03 TeV without it, both on the seed [2] print — a factor 2.53.

The quantum does the work twice. The unconstrained ceiling sits *on* $8D = 1$, and the escape's bill is exactly what makes that rung unaffordable: the content that generates the largest hierarchy is the one that cannot pay. The enumeration respects the bound — of the 1097 window contents at $N \leq 14$, 857 can still afford it, and the best of those reaches 3017 GeV. Scope, and it has to be said whole: this bounds the content *before* the donation, the row that pays. It does not bound the world after, whose bulk needs only $8D \geq 1$.

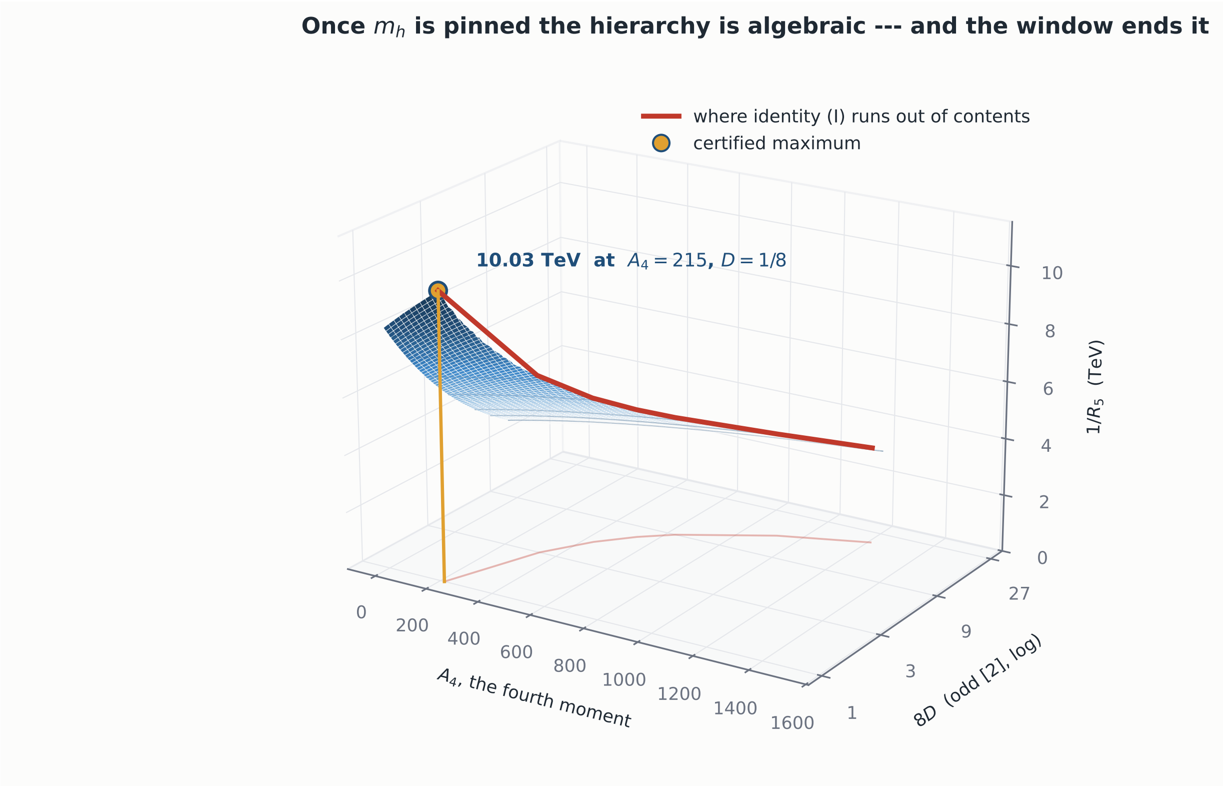
And it answers a question Part VI could only state. Part VI asks, in its own list of what it does not claim, whether a row that pays for the escape still lands in the Higgs window afterwards — and says the modified potential is not recomputed there, because doing so needs a closed form for $\alpha_{\min}$ it does not have. This paper has one, so the question is now arithmetic. Donating one $\mathbf{84}^{(+,+)}$ from each published row and re-minimising through (22):

row	$8D$ after donating	m_h after
(1)	5	145.6 GeV
(2)	19	114.7 GeV
(3)	−1	no interior minimum
(4)	1	169.0 GeV
(5)	5	145.6 GeV

None of the five lands in 125–127 GeV once the host is donated, and case (2) — the row Part VI selects as the only one that can *pay* — comes out lowest of the four that still break. Paying for the escape and keeping the Higgs mass are therefore two different demands, and the five published rows satisfy them one at a time. Every m_h here is a *measured* number and carries the anchor residual of §2; what does not is the $8D$ column, which is the arithmetic of §7.

Scope of the ceiling. No further physical condition is imposed on the lattice — not six-dimensional anomaly freedom, not perturbativity. That is the safe direction: any such criterion can only *remove* contents, so the bound stands, while the champion content would move. Figure 11 is that ceiling drawn as an explicit surface over the two quantised moments.

A bound is only worth stating if something can be done with it — and the same two integers that produced it also say where, inside the bound, the answer is allowed to sit.



Surface: $1/R_5 = 2\pi m_W \sqrt{(6\mu + A_4)/(12\zeta(3)D)}$, identity (II), at $m_h = 127$ GeV. Beyond the red edge no content satisfies identity (I).
8D is odd on the seed [2] print; on the candidate seed of the open-questions section it is even, and the staircase starts one step higher.

Figure 11: The ceiling's geometry. Once m_h is pinned, identity (II) makes $1/R_5$ an explicit surface over the two quantised moments, with no minimisation left in it. The red curve is where identity (I) runs out of contents: beyond it the lattice cannot supply a G small enough. The maximum of the surface over the feasible region is the certified ceiling, and it sits on the smallest positive rung the arithmetic admits.

9 What this predicts

Anchor-free, and arithmetic — but not equally. The congruence $8D - 2A_4 \equiv 0 \pmod 6$ is a property of the matter lattice alone: no normalisation, no loop order, no scheme, and no gauge sector either. The *parity* of the total $8D$ is a different animal: it is fixed by the gauge base point, and §13 is about which base point that is. Both are still tests a reader can run on someone else's table in a minute; only the first is a test of the content rather than of the seed. Their own five rows pass both.

A correlation, not two numbers. $\alpha_{\min}$ and m_h are functions of the same two moments, so they are not independent data. Pinning m_h to the measured value *fixes* $1/R_5$ from the content through (II), with no minimisation — which is what makes the ceiling computable at all.

And the Kaluza-Klein scale is not a continuum: it sits on an arithmetic comb. Cross (II) with Theorem 2 and something falls out that neither says alone. Writing $k = 8D$ and $M_{\mathrm{KK}} = 1/R_5 = 2m_W/\alpha_{\min}$, (II) reads

$$M_{\mathrm{KK}}^2 = \frac{8\pi^2 m_W^2}{3\zeta(3)} \frac{6\mu + A_4}{k}, \tag{46}$$

and Theorem 2 fixes A_4 modulo three once k is fixed, so at fixed k and fixed m_h the admissible A_4 advance in steps of three and

$$M_{n+1}^2 - M_n^2 = \frac{8\pi^2 m_W^2}{\zeta(3)\, k}, \tag{47}$$

independent of the content and of the Higgs mass. **The exactly-spaced quantity is M^2, not M**, and a spacing in mass is $\Delta M = \Delta M^2/2M$, so it is meaningless without saying at which M — and each rung has its own ceiling, so quoting them all at one mass quotes most of them where they cannot reach. At $k = 1$ the step is 0.42 TeV^2, which at that rung's own ceiling of 10 TeV is 21 GeV in mass; at $k = 11$ the step is 0.039 TeV^2 and the ceiling is 3.97 TeV, so 4.9 GeV. The statement is *necessary and not sufficient* — every realisable content lands on a tooth, not every tooth holds a content — and the *position* of the comb carries the anchor residual and the choice of g_4 while its *spacing* does not, being arithmetic. Read as a test it is unusual: given m_h, m_W, g_4 and a candidate Kaluza-Klein resonance, one may ask whether any admissible pair (k, A_4) reproduces it. **Which pairs are admissible belongs to the seed and not to the lattice**: on the seed [2] print the test reads k odd, $A_4 \in \mathbb{Z}$ and $A_4 + k \equiv 0 \pmod 3$, while on the candidate seed k is even and $A_4 \in \mathbb{Z} + \frac{1}{2}$. What survives either is Theorem 2 written so that every entry is an integer on both seeds — A_4 is half-integral on the candidate one, $2A_4$ is not — namely $k - 2A_4 \equiv 3 \pmod 6$ with $(2A_4, k) \in \mathbb{Z}^2$. The spacing is the matter lattice's and the offset is the affine coset's, which is the same split §13 turns on. Verified against the certificate row of §8 in `crossed_equations.py`.

And the ceiling has collider teeth. A bound on $1/R_5$ is a bound on a compactification scale and not on a single resonance mass. [2]'s eqs. (18)–(19) expand the mixed-parity components in $\cos\big((n+\frac{1}{2})x_5/R_5\big)$, so those towers are half-integer-moded and their lightest state sits at $1/(2R_5)$, while the unbroken gauge bosons keep integer levels n/R_5. Which of them a given search bounds, with what coupling to the fixed-point quarks and what width, is fixed by the same parities and is derived in §10; the short answer is that the half-integer tower is a colour singlet and the only coloured state below the Planck scale is the gluon tower at n/R_5. Carrying the anchor residual of §2 through:

	our normalisation	$\times$ the largest measured ratio
content of their own size (≤ 6 multiplets)	2.18 TeV	4.53 TeV
ceiling over *all* contents	10.03 TeV	20.82 TeV†
ceiling once the vacuum must be the true one	9.22 TeV	19.14 TeV†
ceiling for contents that can afford the escape	3.97 TeV	8.24 TeV†

At the mass the Higgs actually has, rather than at the top of [2]'s window, the highlighted row reads 9.09 TeV — (37), and the benchmark to quote at the physical mass, carrying the same two caveats as every theory-side scale here. The right-hand column is not carried across for it, being an extrapolation already.

The highlighted row is the physical one: the row above it allows contents whose electroweak point is stationary without being the minimum, and §8 shows the content that saturates it is exactly of that kind. The 10.03 is kept because it is the bound the certificate proves for arbitrary content, and because the difference between the two rows — eight hundred GeV — is what one physical condition is worth.

†*Extrapolations, not bounds.* The right-hand column multiplies by 2.076, the largest ratio measured on the five published rows. That is legitimate for the first row, whose contents are of their own size and sit inside the anchored range. It is not for the other two: by §2 the ratio is a trend rather than scatter, and the two ceilings sit outside the range that measures it — the first in D ($1/8$ against $[1.125, 3.625]$), the second in A_4 (727 against $[189, 271]$). The trend rises towards the first of those corners, so 20.8 is if anything an under-estimate of where the rescaled ceiling would sit; but it is an extrapolation of five points, and it is written here as one.

Against published exclusions — a Kaluza-Klein gluon below roughly 4 TeV, a sequential Z' below roughly 5 TeV — the small-content corner of the class, which is the corner their Table 1 populates, is already disfavoured, and that is a consequence of the closed form rather than of any new measurement. The ceiling then caps the whole class. And the escape bound of (45) bites harder: 3.97 TeV sits in a region dijet data are highly sensitive to, and the rescaling that would lift it to 8.24 TeV is the extrapolation flagged above, not a bound, so the sensitivity is not evaded by appealing to it. **And the recast is now done, which is what puts a number on the other side at all.** It could not be taken from the generic Kaluza-Klein gluon limits quoted above — those are for a *narrow* resonance, while §10 derives this object's own width, $\Gamma(G^{(n)} \to q\bar{q})/M_n \simeq 0.16$, and the s-channel rate would additionally need $\mathrm{BR}(G_1 \to jj)$, which the model does not fix. The spacelike form factor (52) needs neither, and carried through [39]'s published covariance it gives a nominal recast sensitivity of $1/R_5 > 10.86$ TeV, rising to 13.75 TeV on the mass range [39] themselves use for contact interactions. **So the branch that can also pay for Part VI's escape from proton decay, at 3.97 TeV, sits a factor of three below where this recast loses sensitivity.** §10 says how, why that is not yet the word *excluded*, and what would make it one.

> The prediction is bounded *above*, which is the useful direction: **the model cannot retreat to higher scales by adding matter.** Adding matter raises A_4, and a larger A_4 raises $1/R_5$ only until the Higgs window closes it off — which is precisely what the integer program measures. An analysis with demonstrated sensitivity to *this* tower across the algebraically allowed mass range could therefore test the three-rung class — a weaker claim than saying a machine of some beam energy settles it, and the right one: §10 says which resonance, why reach and beam energy are not the same number, and what still stands between a sensitivity and an exclusion.

A prediction inside someone else's framework. Equation (20) says that the orbifold-stability criterion of [8], applied to this model, is never marginal. That is a statement in their language, about their question, checkable against their own code.

A ceiling on a compactification scale is not a ceiling on anything a detector records. Which state would a search actually be bounding, with what coupling and what width — and is any of that ours to choose?

10 Which state a search bounds

A bound on $1/R_5$ is a bound on a compactification scale, not on the mass of any one particle. A collider does not measure compactification scales; it measures resonances, each with a production rate and a width. Getting from one to the other looks like it needs a phenomenological study, and it does not: it is fixed by the three parity matrices of [2] and the mode expansions that follow from them, all of which are already published. This section reads them.

The parity that separates colour. Of the three $\mathbb{Z}_2$ matrices, the one acting in x_6 is

$$P_6 = \mathrm{diag}(+,+,+,-,-,-,-), \tag{48}$$

[2]'s (11), which splits the seven-dimensional index into the colour triplet and everything else. A gauge component along E_{ij} carries the *product* $P_6^{(i)} P_6^{(j)}$, so (48) has an immediate consequence that is arithmetic rather than dynamical:

$$P_6(i,j) = -1 \quad \Longleftrightarrow \quad \text{exactly one of } i,j \text{ is a colour index,} \tag{49}$$

that is, precisely when the generator changes colour. There are 24 such components, and there is no colour-changing component with $P_6 = +1$.

Theorem 4 (colour lives on the small circle)**.** *Every colour-changing gauge boson of [2] has $P_6 = -1$. By its (21)–(24) each such component expands in $\sin(mx_6/R_6)$ with $m \geq 1$. Hence its lightest level is at least $1/R_6$, and its wavefunction* vanishes *at $x_6 = 0$.*

Proof. (49) is the statement that $P_6^{(i)} P_6^{(j)} = -1$ iff one index lies in the colour block, which is immediate from (48). The four P_6-odd expansions of [2], its (21)–(24), all carry $\sin(mx_6/R_6)$ and all start at $m = 1$; the sine vanishes at the origin and the level is bounded below by the $m = 1$ term. □

This matters because [2] p. 3 fixes $R_5 \gg R_6$ and calls $1/R_6$ "an arbitrary parameter, which is the Planck scale for instance". Theorem 4 then says the coloured exotics have no tree-level coupling to the brane quarks at all, and that in the $R_5 \gg R_6$ hierarchy the model intends they lie far beyond collider reach. Neither statement is that they are unobservable in principle: $1/R_6$ is called arbitrary rather than fixed, and a coloured X_m is still a bulk gauge field, so the vertex $g^{(0)} X_m X_m$ carries $\int dx_6 \sin^2(mx_6/R_6) \neq 0$ and QCD pair production survives the vanishing of the single-X vertex. What the theorem removes is the direct channel, which is the one that matters for proton decay.

So the brane quarks see exactly two towers. [2] p. 22 introduces "quarks q_L, u_R, d_R as the 4D fermions on the brane at the origin in a compactified space $(0, 0)$". A 4D field there couples only to components whose mode function is non-zero at that point — that is, to components whose x_5 *and* x_6 factors are both cosines. Of the eight parity classes only two qualify, and both are colour-diagonal:

parity	eq.	x_5 mode	x_6 mode	at $(0, 0)$	lightest level
$(+,+,+)$	(17)	$\cos(nx_5/R_5)$	$\cos(mx_6/R_6)$	**non-zero**	0, then $1/R_5$
$(+,+,-)$	(18)	$\cos((n+\frac{1}{2})x_5/R_5)$	$\cos(mx_6/R_6)$	**non-zero**	$1/(2R_5)$
$(+,-,+)$	(19)	$\sin((n+\frac{1}{2})x_5/R_5)$	$\cos(mx_6/R_6)$	zero	$1/(2R_5)$
$(+,-,-)$	(20)	$\sin(nx_5/R_5)$	$\cos(mx_6/R_6)$	zero	$1/R_5$
$(-,\cdot,\cdot)$	(21)–(24)	—	$\sin(mx_6/R_6)$	zero	$\geq 1/R_6$

The $(+,+,-)$ class is exactly the two positions $(6, 7)$ and $(7, 6)$ — a colour singlet. So the half-integer tower, whose lightest state sits at $1/(2R_5)$ and which §9 correctly flagged as the lowest level in the model, is *not* something a dijet search can see; and it is not a clean prediction either, since $\langle W \rangle$ of [2]'s (77) acts in the $(5, 7)$ block and shifts it with α. The one coloured object below the Planck scale is the gluon tower.

And that answers a question [2] left open. Their conclusions read: "Since the mass of X, Y gauge bosons is small in our model, this causes rapid proton decay *if they can couple to quarks and leptons*", and they leave the question to future work. By Theorem 4 they cannot: the X and Y bosons are precisely the colour-changing components, and their wavefunction is zero at the fixed point where the same paper puts the quarks. The gauge-exchange channel is shut by the parity assignment itself.

The scope of that is narrow and worth stating. It is a tree-level statement about gauge exchange, for the quarks localised as in [2] p. 22. It says nothing about the brane Dirac masses that same paper introduces on p. 16 to lift its unwanted zero modes, and it is *not* the escape of Part VI [33], which is a family-dependent charge hosted by a donated $\mathbf{84}^{(+,+)}$ — a different mechanism, addressing a different operator. In particular (45) is untouched by it.

The coloron's mass and quark coupling are fixed — and neither is new. Three facts fix it, and all three are results of other people applied here rather than found here. The mass: [2] p. 16 states that the $SU(3)_C$ generators "are commutable with $\langle W \rangle$ irrespective of the VEV $\alpha_{\min}$", so the colour tower takes no Wilson-line shift and sits at n/R_5 exactly. The coupling: their (17) carries $\sqrt{2}$ in front of every $(n, 0)$ mode where the zero mode carries 1, and $\cos(0) = 1$, so at the brane the ratio is $\sqrt{2}$ — and it is flavour-universal, because q_L, u_R and d_R sit at the same point and see the same mode function. The width then follows from the standard colour-octet result $\Gamma(C \to q\bar{q}) = n_f\, \alpha_s \cot^2\theta\, M/6$ with $\cot^2\theta = 2$ and $n_f = 6$:

$$M_n = \frac{n}{R_5}, \qquad \frac{g_{G^{(n)}q\bar{q}}}{g_{gq\bar{q}}} = \sqrt{2}, \qquad \frac{\Gamma(G^{(n)} \to q\bar{q})}{M_n} = 2\alpha_s(M_n) \simeq 0.16. \tag{50}$$

Whose result each of those is. The $\sqrt{2}$ of a Kaluza-Klein gauge boson on a brane-localised fermion, and the width $\Gamma_n = 2\alpha_s m_n$ that goes with it, are [35], whose Lagrangian — gauge fields in the bulk, quarks held at $y = 0$ by a delta function — is structurally the setup of [2]. The colour-octet Lagrangian and the width formula are [36]; the production cross section at next-to-leading order is [37]; and a Kaluza-Klein gluon confronted with a dijet distribution in a gauge-Higgs model is [38], who also note that $\sqrt{2}g_s$ is the *largest* coupling available and carry a free parameter below it. What is specific here is only that [2] puts its quarks exactly at the fixed point, so the bound is saturated rather than assumed — and that (24) caps the scale from above, which is what makes any of these numbers testable at all. Carried across the rows of §9:

	$1/R_5$	α_s	Γ/M	Γ
ceiling at the measured m_h	9.09 TeV	0.0735	0.147	1337 GeV
ceiling once the vacuum must be the true one	9.22 TeV	0.0734	0.147	1354 GeV
ceiling over *all* contents	10.03 TeV	0.0729	0.146	1463 GeV
contents that can afford the escape	3.97 TeV	0.0789	0.158	626 GeV

Which changes what may be quoted against it. A width of one sixth of the mass is not a narrow resonance, so the narrow benchmark of a dijet search does not bound this object: CMS's 6.6 TeV for a narrow axigluon or coloron at 137 fb^{-1} [34] is a limit on a different particle. What does apply is the model-independent limit on $\sigma\, B\, A$ that the same analysis publishes as a function of mass *and* width, read at $\Gamma/M \simeq 0.16$. The previous version of this section compared (45) to a generic Kaluza-Klein gluon exclusion; (50) is what makes that comparison a statement about this model rather than about a benchmark, and it is the reason the comparison is worth making at all.

A correction to how the reach was stated. An earlier version of this paper wrote that a machine reaching 10–20 TeV settles the class. That conflates the energy of a collider with its reach in resonance mass, and the two are not the same: producing a 9 TeV resonance at $\sqrt{s} = 13$ TeV requires partons at values of x where the parton densities are negligible. The correct statement is about mass reach in the relevant channel, and it makes the escape branch far more interesting than the ceiling — 3.97 TeV sits where Run 2 has real sensitivity, while 9.09 TeV does not.

The tower also acts below the first resonance. Since every $(n, 0)$ mode has the same $\sqrt{2}$ at $x_5 = 0$, integrating out the whole tower gives a colour-octet four-quark operator with a coefficient that is a pure number:

$$\sum_{n\geq 1} \frac{g_n^2}{2M_n^2} = g_s^2 R_5^2 \sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}\, g_s^2 R_5^2, \qquad \text{i.e.} \qquad \Lambda_8 = \frac{\sqrt{3}}{\pi\sqrt{\alpha_s}}\,\frac{1}{R_5}, \tag{51}$$

which is 7.79 TeV on the escape branch and 18.48 TeV at the measured-m_h ceiling. **Two things must be said about those two numbers before anyone compares them with a published limit.** First, (51) is the leading term of an expansion in $\hat{s}/M_1^2$, and the dijet data that would test it sit at $M_{jj} \sim 2$–8 TeV — which is *on* the tower when $1/R_5 = 3.97$ TeV, not below it. There the contact operator is the wrong object and the exact tree-level statement is the replacement of the gluon propagator by $1/q^2 + \sum_n 2/(q^2 - M_n^2 + iM_n\Gamma_n)$ of [35], which is what `kk_dijet_lo.py` evaluates. **Second, and it replaces a correction an earlier version of this section carried**: that propagator does not need to be truncated in the spacelike channels, and it must not carry a width there. For $t < 0$ every mode sits below its two-parton threshold, so the self-energy has no absorptive part and the propagator is real; and with all c_n equal — which is Theorem 4 together with [2]'s localisation of the quarks at the fixed point, and is what makes the sum $\sum_n (n^2+a^2)^{-1}$ rather than an arbitrary one — Euler's expansion does it in closed form. Writing $a = R_5\sqrt{-t}$,

$$\frac{1}{q^2}+\sum_{n\geq 1}\frac{2}{q^2-n^2/R_5^2} = \frac{\mathcal{F}(q^2)}{q^2}\,, \qquad \mathcal{F}(q^2) = \begin{cases}\pi a\coth(\pi a)\,, & a = R_5\sqrt{-q^2}\,, \quad q^2<0\,,\\ \pi b\cot(\pi b)\,, & b = R_5\sqrt{q^2}\,, \quad q^2>0\,.\end{cases} \tag{52}$$

so the whole coloured tower is one *form factor* on the gluon propagator. The resummation is standard — it is how a brane-to-brane propagator is written, and the identity goes back to Euler; what is specific here is that the universal $\sqrt{2}$ makes it available at all. Three things follow at once. Its expansion $\pi a \coth \pi a = 1+\pi^2 a^2/3 - \pi^4 a^4/45 + \cdots$ returns $-\pi^2 R_5^2/3$, which is (51): the contact operator is the first term of (52) and not an independent statement. On a subprocess with only t-channel exchange the parton-level ratio is exactly $\mathcal{F}^2$, and with $|t| = M_{jj}^2/(1+\chi)$ that is a closed form in the two variables the measurement is binned in. And *it does not depend on* $\mathrm{BR}(G_n \to jj)$, because nothing is produced and decayed — which is the one thing the model does not supply, below. **The 2.55 % width correction to Λ_8 that stood here is therefore withdrawn**: it came from inserting a fixed $iM_n\Gamma_n$ where there is no absorptive part to insert, and the giveaway is that the shift does not switch off as $t \to 0$ but tends to $1/(1+(2\alpha_s)^2)$, which no self-energy below threshold may do. The zero-width $\pi^2/6$ is the right coefficient, not an approximation to a better one. Measured in `kk_resummation.py`.

And the second line of (52) collapses this section into one statement. Nothing in the sum required $q^2 < 0$; that was only where the width could be dropped. The same expansion continues to the timelike side by $\coth(ix) = -i\cot x$, so a single analytic function covers the whole tree-level tower. Then the three things this section has been saying separately are three regions of it. The contact operator (51) is its expansion about $b = 0$. The angular distortion of Figure 12 is its spacelike branch. And *the resonances are the poles of the cotangent*, at $b = 1, 2, 3, \ldots$, that is $\sqrt{q^2} = n/R_5$ — they are not put in by hand and no truncated sum is needed to find them; they are where the closed form diverges, and they are precisely and only where a width must replace the pole. So the amplitudes read $\mathcal{M}_{qq\to qq} = \mathcal{M}_{\mathrm{QCD}}[\mathcal{F}(t),\mathcal{F}(u)]$ and $\mathcal{M}_{q\bar{q}\to q\bar{q}} = \mathcal{M}_{\mathrm{QCD}}[\mathcal{F}(s),\mathcal{F}(t)]$, and the width enters exactly one of the four arguments. Both branches checked against the sum in `kk_resummation.py`.

And the second circle is genuinely UV-sensitive, so what has to be quoted is its coefficient and not its absence. An earlier version of this paragraph said the logarithmic divergence one would fear from a codimension-two brane "never switches on". It does: a two-dimensional tower is logarithmically sensitive to the cut-off and no hierarchy removes that. What the hierarchy suppresses is the coefficient, and the same Euler identity gives it. Summing n first at fixed m, $\sum_{n\in\mathbb{Z}}(n^2/R_5^2 + m^2/R_6^2)^{-1} = \pi R_5 (R_6/m)\coth(\pi m R_5/R_6) \to \pi R_5 R_6/m$ for $R_5 \gg R_6$, so

$$\frac{\Delta C_{m\geq 1}}{C_{m=0}} = \frac{6}{\pi}\,\frac{R_6}{R_5}\,\ln(\Lambda R_6)\,, \tag{53}$$

logarithmic in the cut-off with one power of R_6/R_5 in front — 1.3 % at $R_5/R_6 = 10^3$ and a cut-off a decade above $1/R_6$, and far less at the $1/R_6$ [2] p. 3 contemplates. Verified against the direct double sum in `kk_resummation.py`. This is a collective prediction of the tower and does not require resolving any resonance; it cannot, however, be compared directly with published compositeness scales, whose operator has a different colour and chiral structure.

And one number the model does not supply. [2] p. 16 lifts its unwanted zero modes by introducing "4D fermion conjugate to the representations and charges of the massless fermions and their Dirac mass terms on the fixed points", and does not fix those masses. If any coloured exotic sits below $M/2$ then $G^{(1)} \to \Psi\bar{\Psi}$ opens, the width grows and $\mathrm{BR}(G^{(1)} \to jj)$ falls by an amount nothing here determines. So

$$\mathrm{BR}(G^{(1)} \to jj) = 1 \quad \text{if every coloured exotic lies above } M/2, \tag{54}$$

and less otherwise. That single unknown is what stands between the arithmetic ceiling and a model-specific exclusion, and a limit quoted without it would be model-independent in appearance only. It is stated here rather than absorbed.

And that unknown points at a different measurement than one would expect. A resonance search looks for a bump, and a bump is only the first rung of (50) — so its rate carries precisely the branching ratio that (54) declines to fix. But the quarks of [2] sit at a point, so they see *every* rung with the same $\sqrt{2}$, and what the whole tower does is not to add a peak: it replaces the gluon propagator by $1/q^2 + \sum_n 2/(q^2 - M_n^2 + iM_n\Gamma_n)$ — in *every* channel, t and u as well as s, which is how [35] write it. Well below the first rung the change is a constant, $-\frac{\pi^2}{3}R_5^2$, so the *relative* change grows with $|q^2|$; and at fixed dijet mass $|t| = M_{jj}^2/(1+\chi)$ is largest at small χ. The effect is therefore central, and the observable is the angular distribution rather than the mass spectrum.
There is a measurement built for exactly that. [39] reports χ_{dijet} at $138\,\text{fb}^{-1}$ *corrected for detector effects* and compared with QCD at next-to-next-to-leading order, quoting a total theoretical uncertainty of 3.5% in its 3.0–3.6 TeV mass bin. Being unfolded, it can be confronted with a calculation directly. Evaluating (52) in the lowest χ bin of that same mass range gives a ratio to QCD of 1.36 at the measured-m_h ceiling and 3.16 on the escape branch, against the 1.3 and 3.4 the all-channel calculation of `kk_dijet_lo.py` returns at the same point — the two agree to a per cent and a half where they should; Figure 12 draws the whole shape, and the shape is the point — the departure is concentrated at small χ and dies away by $\chi = 16$, which is what an angular analysis is built to see and a mass-spectrum search is not. Those are partonic numbers and therefore an over-estimate — the gluon-initiated subprocesses are untouched by the tower, by the same five-momentum argument of [35] that removes it from $q\bar{q} \to gg$, and they dilute it. The comparison with 3.5% is not a significance either, and it is not even the right 3.5% everywhere: that figure is [39]'s total theoretical uncertainty in the 3.0–3.6 TeV bin alone, and above 7 TeV theirs is 12.2% — scale, m_{jj} against $\langle p_T\rangle$, parton densities and the NNLO generator, in that order of size. A deformation that grows across the mass bins must therefore be read against an uncertainty that grows with it, and against the correlations between bins, which is what the recast below does and what the size of a partonic effect cannot.
And the published limit is not the one to read off. [39]'s contact-interaction scenarios are stated to have colour-singlet couplings between quarks, while the tower's operator is a colour octet, and the two interfere with QCD differently. The octet carries the colour flow of one-gluon exchange, so in the t-channel it interferes at full strength while the singlet does not, by $\mathrm{Tr}\,T^a = 0$. That contraction is not the whole story: for *identical* quarks the crossed t–u terms restore the singlet's interference and its colour weight there is three times the octet's. Both are verified with explicit generators in `colour_octet_recast.py`, and the simplest check needs none of that algebra — [39] quotes 17 TeV for destructive and 37 TeV for constructive interference, and a vanishing interference could not make the limit depend on the sign. So which operator a given dataset constrains more tightly is not decidable from colour factors: it turns on the flavour composition of the sample. What does follow is that the published scale cannot be carried across. The translation is a calculation, not a rescaling, and it is performed below.

The recast, and what one number of it may be quoted. [39] publish the unfolded χ_{dijet} distribution in seven mass bins together with a 77×77 correlation matrix and their own QCD NNLO + EW NLO prediction at *both* of their scale choices, so the Standard-Model baseline is

Figure 12: Why the angular distribution and not the mass spectrum — and it is one function. The exact t-channel ratio $\mathcal{F}^2$ of (52), with no truncation, no width and no $\mathrm{BR}(G_n \to jj)$. At fixed dijet mass $|t| = M_{jj}^2/(1+\chi)$, so the effect is largest where χ is smallest, which is exactly where t-channel QCD is weakest. *Left:* the branch that can also pay for Part VI's escape from proton decay. *Middle:* the whole class, capped at the measured Higgs mass. *Right:* why those two are the same picture. $\mathcal{F}$ sees M_{jj}, χ and $M_{\rm KK}$ only through $a = M_{jj}/M_{\rm KK}\sqrt{1+\chi}$, so every curve in the other two panels *is* $(\pi a \coth \pi a)^2$ and they collapse onto one line. The hollow markers are the superseded all-channel calculation of `kk_dijet_lo.py` — tower truncated, widths on — which meets the closed form to 1.4 % away from the tower and departs from it upward only where the s-channel opens; that departure is exactly the part $\mathcal{F}^2$ does not describe, and it is drawn rather than described. The grey band is the 3.5% total theoretical uncertainty [39] quotes for its normalised $\chi_{\rm dijet}$ in the 3.0–3.6 TeV bin — drawn to give the vertical axis a scale, **not as an exclusion threshold**, which would need the full likelihood over correlated bins. **And the curves are an over-estimate**: they are one subprocess at parton level, gluon-initiated ones do not see the tower at all by the same five-momentum argument that removes it from $q\bar{q} \to gg$, and every CMS distribution is normalised per mass bin. What the picture establishes is the size and the shape — which is a reason to do the recast, not a result of one. Drawn by `make_fig_tower.py`.

theirs and only the relative deformation is ours. The deformation is (52) in t and u alone — no width, no $\mathrm{BR}(G_n \to jj)$ — folded with parton densities over every subprocess, and fitted with the scale choice profiled between their two columns as a nuisance, because their own small tension below 4.8 TeV moves between them. Three things are checked before the fit is read. The hadron-level layer reproduces CIJET's leading-order colour-octet interference to 0.2 % bin by bin, absolutely and in picobarns, on a matching with no free factor. [39]'s own NNLO put through this χ^2 gives $\chi^2/\mathrm{ndf} = 0.81$. And, aimed at the published contact-interaction limit for the standard colour-singlet operator — and run on the mass range they set it on, $M_{jj} > 3.6$ TeV, which is not the full seven bins — the machinery returns 17.5 against their 17 TeV and 29.5 against their 37 TeV. On all seven it returns 20.5 and 25.5 instead, and the difference is the point: a closure test has to close on the selection it is closing on.

The statistic, written out, because a limit whose χ^2 is only in the code is not reproducible from the paper. Let d be the 77 unfolded values, laid out in seven mass blocks B_m of sizes $(12, 12, 12, 12, 12, 12, 5)$, and let

$$\begin{aligned} t_i(M, a) \;&=\; \mathcal{N}_m \Big[(1-a)\, T_i^{\mu = m_{jj}} + a\, T_i^{\mu = \langle p_T \rangle} \Big] \big[1 + \delta_i(M) \big], \\ \chi^2(M) \;&=\; \min_{a \in [0,1]} \; r^{\mathsf{T}} C^{-1} r \,, \qquad r = d - t(M, a)\,, \end{aligned} \tag{55}$$

with $\delta_i(M)$ the tower's deformation of (52) in t and u, and $\mathcal{N}_m$ renormalising each mass block to unit area, as the measurement is. The nuisance a interpolates linearly between [39]'s two published scale choices and is profiled by minimisation with *no* penalty term — it is a choice between two of their own columns, not a measured parameter, so a Gaussian prior on it would be invented. The quoted number is the M at which $\chi^2(M) - \chi^2_{\rm SM}$ falls through 3.84, with a re-profiled at every M.

That is an SM-referenced criterion and not Wilks' 95 % point, and the difference is not academic here. Wilks measures $\Delta\chi^2$ from the *minimum*, which equals $\chi^2_{\rm SM}$ only if the Standard Model is the best fit — and it is not. On the full seven bins χ^2 bottoms out at 58.46 at $1/R_5 = 16.0$ TeV against 62.57 for no tower at all: **the data weakly prefer a finite tower**, by $\Delta\chi^2 = 4.12$. That is a nominal $2.0\,\sigma$ for one degree of freedom and it should be read as almost nothing, for a reason the same scan supplies: restrict to $M_{jj} > 3.6$ TeV, which drops the 3.0–3.6 TeV bin, and the preference collapses to $\Delta\chi^2 = 0.29$. *The preference is [39]'s own* $2.0\,\sigma$ *local excess in that bin, seen through our template, not information this recast adds.* There is no look-elsewhere correction here, no detector-level likelihood, and the $\theta = 1/M^2 \geq 0$ boundary makes the asymptotic distribution a Chernoff mixture rather than a χ^2_1; we report the number because hiding it would be worse, and claim nothing from it.

So the criterion is varied instead of defended, and the spread is what the reader should look at:

what is varied	points	covariance	criterion	$1/R_5$ [TeV]
— (the reference point)	70	raw C	$\Delta\chi^2_{\rm SM} = 3.84$	11.20
reference of $\Delta\chi^2$	70	raw C	$\Delta\chi^2_{\rm min} = 3.84$	12.18
reference and one-sidedness	70	raw C	$\Delta\chi^2_{\rm min} = 2.71$	12.60
scale profiled *discretely*	70	raw C	$\Delta\chi^2_{\rm SM} = 3.84$	11.19
normalisation projected out	70	JCJ^{T}	$\Delta\chi^2_{\rm SM} = 3.84$	11.29
NNPDF3.1 [50], [39]'s baseline	70	raw C	$\Delta\chi^2_{\rm SM} = 3.84$	11.62
CT14nlo [51]	70	raw C	$\Delta\chi^2_{\rm SM} = 3.84$	11.46
quoted in (56)	**77**	raw C	$\Delta\chi^2_{\rm SM} = 3.84$	**10.86**

The columns matter: the spread is *not* all statistical redefinition. Seven of the eight rows sit on the same 70-point χ^2 and differ only in what they vary; the last drops that choice instead and keeps all 77, which is where the extra 3 % comes from. Nothing in the table mixes two changes at once.

Three of those rows answer objections rather than tune. The discrete row profiles over [39]'s two scale columns *without* interpolating: their difference is larger than the usual six-point scale variation, so the interpolants between them are not themselves computed predictions, and a limit leaning on one would have bought freedom rather than measured it. It gives 11.19 against 11.20, so the continuous nuisance buys nothing and is validated empirically. The two parton-density rows answer the objection that CT10nlo is not [39]'s baseline and that at $M_{jj} = 5$–8 TeV we sit at high x, where the $uu\!:\!ud\!:\!dd\!:\!qg\!:\!gg$ mixture is set-dependent and the deformation touches only quark channels: on the 70 points of the table, NNPDF3.1 gives 11.62 TeV and CT14nlo 11.46 against CT10nlo's 11.20 — **a spread of 3.7 %, with CT10nlo the weakest of the three**. **And the same comparison repeated in the configuration that actually produces** (56) — all 77 points — gives 11.27, 11.11 and 10.86: the same 3.7 % spread, the same ordering, CT10nlo still weakest. So staying on CT10nlo is the conservative choice in both, and the systematic does not depend on which of the two χ^2 the comparison is made in. *Two earlier versions of this sentence got that number wrong in opposite directions, both by comparing across configurations: first* 3.8 % *from NNPDF3.1 at* 70 *points against the quoted* 10.86 *at* 77*, then "*1.4 %*" from a mixture of both. Quoted consistently it is* 3.7 % *either way.* CT10nlo also stays in the CIJET control, because it is the set their grid was generated with and a control run with another set is not a control. The projected row is explained below. Every criterion lands above the quoted 10.86 TeV, which is therefore the weakest of them and the one it is safe to carry.

And that 3.7 % is the spread *among* fits, not the uncertainty of what they assume. All three sets are fitted to overlapping data under the same hypothesis — that the proton has no substructure at these scales — and this recast's whole sensitivity sits at high x and high M_{jj}, which is where that hypothesis is least constrained and where a mismodelled parton density and a four-quark contact operator produce the *same* signature. That degeneracy is why compositeness searches are hard, and it is not covered by varying the set. It does not move (56) — nothing here can say by how much it would — but a reader should know it is a floor under the systematic and not something the three rows above have measured.

Three facts about $C = \sigma\sigma^{\mathsf{T}} \circ R$ are measured rather than assumed, and one of them corrects an earlier claim of ours. C has *full rank* 77, condition number 1.2×10^4: no pseudoinverse is used or needed anywhere. The data do obey the unit-area constraint block by block to 2×10^{-5} — *but C does not know it*: on the constraint direction w (the bin widths inside a block) $w^{\mathsf{T}} C w$ runs from 0.13 to 0.83 of that block's own scale rather than vanishing, so the published matrix is not the covariance of the normalised distribution alone. An earlier version of this work dropped the last bin of each block and justified it as the remedy for a singular matrix; that justification was wrong, and the drop is a *choice*. So it is made both ways: on 70 points the fit gives 11.20 TeV and on all 77 it gives 10.86 TeV, a spread of 3.2 %, and what is quoted is the weaker. On [39]'s own contact-interaction mass range those two give 13.95 and 14.10 TeV, a spread of 1.1 % — there the weakest reading is neither of them but the projected covariance below, at 13.75 TeV, which is why that is the number quoted for that selection.

And that full rank has a documentary explanation which is also the sharpest objection to this fit, so it is stated in the record's own words. The HEPDATA record describes the matrix as "the correlation matrix of the maximum likelihood estimators of the signal strength modifiers", obtained "after the fit to the data" — *not* as the covariance of the normalised unfolded points. If the unit-area normalisation is applied *after* that fit, then $d_i = x_i / \sum_j w_j x_j$ mixes bins and the covariance of d is JCJ^{T} with $J_{ij} = \delta_{ij} - d_i w_j$, which does have the constraint direction as an exact null mode — so for that row, and only that row, one bin per normalised block is removed and the χ^2 is evaluated on the resulting 70-dimensional non-singular representation. There too no pseudoinverse is used: on the raw matrix because none is needed, here because the singular direction is projected out by construction rather than inverted through. A reader is entitled to ask why $\sigma_i \sigma_j R_{ij}$ may be used when the two objects may live in different parametrisations, and we cannot settle it from the record: it is not documented either way. What can be done is to compute the limit *under the unfavourable reading* and see whether the question has any numerical force. It does not: through JCJ^{T} the limit is 11.29 TeV against the 11.20 of the raw matrix, eight parts in a thousand, and in the direction that strengthens it. **The objection is real and its consequence is below a per cent**, which is a better answer than an argument would have been. The result is

$$\frac{1}{R_5} \; > \; 10.86 \text{ TeV}\,, \tag{56}$$

and 13.75 TeV on the mass range [39] themselves restrict their contact-interaction limit to, $M_{jj} > 3.6$ TeV. That the number *rises* when the 3.0–3.6 TeV bin is dropped is worth saying: that bin carries their largest local excess, 2.0σ, in the same direction our signal would push, so the limit is not leaning on the one place the data are least quiet. Figure 13 draws the profile those criteria summarise, and adds the one thing a table cannot: where the model's own allowed scales fall on it.

And that last panel is the question this paper can ask and a continuous lower limit cannot. A limit says "above 10.86 TeV". But $1/R_5$ is not continuous here: $8D$ is an odd integer on the seed [2] print, so the per-rung ceilings form a *comb* of their own, and (25) gives its teeth — 10.03 TeV at $8D = 1$, then 6.27, 5.14, 4.56, falling to the Lambert asymptote. **And it is not the arithmetic comb of §9**: that one lives *inside* a rung, with an exact step in M^2, and is the set of masses that rung reaches; this one has a single tooth per rung, and the tooth is the most that rung reaches rather than all of it. A content may sit anywhere at or below its own. Reading the recast at each tooth instead of at an arbitrary mass gives

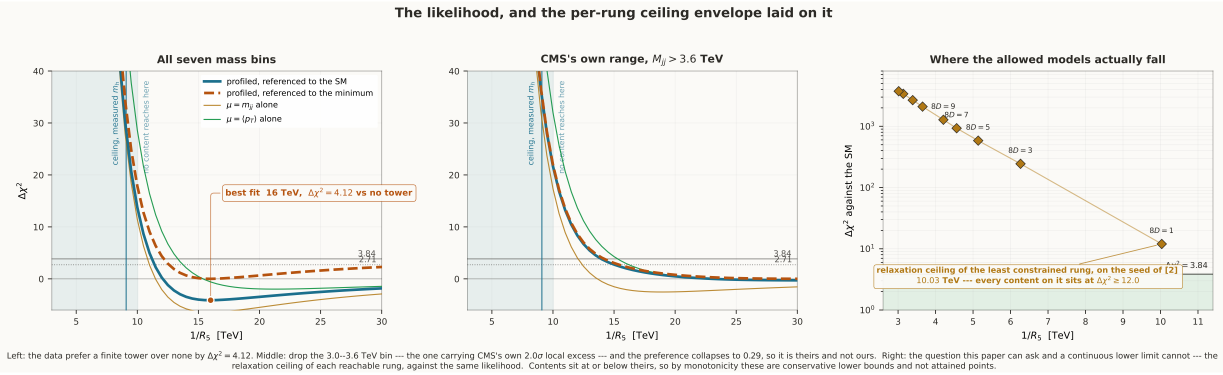


Figure 13: The likelihood, and the per-rung ceiling comb laid on it. The recast χ^2 of (55) against $1/R_5$, with both references drawn — solid against the Standard Model, which is what (56) uses, and dashed against the *minimum*, which is what Wilks' 95 % point actually means. They differ because the Standard Model is not the best fit. *Left, all seven mass bins:* the curve bottoms out at a **finite** $1/R_5 = 16$ TeV, $\Delta\chi^2 = 4.12$ below no tower at all — a nominal $2.0\,\sigma$ for one degree of freedom. The two thin curves are [39]'s two scale choices taken separately, not interpolated: they bracket the profiled result and show the same shape, so the minimum is not an artefact of the nuisance. *Middle, the same on $M_{jj} > 3.6$ TeV*, which drops the 3.0–3.6 TeV bin: the minimum flattens and the preference collapses to 0.29. **So the preference is [39]'s own $2.0\,\sigma$ local excess in that one bin, seen through our template, and not information this recast adds** — which is why no significance is claimed from it. *Right, the question a continuous lower limit cannot ask.* The diamonds are the per-rung ceilings of §8, (25) — the *largest* $1/R_5$ the integer programme allows at each quantised $8D$ — read against the same likelihood, with $\Delta\chi^2$ on a logarithmic axis because *it* is what spans decades: the teeth themselves run only from 10.03 down to 3.02 TeV, while their $\Delta\chi^2$ runs from 12.0 to some thousands. **They are relaxation ceilings and not attained contents**: the $8D = 1$ diamond sits at 10.03 TeV while the lattice reaches only 10.01 —both on the seed [2] print— and a content sits at or below its rung's ceiling on the discrete set the arithmetic comb allows. **That does not weaken the reading; it is what makes it a bound.** Since $\Delta\chi^2$ decreases monotonically with $1/R_5$ across this range — measured, and true up to 15.5 TeV, well past the highest tooth — a content at or below its ceiling sits at or *above* that ceiling's $\Delta\chi^2$. So each diamond is a conservative lower bound on the $\Delta\chi^2$ of every content on its rung, and the minimum over the diamonds bounds the minimum over all contents from below without needing any diamond to be attainable at all. The band below $\Delta\chi^2 = 3.84$ is **empty**: the comb's highest tooth is the $8D = 1$ ceiling at 10.03 TeV on the seed [2] print, and even that sits at 12.0. **That panel is read on the 77-point χ^2**, the one (56) is quoted from, and not on the 70-point curve the other two panels draw — which is why its $8D = 1$ point does not lie on the solid line to its left. On that curve the same tooth would read 13.4; the weaker of the two is the one quoted. Drawn by `make_fig_chi2.py` from `scan_mkk.py`'s and `ceiling_ilp.py`'s archived runs.

$$\min_{k\ \text{odd}}\ \Delta\chi^2\big(M_{\max}(k)\big)\ =\ 12.0 \qquad \text{at } 8D = 1,\ 1/R_5 = 10.03\ \text{TeV}, \tag{57}$$

against a threshold of 3.84; the next tooth is at $\Delta\chi^2 = 244$ and the escape branch's best, 3.97 TeV, beyond a thousand — all three are $\Delta\chi^2$, not masses. **Two things about that number are choices and are made the unfavourable way.** It is read on the 77-point χ^2 — the one (56) is quoted from because it is the weaker — and not on the 70-point one, which would give 13.4 instead; and the teeth are the *truncated* per-rung ceilings, 10.034 TeV at $8D = 1$, rather than the untruncated 10.037, which sits three GeV higher and so a shade lower in $\Delta\chi^2$. Neither is worth a decimal on its own; both are stated because a reader cannot otherwise tell which of several available numbers was picked.

And now the question that matters more than either: why does *one* tooth carry all of this? Because the per-rung ceiling falls with k — A_4 cannot keep pace, its cap per rung running $215, 112, 87, 76, \dots$ per unit of $8D$ — so the least-constrained tooth is always the *smallest* rung, and the smallest rung is $8D = 1$ only because Theorem 1 says $8D$ is an odd *integer* wherever its

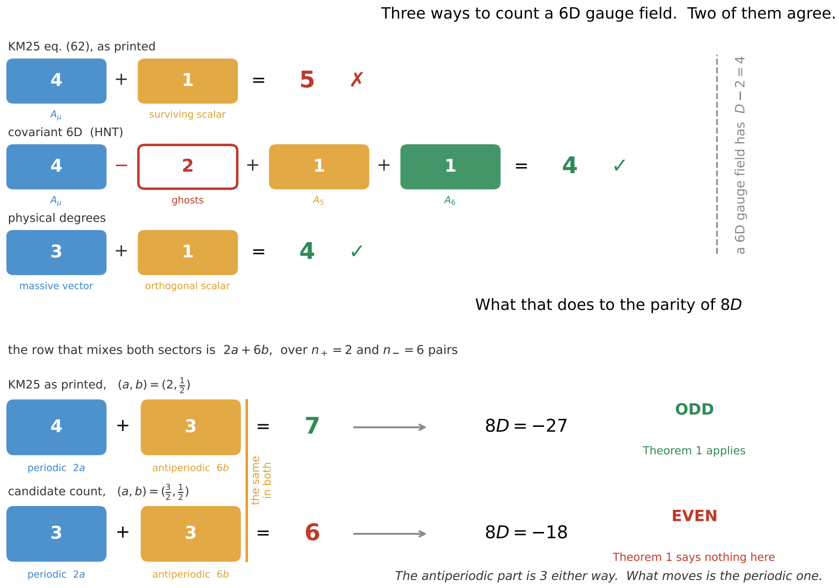


Figure 14: Where the parity comes from. *Top:* three ways to count the degrees of freedom of a six-dimensional gauge field. [2] take 4 for A_μ and 1 for the surviving scalar; the covariant six-dimensional count of [48] subtracts the two ghosts and keeps both extra-dimensional components, consistent with [17]'s general $-(D-2)$ rule; the physical count uses the three polarisations of a massive vector. Two of the three give $D-2=4$, and the one that does not is the one this paper inherits. *Bottom:* what that does to the row mixing the two P_6 sectors, whose coefficient is $2a+6b$ over $n_+=2$ and $n_-=6$ pairs. The antiperiodic part is $6\times\frac{1}{2}=3$ either way — an *integer*, and it does not move. What moves is the periodic part, from 4 to 3, and with it the parity: 7 against 6, hence $8D=-27$ against -18. So the parity change cannot be put down to the antiperiodic half-integer alone — the number §13 used to name as the open one: what changes is the parity character of the complete mixed coefficient $2a+6b$. The figure is drawn through the same five-coordinate map as the text, and refuses to draw at all unless it returns the published -27.

hypothesis is met, which on the seed [2] print it is. **So the margin behind** (57) **is not** 12.0 **against** 3.84**. It is the integrality of** $8D$, and that can be measured rather than admired: at fixed A_4 the ceiling scales as $k^{-1/2}$, so a quantum half as large would put the top tooth at 14.19 TeV, where this recast returns $\Delta\chi^2=-6.0$ — *below* the Standard Model, inside the region the data mildly prefer. Halve the quantum and the conclusion does not weaken; it changes sign. That is worth following one step further, because it ends somewhere specific and the ending is not a weakness. On the seed [2] print, $8D$ is odd because matter contributes an even amount and the gauge sector an odd one. It is tempting to put that oddness down to a *single half-integer* — the weight $\frac{1}{2}$ of each $P_6=-1$ adjoint pair, against 2 for each $P_6=+1$ pair — but that is not what carries it, and the distinction turns out to decide the whole chain. Over the $n_+=2$ and $n_-=6$ pairs of the row that mixes them, the coefficient is $2n_+ + \frac{1}{2}n_- = 4+3=7$: the antiperiodic term contributes 3, an *integer*, and the oddness of the sum needs the periodic term to be even at the same time. **Two conditions, not one, and neither weight decides alone** (Figure 14). The antiperiodic $\frac{1}{2}$ is not read off [2]'s eq. (68) — it is *inferred*, and by a fit that had a way to fail. Part VI [33] builds all 48 adjoint generators, reads (q, P_5P_5', P_6) off each, and fits *two* weights to *three* coefficients: the *two* rows with no $P_6=-1$ pairs fix the periodic weight on their own, and they agree; the remaining row then forces the antiperiodic one to be exactly $\frac{1}{2}$, reproducing their 7 as $2\times2+6\times\frac{1}{2}$, the split their text states and never derives. **But the redundancy is not where it matters.** Three coefficients against two unknowns is overdetermined as a *system*, and it is tempting to report it as a test that could have failed. It could — but only of the periodic weight, which is the one both spare rows constrain. The antiperiodic weight is fixed by one equation and checked by none. What agrees is not what decides.

And then the half does not survive, which is the part worth understanding. The gauge weight — written ω_g, because W is already the stability functional of §8 and the Lambert branch,

and three objects to a letter is one too many — is $\omega_g = 2n_+ + \frac{1}{2}n_-$ over the $P_6 = \pm 1$ adjoint pairs, and a census of all 4096 admissible diagonal assignments returns, with no counterexample, $n_- = 2m$ — where m counts the indices doubly flipped relative to the Wilson-line pair. So $\omega_g = 2n_+ + m$ *is an integer*, and it is odd exactly when m is. In [2]'s assignment $m = 3$, and the three indices that make it up are the colour indices of their eqs. (78)–(79). *So what makes* $8D$ *odd is not a half at all: it is the parity of a count of colours.* The $\frac{1}{2}$ is the bookkeeping that counts each pair once, and because pairs come in twos it always reconstitutes an integer. The chain is then: $m = 3$ is odd, so ω_g is odd, so $8D$ is odd, so $|D| \geq \frac{1}{8}$, so $1/R_5$ is capped, so the class is finitely testable — and the ceiling this paper computes is that fact read at the other end.
Every link of that chain is sound; its first link is inherited. It begins with the weights $(2, \frac{1}{2})$ read off [2], and §13 shows what the chain does if they are $(\frac{3}{2}, \frac{1}{2})$ instead: it does not break, it moves down one rung.
One link in that chain is inferred and not derived, and the difference is one of direction. What Part VI establishes independently is the *table*: over the 48 adjoint generators, (q, P_5P_5', P_6) is group theory and cannot move, which rules out any error in [2]'s multiplicities. What it does *not* establish is the weights — and it is worth being exact about which of the two, because "three coefficients, two unknowns, overdetermined" does not mean what it sounds like here. **Two** of the three rows contain no $P_6 = -1$ pair at all, so each fixes the first weight on its own, and their agreement *is* the redundancy. **One** row contains the $P_6 = -1$ pairs, so the second weight is fixed by a single equation with nothing to check it against. *So the redundancy tests the periodic weight and only that one.*
And it is worth saying which of the two decides, because an earlier version of this paragraph had it backwards. The row that mixes them is $2n_+ + \frac{1}{2}n_-$, and for that to come out odd **two** things must hold at once: the periodic part even *and* the antiperiodic part odd. The antiperiodic part is $6 \times \frac{1}{2} = 3$, odd either way, so it is safe. The periodic part is the one a ghost subtraction would move, from 4 to 3. *So the redundancy tested precisely the weight that decides, and tested it in the one direction that was never in doubt.* And in either case the inference runs from their numbers back to the weights, not from the geometry forward to their numbers. The causal order is: six-dimensional field content $\to$ mode decomposition $\to$ weights $\to$ coefficients, and every route in this series enters that chain at the right-hand end. **So $(2, \frac{1}{2})$ is what the coefficients require, not what the compactification predicts**, and no computation anywhere in Parts I–VII derives it the other way. What would: a Coleman–Weinberg sum over the six-dimensional Kaluza–Klein tower with these orbifold parities, computing the weight of a $P_6 = -1$ pair from its antiperiodic moding rather than reading it off. That is a standard calculation and it is not done here; §13 carries it. Everything downstream of the weights is proved.
An attempt to close it forward, and why it failed — which is worth recording because the failure is instructive. [17] state their counting rule in words: the potential is [fermionic d.o.f. − bosonic d.o.f.] times the potential of one degree of freedom, which for gauge plus ghosts is $-(D-2)$; and their eq. (3.10) gives the one-d.o.f. adjoint potential as weight $\frac{1}{2}$ per charge pair, with one pair at $c = 2$ and two at $c = 1$ by their eq. (3.8). At $D = 5$ that returns their own eq. (3.20), $(\frac{3}{2}, 3)$. Carried to $D = 6$ it returns $(2, 4)$, the periodic part of [2]'s eq. (68), and we briefly took that for an out-of-sample derivation. **It is not one.** In six dimensions $D - 2 = 4$ coincides numerically with the number of A_μ components, which is four in *every* dimension; the agreement therefore does not distinguish the two bookkeepings, and in five dimensions — where they differ, 3 against 4 — it is $D - 2$ that [17] use. Worse, $D - 2$ is the total over *both* parity sectors, and assigning it whole to the periodic one is precisely the step that a second parity makes illegal. So the periodic weight remains inferred, and the caveat above stands unweakened.
And the same arithmetic, read the other way, is a live question about [2]'s bookkeeping that this paper cannot settle and should not hide. Their eq. (62) assigns $N = 4$ to A_μ and, after removing the Goldstone combination, $N = 1$ to the surviving $A_{5,6}$ scalar. With our own generator table — one pair at $c = 2$, two at $(c{=}1, s{=}{+}1)$, and two $P_6{=}{+}1$ plus six $P_6{=}{-}1$ at $(c{=}1, s{=}{-}1)$ — those two numbers reproduce eq. (68) exactly: 2, 4, and $(2 \times 4 + 6 \times 1)/2 = 7$.

But $4+1=5$, while a six-dimensional gauge field has $D-2=4$ physical polarisations. [48] do the covariant count on $T^2/\mathbb{Z}_2$ and state it plainly — "the mass matrix for ghost fields is the same as that for A_μ", so A_μ and ghosts give $4-2=2$, and the two extra-dimensional components give 2 more, totalling four. Their orbifold is not this one — [2] use the product $S^1/\mathbb{Z}_2 \times S^1/\mathbb{Z}_2$, with two independent reflections — so what they fix is the total, not how a second parity divides it. Carried over to the product orbifold at $R_5 \gg R_6$, a parity-resolved covariant count *suggests* that the four split 3 on the $P_6{=}{+}1$ side and 1 on the other, rather than 4 and 1. Were that right, substituting $4 \to 3$ would turn eq. (68) into $(\frac{3}{2}, 3, 6)$, the gauge sector's $8D$ would be -18 rather than -27, *even*, and **the hypothesis of Theorem 1 would not be satisfied** — which is a different statement from the theorem failing, and the difference is the whole point. **We do not assert that [2] are wrong**: no Faddeev–Popov determinant is displayed in their one-loop bookkeeping, but a gauge-fixing term is, and a cancellation could be implicit in how their spectrum is built. What settles it is a computation this paper does only in part (§13) — the background-gauge determinant for a *single* charged adjoint orbit, $\frac{1}{2}\log\det\Delta_{A_\mu} + \frac{1}{2}\log\det\Delta_{A_5} + \frac{1}{2}\log\det\Delta_{A_6} - \log\det\Delta_{\mathrm{FP}}$ at $R_5/R_6 \to \infty$, which needs neither $SU(7)$ nor forty-eight components. *Until that is done, every consequence here that needs the published gauge base point inherits this question — which is not every theory-side scale, since the asymptote (27) carries no gauge quantity at all — and it is a sharper question than the anchor residual it sits beside.*

And the exposure is narrower than "one inferred number" makes it sound, which is worth saying because it cuts the other way. The natural worry — what if the weight is not exactly $\frac{1}{2}$? — is the wrong question. Since $n_- = 2m$ always, the gauge weight is $\omega_g = 2n_+ + 2b\,m$. **An earlier version of this paragraph read that as a robustness result about b, and had the emphasis exactly backwards.** On the row that carries the physics, $n_+ = 2$ and $n_- = 6$, the whole question is this:

$$\boxed{\begin{aligned} \omega_g \;&=\; 2a + 6b \;=\; (2a) + 3\,(2b)\,, \\ \omega_g \text{ odd} \;&\Longleftrightarrow\; (2a) + (2b) \text{ odd} \qquad (\text{for } 2a, 2b \in \mathbb{Z})\,. \end{aligned}} \tag{58}$$

Both weights enter, symmetrically, and neither decides alone:

$$(a,b) = \left(2, \tfrac{1}{2}\right):\ 4+3=7 \text{ odd}; \qquad (a,b) = \left(\tfrac{3}{2}, \tfrac{1}{2}\right):\ 3+3=6 \text{ even}.$$

The antiperiodic weight is the *same* in both, and the parity still flips. So the chain does not hang on b, and it was never b that was in danger: what §13 puts in question is a, and (58) says why that is enough on its own. Measured in `gauge_weight_origin.py` and `gate_km_ghosts.py`. What §13 records is a different and narrower gap: no *published* computation tests that weight. Every anchor a curvature can be checked against sits at $B=0$, where the parity-weighted and parity-blind moments coincide; the parity-split results that *do* carry an antiperiodic sector — [8]'s among them — live at the value rung $p=5$ and never reach the curvature. Our own chain has two legs; the outside world has none. A reader deciding how much weight (57) can bear should know that this is where it lives — not in the statistics. **So it is not that the allowed region is partly disfavoured: no bulk content the arithmetic of the seed [2] print permits reaches any $1/R_5$ at which this recast's $\Delta\chi^2$ falls below 12.0.** The caveats of the previous paragraphs all still apply to that number, and none of the *quantified* collider-side variations above comes near the factor of three and a half in $\Delta\chi^2$ that erasing the margin would take — the unmeasured NLO deformation and the theory-side anchor residual sit outside that statement, and are not claimed to be small. And it is a nominal figure from a Gaussian recast, not a CL_s, and it is stated as one. What makes it worth stating is that it is the two halves of this paper meeting: the integer programme says which scales exist, and the likelihood says what the data think of them.

And the channel that was left out, put back. (56) dresses t and u only, and the reason given for that is good — there the resummed propagator is real, so neither the width nor $\mathrm{BR}(G_n \to jj)$ enters, and the limit does not depend on exotic masses the model never fixes. **But that argument does not show the omission is conservative**, and it is not: amplitudes interfere, and in $q\bar{q} \to q\bar{q}$ the timelike term can add to the t/u deformation or subtract from it. So it is computed. Only two matrix elements carry a timelike gluon — $q\bar{q} \to q\bar{q}$, through its s piece and the s–t interference, and $q\bar{q} \to q'\bar{q}'$, which is pure s; the gluon final states stay undressed by the same five-momentum argument of [35] used above. The pole is regulated by the *total* width $\Gamma_{\text{tot}} = \Gamma_{q\bar{q}} + \Gamma_{\text{exotic}}$ with $\Gamma_{q\bar{q}}/M = 2\alpha_s$ and $\Gamma_{\text{exotic}} \geq 0$ unknown, so Γ_{exotic} is profiled rather than chosen.
Two things come out, and the first makes the second easy. *The shift shrinks as the width grows* — at $1/R_5 = 9.09$ TeV the ratio to QCD moves by $-0.47\,\%$ at $\Gamma_{\text{exotic}} = 0$ and by only $-0.18\,\%$ at $\Gamma_{\text{exotic}}/M = 1$ — so the profile over $\Gamma_{\text{exotic}} \geq 0$ is bounded at the *boundary*, and the worst case is a width we know rather than one we have to scan to infinity. And *the sign is negative*, as the objection suspected: dropping the s channel was not conservative. What it is worth is small. That $-0.47\,\%$ on the ratio is $-6\,\%$ on the *deformation*, ratio minus one, which is what the fit sees; recomputing the limit rather than rescaling it — the deformation does not scale exactly, the bins are correlated and the scale is profiled — gives

$$\frac{1}{R_5}\bigg|_{s+t+u,\ \Gamma_{\text{exotic}}=0} = 12.86\ \text{TeV} \qquad \text{against} \qquad \frac{1}{R_5}\bigg|_{t/u} = 12.99\ \text{TeV}\,, \tag{59}$$

both at leading order on the same grid, so that the $-1.03\,\%$ between them is the s channel and nothing else. That suggests an $\mathcal{O}(0.1\ \text{TeV})$ effect on (56), and both readings stay above 10.03 either way; the exact shift needs the final 77-bin pipeline rerun with $s + t + u$, and carrying a leading-order per cent across to it is the rescaling this paper declines to do everywhere else. **So the omission is real, its direction is the unfavourable one, and its size is one per cent** — which is worth saying in that order. Measured in `s_channel_width.py`, whose first control rebuilds the whole calculation from scratch and, with the s channel switched off, reproduces the archived t/u deformation to 6×10^{-16}; without that the difference could not be attributed to the channel rather than to the rebuild.

And what that number is not. It is **not** a CMS-equivalent 95 % CL exclusion, and the difference is not pedantry. [39] set limits with detector-level Poisson likelihoods, profiled nuisance parameters and an asymptotic CL_s construction; this is a Gaussian χ^2 over their published particle-level covariance with one nuisance. The two closure benchmarks measure that gap — and they measure it *differently*, 1.03 on the destructive sign against 0.80 on the constructive one, because those two are not two draws from one uncertainty. **Running the same closure on the *expected* limits says which of the two explanations that is.** An Asimov dataset — [39]'s own Standard-Model prediction put in place of their data, at its best-fit scale, with the covariance untouched — returns 20.0 and 29.0 TeV against their published expected 18 and 34, that is 1.08 and 0.85. Those are the observed ratios again to within five points. *So the gap is a property of the method and not of a fluctuation in this particular dataset*, which is what an observed-only closure could never have separated. It places the *benchmark-dependent closure discrepancy* on the matched contact-interaction selection at the ten to twenty per cent level in Λ, and larger over all seven bins — which is not a bias estimated over an ensemble, because no ensemble was built; it is the disagreement measured on two templates, its size depends on which selection it is measured on, and it is not a band that transfers to (56). Changing the sign of the interference changes the shape, can cancel against the $1/\Lambda^4$ term and moves which bins dominate; read as coefficients of $1/\Lambda^2$ the same two ratios become 0.94 and 1.56, which is template dependence and not a calibration. **So they are diagnostics, and this paper does not turn them into a band on (56).** What may be said is that the recast finds enough tension that closing the likelihood properly is the priority, and that 3.97 TeV sits a factor of three below where the recast loses sensitivity — with no width and no $\mathrm{BR}(G_n \to jj)$ anywhere in it, which is the whole reason for using the spacelike channels alone. What is *not* settled is the ceiling: 10.86 and 13.75 both clear 9.09 and 10.03 TeV, but a nominal recast sensitivity is not the instrument that may retire a model, and §13 says what would be. Measured in `hadronic_chi.py`, `cijet_control.py`, `fetch_hepdata.py`, `cms_ci_closure.py` and `scan_mkk.py`, with parton densities from [47] through [46] and the contact-operator coefficients from [52].

So the two calculations already cross *numerically*: the algebraic ceiling of §8 approaches $1/R_5$ from *above*, at 9.09 TeV, and the nominal collider sensitivity from *below*. Promoting the second to an experimental lower bound requires the detector-level likelihood, and §13 says so; the crossing is what makes that worth doing rather than a result of having done it. **And read at the comb rather than at a continuous scale, the crossing is not marginal**: (57) puts every reachable $1/R_5$ at $\Delta\chi^2 \geq 12.0$ — on the seed [2] print, whose comb this is. The candidate seed's highest tooth is *lower*, 7.38 TeV, and $\Delta\chi^2$ falls monotonically with $1/R_5$ across this range (Figure 13), so that branch is more disfavoured and not less — though it has not been read tooth by tooth. What is already true without any of it is the asymmetry: every published constraint on a compactification scale bounds from below, and a lower bound alone can never exclude a class — the scale can always be raised. It is the ceiling that makes the question finite, and the quantisation that makes it sharp.

The dictionary is derived, the one free parameter is named, the measurement has been identified and then made, and the comb has been read against it — which leaves only the arithmetic instrument itself to be checked against someone else's table.

11 A second anchor

Part VI has one anchor and it fails. With one, a failure localises nothing: it could be the potential, the expansion, the minimiser, or the model-specific dictionary.

Von Gersdorff, Irges and Quirós [7] supply a second, independent in group, dimension, decade and authorship. Their model is five-dimensional on $S^1/\mathbb{Z}_2$ with a *single* parity, so every tower is integer-moded, there is no antiperiodic sector, $B_2 = 0$, and our $D = A_2 - \frac{3}{4}B_2$ collapses to $D = A_2$: pure Dynkin index. **Their setting is the $B = 0$ corner of** (18), and that is what is tested, not assumed. It is worth being exact about which corner, because $B = 0$ is not $\Delta = 0$: on a single parity $\Delta_k = \mathcal{A}_k = S_k$, as far from zero as the moment goes. The two coincide nowhere except at the origin, and §13 turns on telling them apart. Their conventions are fixed by their own footnotes rather than guessed — their Kaluza-Klein gauge boson masses are "n^2, $(n+2\omega)^2$, $(n-2\omega)^2$", so a state of Wilson-line eigenvalue λ shifts by $q = 2\lambda\omega$; and their footnote identifying $\alpha = 2\omega$ makes their phase ours.

quantity	theirs	ours
adjoint charges of $SU(2)$	$n^2, (n \pm 2\omega)^2$	$c = -2, 0, 2$
$C_2(G)$, C_R	textbook	$2, 3$ and $1/2$, from generators *and* from Sage
their criterion, eq. (5.3)	$C_G/C_R < 4/3N_f$	*is* $D > 0$
critical N_f, $SU(2)$ fundamental	3	3
critical N_f, $SU(3)$ fundamental	9/2	9/2
critical N_f, adjoint	3/4	3/4
minimum, $SU(2)$ adjoint	1/4	1/4, exactly
bracket of their eq. (5.2)	$3C_2(G) - 4C_R N_f$	exact, the 3 : 4 included
minimum, $SU(3)$ adjoint	0.29, 0.71	1/3 *exactly*; open, below

The three critical flavour numbers come out twice: from our D, and from a numerical second derivative of *their* potential, agreeing to 10^{-6}. What is left over in the prefactor is $2/\pi^2$ to eight digits — a pure number with no group theory and no zeta in it, being their g^2 and their A_5-to-h convention, with the π's a Kaluza-Klein reduction puts there.

One disagreement, and the control that decides it. For adjoint fermions they publish two degenerate minima at $\omega \simeq 0.29$ and 0.71 in $SU(3)$. We get $\omega = 1/3$ *exactly*, and for a reason: stationarity is $\mathrm{Sl}_4(\theta) + \mathrm{Sl}_4(2\theta) = 0$, and at $\theta = 2\pi/3$ the Clausen function is odd about 2π, so $\mathrm{Sl}_4(4\pi/3) = -\mathrm{Sl}_4(2\pi/3)$ and the sum vanishes identically. With adjoint matter the potential sees only the centre, so the minimum must be the centre-symmetric holonomy — the textbook statement of [6]. The Polyakov loop in the fundamental vanishes at our 1/3 to 3×10^{-31} and equals 0.168 at 0.29. The evidence is on our side; we cannot prove their two-digit number is a slip, so it is recorded as open.

> Either way the anchor does its job. **The potential, the expansion, D and the minimiser are validated, in the periodic $B = 0$ sector, against an independent published one-loop computation**, so Part VI's 1.94/1.20 residual against [2] cannot be blamed on their implementation there. What is left is the model-specific dictionary — their charge and parity assignment, or the α-to-$1/R_5$ relation — *or* the antiperiodic sector, which this anchor does not reach because their model has none. Either way, a far smaller place to look.

Scope: this anchors the $B = 0$ corner. The antiperiodic sector, which is what makes this paper what it is, is not anchored by it, because their model does not have one.

What is claimed, what is not, and what was withdrawn along the way.

12 Novelty, scope, and what this does not claim

Contribution, stated plainly. On the arithmetic side: the closed form (11) and its Lambert-W solution (12) with the two branches and the fold, the cancellation (15), the two logarithm-free identities (22), Theorems 1 and 2 — the second on *either* gauge seed, the first with its hypothesis carried in its own statement, the geometric mode law (8) and its closed form (18)–(19), the completeness of the five coordinates (28), the ceiling (24) with its certificate and its untruncated companion (39), the escape bound (45), and the anchor of §11.
On the collider side, where less is ours and the line matters more: the *reading* of [2]'s three parity matrices that puts every colour-changing gauge boson at $1/R_6$ with a wavefunction vanishing on the quark brane, and the observation that this answers for the gauge channel the proton-decay question they leave open; the recast (55)–(56) of [39]'s unfolded distribution onto the tower's colour-octet operator, with its closure tests; the coefficient (53) of the second circle's logarithmic sensitivity; and (57), reading that likelihood at the teeth of the per-rung ceiling comb rather than at an arbitrary mass, which is the one place the two halves of this paper actually meet. **What is *not* ours there**: the resummation (52) is Euler's identity and the all-channel propagator is [35]'s; the coloron mass, coupling and width formulas are [36]'s; the measurement, its unfolding, its covariance and its Standard-Model baseline are [39]'s; the contact-operator coefficients are [52]'s; and the parton densities are [47, 50, 51]'s through [46].

Relation to [8], and two claims withdrawn. The parity-split architecture — the sum split by boundary condition, weighted by $\eta(p)/\zeta(p)$, with the sign deciding symmetry breaking — **is published**, at $p = 5$ and at the level of the potential's value. We do not claim it. Neither do we claim the **orbifold-stability criterion** of §8: that one evaluates the potential at the two symmetric points and takes the smaller, that $F_-(0) = -\frac{15}{16}F_+(0)$ and $F_-(1/2) = F_+(0)$, and that $(\pm,\pm)$ components stabilise while $(\pm,\mp)$ destabilise, are their §2.2, their eqs. (2.12)–(2.13) and their own conclusion. (34) is that summed over a bulk content and rewritten in the notation of (2); the content of the rewriting is that the sum is *linear* in the multiplicities, which is what lets it enter an integer program — and [8] compute no compactification scale, so no such program appears there. What is ours is the same parity structure at $p = 3$, the curvature, which their method does not compute and which is the rung the interior small-α vacuum needs; its packaging as two group-theory traces rather than component counts; and every arithmetic consequence in §7–§8, the two ceilings included.

Relation to the brane-localised kinetic term literature. The bulk half of the correspondence is also published: [18] states that the same Dynkin-index factor that runs the gauge coupling feeds the Higgs mass term. Parity-inserted traces are standard in the anomaly literature [19, 20], which — measured over 94 pages — never mentions the effective potential. What we cannot find stated anywhere is the bridge: nobody writes the curvature as a combination of $T(R)$ and $\mathrm{Tr}_R[PQ^2]$, and the weight $\eta(3)/\zeta(3)$ does not appear in this context.

The null is explained rather than merely reported. On $S^1/\mathbb{Z}_2$ a hypermultiplet induces *no* renormalisation of the brane gauge couplings, and [21] shows this is not a supersymmetric cancellation — the bosonic and fermionic contributions vanish separately — so in the case everyone studies there was nothing to bridge; [7] find the same for the brane contributions to the Higgs mass. In six dimensions the brane couplings do renormalise on general $\mathbb{Z}_N$ orbifolds, but *not* at the $\mathbb{Z}_2$ fixed points of an even-ordered one [21], and a bulk *fermion* generates no localised gauge coupling at all: the parity-odd part of its self-energy carries an ϵ tensor and integrates to zero [22]. **We therefore do not identify $\mathrm{Tr}_R[PQ^2]$ with a brane-localised gauge coupling.** It sits at the T^2 slot of the family of parity-inserted traces — tadpole, Fayet-Iliopoulos, localised gauge kinetic term, localised anomaly — but for the bulk fermions of this model that slot is empty. Its role here is Part IV's: the index that can cancel, set against the graded dimension that cannot.

What the two computations do share, and it is the function itself. By the Hurwitz formula [23, 24], $\zeta(1-s,a) = \Gamma(s)(2\pi)^{-s}\big[e^{-i\pi s/2}\mathrm{Li}_s(e^{2\pi i a}) + \text{c.c.}\big]$, the polylogarithm of a Wilson phase *is* a Hurwitz zeta with the phase in its second argument — the slot occupied there by the discrete momentum shift. Ghilencea, Lee and Schmidt-Hoberg [22] trace their bulk divergence to the simple pole of that function at $z = 1$, and our ladder reaches the same pole: $\zeta[z,1]$ is the Riemann zeta, and $p = 5 - 2k$ ends at $p = 1$. Which half feeds it is then fixed by (18): at $p = 1$ the antiperiodic coefficient $2 - 2^p$ vanishes, so the pole is fed by the periodic moment A_4 alone. That the surviving moment is the momentum-conserving one is suggestive of what they call bulk; identifying it with the bulk-localised divergent structure, and the complementary sector with boundary terms, needs the explicit counterterm calculation §13 asks for, and is not claimed here. Their selection rule is ours in their variables: the two Hurwitz terms enter as $\zeta(z, 1+c_1) + (-1)^v\zeta(z, 1-c_1)$ and survive only for even v, which is the reflection $n \to -n$ that makes charges enter our tables in $\pm Q$ pairs. What differs is the Green function, and therefore the consequence: theirs is an ultraviolet divergence of the gauge boson self-energy and needs a counterterm, while ours is a branch point of a *finite* potential — and the $H_4 = \frac{25}{12}$ that resolves it is the one already carried by G in (3). Same function, same pole, same parity selection; in our ladder only the *periodic* half feeds it. Different correlator, different cure — and whether that periodic half is the same bulk object as theirs is exactly the counterterm question above, which is why the sentence stops here.

Scope of the results. One loop. Flat space, not warped. A free Kaluza-Klein tower: no bulk masses and no brane-localised masses, either of which destroys the branch point. A single Wilson phase — with several, stationarity is a system, and that is the two-phase case of Part V; for *this* model it is not a restriction at all, and the next paragraph is why. Small α: the expansion converges for $|\alpha| < 1/c_{\max} = 1/3$ — see §5, this is a radius and not an asymptotic caveat — and its truncation costs 0.13 % near $\alpha \approx 0.02$ and a few per cent by $\alpha \approx 0.25$. And every theory-side scale carries the anchor residual of §2.

The moduli space is the Higgs doublet and nothing else, and that is measured. A reader counting zero modes will ask whether the symmetric point is a minimum along *every* flat direction or only along the one being minimised. The question is not ours either: [17] open with it — the tree-level potential cannot fix the vacuum because the Wilson-line phases are flat, so the one-loop potential is what decides, and *"we can know whether the color is conserved or not"* from it. In a grand gauge-Higgs model that is the sharp form: whether any flat direction is coloured, since a coloured Wilson-line modulus is a scalar leptoquark with a loop-suppressed mass, and if it lowered the potential the vacuum would break colour and every number here would sit at a point that is not the vacuum. The flat directions are the zero modes of A_5 and A_6, and `moduli_space.py` enumerates them from their eqs. (3)–(10): fourteen in A_μ, which is the unbroken $SU(3) \times SU(2)_L \times \mathrm{U}(1)^3$; four in A_5, one colourless $SU(2)_L$ doublet, which is the Higgs; and *none* in A_6. The null is not an accident of counting but of their sign convention: their eq. (4) gives A_6 the opposite sign to $A_{\mu,5}$ under the x_6 reflections, so an A_6 zero mode must be P_6-odd, while eqs. (7) and (9) ask it to be even under P_5 and P_5' — and the P_6-odd block is the off-diagonal 3×4 rectangle, on which those two conditions select disjoint columns. Four real moduli whose generic $SU(2)_L \times \mathrm{U}(1)_Y$ orbit is three-dimensional leave one phase, so minimising in α minimises over all of it. The same script reproduces their eq. (57) state by state, all 48 generators, which is what makes a null worth quoting.

One precondition is inherited rather than chosen, and it belongs on this list because it is not ours to relax. The potential of [2] is bulk fermions and gauge only: they drop the Standard-Model fermion contribution outright, on the stated ground that a fixed-point-localised quark gives $m_q(\alpha)^4 \ln m_q(\alpha)^2 \propto \alpha^4$ with none of the enhancement factor the massless bulk fermions carry — *"from this observation, we do not consider the potential from the SM fermion contributions in this paper"*. At $\alpha_{\min} \simeq 0.05$ that is $\alpha^4 \sim 6 \times 10^{-6}$ and the omission is safe at one loop and, at $\alpha^2 \sim 0.25\,\%$ of the curvature, still an order below the two-loop margin of Part VI [33]. It is recorded because every number here is computed on *their* potential, and a reader entitled to ask what is in it should not have to reconstruct the answer from their §5.

And g_4 need not be chosen at all. Every theory-side scale here passes through $K = (\sqrt{3}/2\pi^3) m_W g_4$ of §6, so $\mu \propto g_4^{-2}$ and the ceiling of §8 depends on it. The value used throughout is $g_4 = 0.63$, the Standard-Model $SU(2)_L$ coupling run up to the compactification scale, while the sibling anchor takes it at the weak scale: [3] write $g_4 = 2m_W/v \simeq 0.653$ under their eq. (4.4). Earlier versions of this paper left the two side by side as a declared choice. They are the same coupling at two scales — $g_2(M_Z) = 0.6517$ and $g_2 = 0.630$ at 5.6 TeV — and the choice can be removed, because the scale g_4 should be evaluated at is the answer itself. Imposing

$$g_4 \;=\; g_2(1/R_5)\,, \qquad 1/R_5 = \frac{2\pi m_W}{x}, \qquad x^2 = \frac{12\zeta(3)D}{A_4 + 6\mu}, \qquad \mu = \left(\frac{m_h}{K\pi^2}\right)^2 \tag{60}$$

closes the self-consistency loop — called that, and not a closure, because §10 already spends that word on the recast's benchmarks. The map contracts hard — its derivative at the fixed point is 6.5×10^{-3}, and every start between 0.30 and 1.30 converges to the same place in seven steps — and gives

$$g_4^\star = 0.6275, \tag{61}$$

0.39 % from the value used here, leaving the vertex at $A_4 = 104$ and $1/R_5$ at 9248 instead of 9218 GeV.

But that does not remove the choice — and it is worth saying both why not and how far the saying goes. (60) fixes g_4 *given* that it is evaluated at $1/R_5$, and a matching scale is a convention like any other. Moving it down by a factor of two, to $1/2R_5$, takes $g_4^\star$ from 0.6275 to 0.6310: a spread of 0.56 %, and it does not move the vertex. **And that is the whole of what a four-dimensional running is entitled to say here**, because $1/R_5$ is where the Kaluza-Klein tower starts — §13 uses the same fact in the other direction, that below $1/R_5$ the theory is the Standard Model. Above it, Standard-Model running alone no longer applies, because the Kaluza-Klein thresholds enter; what replaces it is a computation this paper does not do. Pushing the matching scale to $2\pi/R_5$, six levels up, would give 0.6186 and would carry the vertex to $A_4 = 107$; that number is archived as a marker of what is *not* settled and is deliberately not quoted here as a systematic. **So g_4 remains an input — the matching scale is a convention, this does not remove it, and the size of that freedom is not established here.** Establishing it needs the tower's own matching to the bulk $SU(7)$ coupling, which is the second condition the same script records and this paper does not attempt. The paper keeps 0.63, and the sentence it has always carried — both values defensible, the choice not free — stands as written. What the self-consistency loop does establish is smaller and still worth having: given a matching scale the value is a hard-contracting fixed point rather than a fit, and the two numbers in the literature are one coupling at two scales, $g_2(M_Z) = 0.6517$ and $g_2 = 0.630$ at 5.6 TeV. At the weak-scale value 0.653 the vertex moves the other way, to $A_4 = 98$ and 8903 GeV. Measured in `running_g4.py`, with the four-dimensional running imported from `running_gap.py` rather than retyped.

What we tried and withdrew. Three readings died in the making of this paper and are recorded because their deaths are informative. A quantised near-cancellation between the bulk and brane pieces was measured before being written and turned out to be a 10 % effect, not a 90 % one — what survives is that the five published rows sit in the fourth to ninth percentile of that ratio, so the cancellation is what phenomenological viability *selects*, not a mechanism. A reading in which the gauge table's bracket equalled the number of colours, which would have made Theorem 1 and the death of the obstruction at $k = 2$ two factors of one product, holds at their parity assignment and fails on 3872 of the 4096 admissible ones. The instinct there was right and the object was wrong: the two facts *are* one, and what joins them is the coefficient $2 - 2^p$ of (18) — even at $p = 5, 3$, which is the parity protection, and zero at $p = 1$, which removes the *antiperiodic* contribution — on the seed [2] print that is the death of the obstruction, while on the candidate seed it leaves an odd periodic sector carrying it. And an identification of a blind direction of the potential turned out to be supersymmetry, which the source discusses itself.

Which leaves the question a bound is always owed: what would have to be true for the absolute numbers to mean what they appear to mean — and what, in this paper, is still holding them up rather than being held up by them?

13 What this leaves

- **The anchor.** §11 narrows it to the model-specific dictionary but does not close it. The sharpest remaining test is a second published table for a model meeting the five preconditions; a screen of the literature found exactly one, the one studied here, and the reason is structural rather than accidental.
- **Work out how the four gauge degrees of freedom split between the two P_6 sectors.** This is the sharpest thing still open. It is also not the problem an earlier version of this list named, and the correction is worth stating plainly.

  The total is settled. Three published papers agree on it. [17] give the gauge-plus-ghost factor as $-(D-2)$, call the coefficients "numbers of d.o.f.", and say their method "is available for more than 5 dimensional gauge theories". [48] do the count in *six* dimensions, on a $T^2/\mathbb{Z}_2$ orbifold, and show the steps: the ghosts have the same mass matrix as A_μ, so "contributions to the effective potential from A_μ and ghosts are, therefore, $4 - 2 = 2$". The two extra-dimensional components add two more. Four in total. [3] take $N = 4$ for the whole gauge sector of a six-dimensional $T^2/\mathbb{Z}_2$ model. Four, three times over.

And the geometry is exactly why that is the total and not the split. [2] compactify on $S^1/\mathbb{Z}_2 \times S^1/\mathbb{Z}_2$ — each extra coordinate on its own circle, with *two* independent reflections, which is where P_5 and P_6 come from and why this paper has two parity sectors to divide four degrees of freedom between. [48] and [3] work on $T^2/\mathbb{Z}_2$, one $\mathbb{Z}_2$ acting on both coordinates at once. They therefore settle how many degrees of freedom there are, and cannot settle how a second, independent parity would share them out. That is not a gap in their papers. It is the reason the question this paper has to ask is still open.

A short history is worth the space here, because it explains how a settled count can go missing. The idea that the Higgs is an extra-dimensional component of a gauge field is from 1979 [4]; the mechanism that gives it a potential is from 1983 [5]. The one-loop bookkeeping arrived much later and, as it happens, twice in the same season: Haba and Yamashita posted the general five-dimensional formula in January 2004 [17], and Hosotani, Noda and Takenaga posted the six-dimensional count on $T^2/\mathbb{Z}_2$ two months after [48]. One states the coefficient as $-(D-2)$ and leaves it at that; the other writes the subtraction out in words. **The total degree-of-freedom count was settled more than two decades ago. The parity-resolved split this model needs was not**, and saying it that way makes our own contribution sharper rather than smaller.

Neither is in [2]'s reference list. Takenaga is there, for a paper from 2007; Hosotani is there, for 1983 and 1989. We say this as diagnosis and not as complaint — reference lists are finite, and theirs is long — but it is the mechanism, and it is worth seeing plainly. A result that a field settles in two papers in one spring can still fall out of the citation chain that a later model inherits, and once it is out of the chain it can be re-derived differently, in good faith, decades on. *We would not have looked either, had a number come out right.*

The split is what is open, and showing the steps is what makes that useful. §10 objects that $D - 2 = 4$ happens to equal the number of components of A_μ, so a bare total cannot tell the two bookkeepings apart. That is right — but [48] do not write 4; they write $(4-2)+2$, and that does tell them apart. What they carry is one parity, not two, so they fix the total and say nothing about how it divides, and the second parity is what this model turns on. Read as weights per charged pair, [2]'s coefficients are $(2, \frac{1}{2})$: four degrees of freedom in the periodic channel and one in the antiperiodic. That is five. A parity-resolved covariant bookkeeping *suggests* $(\frac{3}{2}, \frac{1}{2})$ instead: three and one, which is four. **We write "suggests" and mean it.** That candidate would correspond to the missing ghost subtraction residing in the periodic sector; what would establish it is the quadratic operator, the boundary conditions on A_μ, A_5, A_6 and the ghost pair, and the determinant they give — and that is the calculation this item exists to ask for. It cannot be both a result and a computation nobody has done. What we can say without it is narrower and still useful: no change of normalisation produces the difference, because a common factor would move all three coefficients together, and these move by $\frac{4}{3}$, $\frac{4}{3}$ and $\frac{7}{6}$.

And the weight that decides is the periodic one. An earlier version of this list said the opposite. The mixed row runs over $n_+ = 2$ periodic pairs and $n_- = 6$ antiperiodic ones, so its coefficient is $2n_+ + \frac{1}{2}n_-$. For that to be odd, two things must hold at once: the periodic part even, and the antiperiodic part odd. The antiperiodic part is $6 \times \frac{1}{2} = 3$, odd either way. The periodic part is the one the ghost subtraction moves, from 4 to 3. So the overdetermination of the fit tested the periodic weight — which is the weight that turned out to matter, and the one now in doubt (`gauge_weight_origin.py`, `gate_km_ghosts.py`).

Closing this takes a calculation, not another check: a background-gauge determinant for one charged orbit, with both parities kept, returning the split and not just the total. Until that exists, everything here that needs the gauge offset to be *odd* is conditional on [2]'s eq. (68) as published.

And part of that calculation is now done, which changes what is open without closing it. With the Wilson line flat, $F_{MN} = 0$, the quadratic operator in a mode of momentum p_M is $\Delta_{MN} = p^2 g_{MN} - (1 - \xi^{-1}) p_M p_N$: a rank-one update of p^2, with one longitudinal eigenvalue p^2/ξ and $D-1$ transverse ones p^2. The ξ term *does* mix the two parity sectors in general, since $\Delta_{6a} \propto p_6$ — so the question was a fair one and not a formality. But the leading Li_5 α-dependence that §2 retains lives at $p_6 = 0$ — the $p_6 \neq 0$ modes depend on α too in the full six-dimensional theory and are simply suppressed in the limit this paper adopts throughout — where that factor vanishes, and where in any case only one parity class is present at all: for a $P_6 = \varepsilon$ generator A_μ and A_5 carry cosines and have a zero mode while A_6 carries sines and has none. What is left, sector by sector, is

$$\varepsilon = +1: \quad \tfrac{1}{2}\big[5 \ln p^2 - \ln \xi\big] - \ln p^2 = \tfrac{3}{2} \ln p^2 - \tfrac{1}{2} \ln \xi, \qquad \varepsilon = -1: \quad \tfrac{1}{2} \ln p^2, \tag{62}$$

that is 3 and 1, summing to the $D-2 = 4$ of [17]. The surviving $\ln \xi$ carries no p, and that is not by itself enough to discard it — summed over a *shifted* tower a p-free constant need not be α-independent, since $\zeta_H(0,a) = \frac{1}{2} - a$ moves with the shift. It is discarded because the two halves of the tower cancel, $\zeta_H(0,a) + \zeta_H(0,1-a) = 0$ for every a, and independently because the four-dimensional integral of a p-free coefficient is scaleless. **So the split does not depend on ξ, and it is $(a,b) = (\frac{3}{2}, \frac{1}{2})$: the candidate seed.** Measured in `gauge_xi_independence.py`, whose decoy — ghosts carrying A_6's parity instead of A_μ's — returns 5 and 1 and fails the $D-2$ test, so the hypothesis is doing work.

What that does not settle is two things and a half, and they are worth separating. The ghosts are *taken* to inherit the parity of A_μ, on the standard ground that the gauge parameter carries no vector index; it is what [48] do, and the decoy shows the answer turns on it — but assuming it is not deriving it. Whether [2]'s spectrum construction hides a cancellation that restores the missing degree is a question only its authors can answer. And the half: all of this is leading order in R_5/R_6, which [2] require to be large without fixing it, so at a moderate ratio the count would have to be redone. **Theorem 1 therefore stays conditional and the front-page warning stays with it.** What has changed is that the fork is no longer symmetric: this paper's own machinery now points at the branch on which that theorem's hypothesis is *not* met. **But it points once, not twice.** (62) and the ghost-subtraction test of `single_orbit_determinant.py` share the assignment that decides the answer — the ghosts carrying A_μ's parity — so they are two implementations of one assumption and not two independent checks. An earlier version of this paragraph called them different routes; they are not, and the distinction is the whole reason the assumption is worth deriving rather than stating.

And it can be derived, in three lines, from gauge invariance rather than assumed. Under the x_6 reflection a generator obeys $P_6 T^a P_6^{-1} = \eta_a T^a$. The gauge variation $\delta A_M = D_M \omega$ then fixes the parity of the parameter and not the other way round: $\delta A_\mu = \partial_\mu \omega + i[A_\mu, \omega]$ carries no x_6 derivative, so ω^a must share A^a_μ's parity η_a; and $\delta A_6 = \partial_6 \omega + i[A_6, \omega]$ is then automatically odd where A_6 is, because ∂_6 flips the sign. The background gauge-fixing functional $\bar{D}_M \delta A^M$ has parity η_a for the same reason, so it respects the boundary conditions rather than imposing new ones. BRST replaces ω^a by c^a in $sA_M = D_M c$, so c^a and $\bar{c}^a$ carry η_a too. **The ghosts therefore inherit A_μ's parity as a consequence of the orbifold action on the gauge parameter, not as a convention**, which is what (62) needed and what the decoy of `gauge_xi_independence.py` shows the answer turns on.

What that leaves open is narrower than it was, and it is worth naming exactly. Whether [2]'s spectrum construction hides a cancellation that restores the missing degree is a question only its authors can answer. And the split above is derived in the reduced $R_5/R_6 \to \infty$ theory — the same limit in which §2 adopts the Li_5 potential, so it is the scope the whole paper already works in, and not a weakening special to this calculation. **Theorem 1 stays conditional and the front-page warning stays with it**, because a derivation of ours is not a statement about what [2] did. We would rather write that down than let a bound keep standing on the branch that suits it.

And it does not explain the other open question either, which is worth measuring rather than assuming. §2's anchor residual and the gauge seed are both about [2]'s coefficients, so a reader is entitled to ask whether they are one problem. They are not. Minimising the exact potential of the five published rows under the candidate seed moves $\alpha_{\min}^{\text{ours}}/\alpha_{\min}^{\text{theirs}}$ from 1.03–2.08 to 1.24–3.20 — worse on all five rows without exception, the mean $|\text{ratio} - 1|$ going from 0.50 to 1.13. **So neither finding supports the other**: the ghost question stands or falls on the determinant and not on that column. Nor could that column arbitrate it, since we reproduce it under *neither* seed, and an instrument that does not reproduce its reference cannot be a witness about anything. Measured in `ghost_seed_vs_anchor.py`, whose falsification scales the gauge sector by 0.2 through 0.8 and watches the residual worsen monotonically — so "worse" is a measurement the test could have contradicted and did not.

But we can say what the other branch costs, and it is less than it looks. Running the same integer program with the candidate seed is a small change — the per-multiplet moments are *differences* against the gauge sector, so they do not move at all; only the base point does. Three things come out of it, and only one is a loss. The mod-6 law survives verbatim: the lattice subgroup is unchanged and only its base point shifts, so $8D \equiv 2A_4 + 3$ still holds, with no violation in 20 000 random contents under either seed. Theorem 1 was its *corollary* under "A_4 integer", and it is A_4 turning half-integral that flips $2A_4 + 3$ from odd to even. The $2W$ theorem survives untouched, because the gauge sector contributes $2W = -3$, odd, under *both* seeds. And the ceiling survives too, lower: the smallest *physically admissible* positive rung goes from $D = \frac{1}{8}$ to $D = \frac{2}{8}$ —on the candidate branch the lattice reaches $D = 0$, as the next paragraph says, and what removes it there is identity (II) and not the arithmetic; on the seed [2] print $8D$ is odd and the arithmetic removes it— and since the per-D ceiling falls monotonically the bound falls with it, from 10.03 to 7.38 TeV (Figure 15). **Those two numbers are the same kind of object and only that kind**: both are the certified LP-dual bound over the moment cone, so 7.38 is the candidate-branch analogue of 10.03 and of nothing else. The published branch has two further levels below it — 10.01 TeV *attained* once the lattice is lifted, and 9.22 TeV once the vacuum is required to be the true one. *Neither has been recomputed on the candidate branch*, and until they are, comparing 7.38 with 9.22 would be comparing a bound with an optimum.

And that table has a structural consequence worth stating, because it runs the opposite way from what a fork usually costs. The candidate branch's 7.38 TeV is an upper bound reached *without* the true-vacuum condition, and imposing that condition only removes contents from the feasible set, so the candidate branch's true-vacuum ceiling can only be lower still. **Between the two seeds this paper considers, and on the three-rung algebraic surface,** 9.22 **TeV therefore remains the conservative bound.** The gauge-seed ambiguity does not open a route to larger compactification scales; it closes one. What it moves is the explanation and the sharpness — which rung the maximum sits on, and whether a theorem or an identity is what forbids $D = 0$ — and not the number a reader is asked not to exceed. Both qualifiers are load-bearing and neither is decoration: a third seed is not costed here, and off the three-rung surface the untruncated question of §8 is open on either branch.

What it costs is not the number but the reason. With the odd-eighths theorem, $8D = 0$ was impossible and nothing had to be argued. Without it, $8D = 0$ *is* reachable on the lattice — nine copies of $\mathbf{28}(+,-)$ get there — and what excludes it is that identity (I) would demand an infinite G, and — more cleanly, and with no appeal to the optimisation —

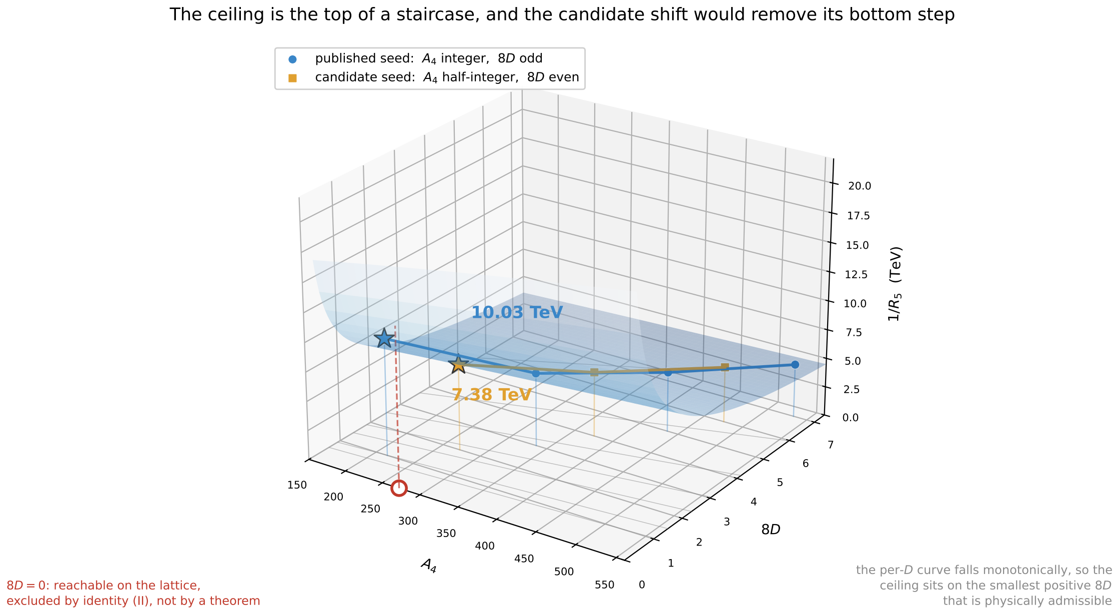


Figure 15: The ceiling is the top of a staircase, and the candidate shift would remove the bottom step. Each marker is the certified upper bound on $1/R_5$ at that value of $8D$ — the LP dual over the moment cone, not an attained optimum; the surface behind them is the objective $1/R_5 = 2\pi m_W/x$, which diverges as $8D \to 0$. The two lattices interleave in the base plane: with [2]'s coefficients A_4 is an integer and $8D$ is odd, so the staircase starts at $8D = 1$ and the ceiling is 10.03 TeV; on the *candidate* parity-resolved seed A_4 would become half-integral and $8D$ even, the $8D = 1$ step would no longer exist, and the ceiling would be 7.38 TeV at $8D = 2$. The bound is not lost on either branch; it is demoted by one step. The open circle at $8D = 0$ is the honest part: the *candidate* lattice does reach it, and what forbids it there is identity (II), since $6\mu + A_4 > 0$ for every content forces $x = 0$; on the published lattice $8D$ is odd and $8D = 0$ is not on it at all. Both curves are computed by the same program, which reproduces the published 10.03 TeV before it is allowed to report anything else.

that identity (II) settles it in one line. It reads $x^2(6\mu+A_4) = 12\zeta(3)D$. Every bulk multiplet contributes a *non-negative* amount to A_4, which is the property the integer program already relies on, and the gauge sector's own A_4 is $-\frac{27}{2}$ on the candidate seed; meanwhile $m_h \in [125, 127]$ GeV with $g_4 = 0.63$ puts 6μ between 481 and 496. So $6\mu + A_4 > 0$ for *every* content, without exception and without a caveat. Then $D = 0$ forces $x = 0$: there is no finite electroweak vacuum on this surface compatible with the measured Higgs mass. That is a physical argument about a limit, not a theorem about integers, and it is weaker. We would rather say so than let a bound keep the standing it had when a theorem was holding it up.

So the honest way to present the whole of it is not as a doubt but as a fork, with both branches costed. The gauge seed enters everywhere through two numbers, the weights (a, b) per charged adjoint pair, and once they are named the rest is bookkeeping:

	seed printed in [2]	candidate (3+1) seed
weights (a,b) per charged pair	$(2,\frac{1}{2})$	$(\frac{3}{2},\frac{1}{2})$
gauge bracket	$(2,4,7)$	$(\frac{3}{2},3,6)$
$8D$ from the gauge sector	-27, **odd**	-18, **even**
A_4 from the gauge sector	-18, integer	$-\frac{27}{2}$, half-integer
$2W$ from the gauge sector	-3, odd	-3, odd
$32F(0)/\zeta(5)$ from the gauge	9, **odd**	18, **even**
parity of the three rungs $k=0,1,2$	odd, odd, **even**	even, even, **odd**
is $A_4=0$ in the coset?	yes, $A_4\in\mathbb{Z}$	no, $A_4\in\mathbb{Z}+\frac{1}{2}$
hypothesis of Thm 1	**satisfied**	**not satisfied**
Thm 2, $8D\equiv 2A_4+3$	holds	holds
$2W$ odd for every content	holds	holds
$\lvert F(0)\rvert\ge\zeta(5)/32$	holds	**not proved**
Lambert closed form, ladder, kernel	holds	holds
smallest positive D the ceiling can sit on	$\frac{1}{8}$	$\frac{2}{8}$
what forbids $D=0$	a theorem	identity (II)
LP-dual ceiling on $1/R_5$	10.03 TeV	7.38 TeV
attained optimum; true vacuum	10.01; 9.22 TeV	not recomputed

Read down the second block and the shape of the result is visible at once. Some of the load-bearing statements are indifferent to which column is right and some are not, and the table is the only place that says which is which. Two rows deserve a second look. The value bound $|F(0)|\ge\zeta(5)/32$ is not merely rescaled on the candidate branch: its *proof* goes, since $32F(0)/\zeta(5)$ reads 9 on one seed and 18 on the other, and an even integer is not kept away from zero by being an integer. And $A_4=0$, which the published branch reaches and rules out on physical grounds, is not even in the candidate coset — there A_4 is half-odd-integral, so the obstruction stops being a physical argument and becomes arithmetic. *Quantised does not mean non-zero;* ***oddly*** *quantised does*, and separating those two took a second seed to see.

And the table is easier to read once the right object is named. It is not the half-integer, and it is not the three colours: both of those are properties of one seed that happen to be true. What the whole table turns on is the *parity character of the affine gauge base point.* On the row that mixes the two sectors,

$$\chi_{\text{gauge}}(a,b)\;\equiv\;(2a)+(2b)\pmod 2\,,\qquad \chi_{\text{gauge}}\big(2,\tfrac{1}{2}\big)=1\,,\qquad \chi_{\text{gauge}}\big(\tfrac{3}{2},\tfrac{1}{2}\big)=0\,. \tag{63}$$

Read the table again with that in hand and every row explains itself. The mod-six law survives because it lives in $8D-2A_4$, whose residue at the base point is 9 on *both* seeds — it never sees χ_{gauge} at all. The oddness of $2W$ survives because $2W$ lives in the *difference* of the two charge-one weights, where a common shift cancels. The oddness of $8D$ does not survive, and neither does the gap below $|F(0)|$, because both read χ_{gauge} directly. *One character, four rows, and no coincidences left over.*

And once the object is named, the whole table is one contraction. Two seeds are two base points of the same affine lattice, so what separates them is a *vector*. In the integer coordinates $(2A_4,8D,2U,V,2W)$ it is

$$\Delta_{\text{seed}}\;=\;(9,\;9,\;9,\;0,\;0),$$

derived from the three numbers of the gauge bracket and nothing else. A statement written as a linear functional ℓ on the content then survives the change of seed exactly when $\ell(\Delta_{\text{seed}})$ vanishes in the modulus that statement is written in — and every row above is that one

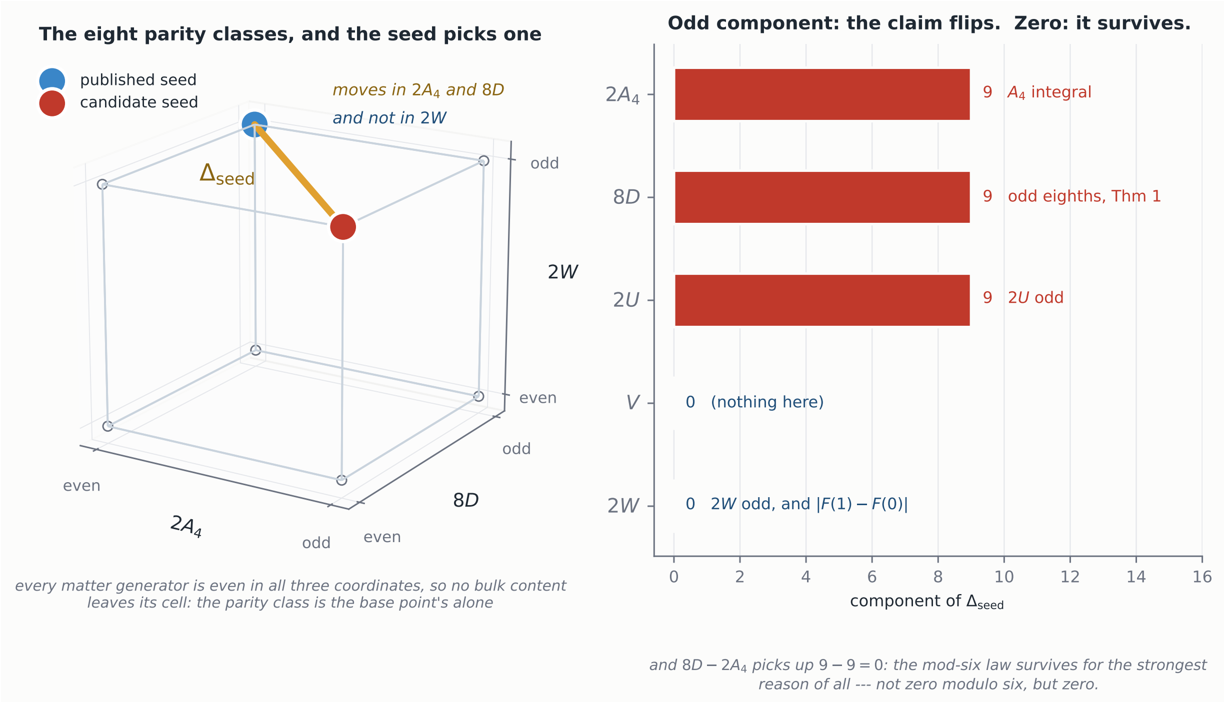


Figure 16: Two seeds, two base points, one vector. *Left:* the parity classes of $(2A_4, 8D, 2W)$ are the eight vertices of a cube, and no bulk content can leave the one its seed picks — every matter generator is even in all three coordinates, so the class belongs to the gauge base point and to nothing else. The seed [2] print sits at (even, odd, odd); the candidate of §13 at (odd, even, odd). The amber segment is Δ_{seed} read modulo two: it moves $2A_4$ and $8D$ and leaves $2W$ where it was, which is the whole of why the hypothesis of Theorem 1 fails on the candidate branch while the $2W$ theorem does not even notice. It is a face diagonal and not an edge, because two coordinates change and one does not. *Right:* the five components of Δ_{seed} and the statement that lives in each — an odd component flips its claim, a zero leaves it standing. Drawn by `make_fig_seedshift.py`, which takes the eight generators from `congruences.py` rather than retyping them, and refuses to draw unless all eight are even in those three coordinates and a fictitious seed that also moves the antiperiodic weight lands on a different vertex.

contraction. $8D$, $2A_4$, $2U$ and $32F(0)/\zeta(5)$ each pick up 9, which is odd, so all four flip. $2W$ picks up 0, because it reads the *difference* of the two charge-one weights and the candidate moves both by the same $\frac{1}{2}$. And $8D - 2A_4$ picks up $9 - 9 = 0$ — not zero *modulo six* but zero, which is why Theorem 2 survives for a stronger reason than the rest. The same contraction settles, with no further proof, that the mod 8 congruence $8D \equiv 2U$ of (64) survives too. **So what a second seed buys is not a second column: it is the vector between them**, and with it the list of which observables are blind to a shift of the gauge base point and which detect it. Computed in `seed_shift_character.py`, which derives the five gauge coordinates from the bracket and only then compares them against the table, and which carries a control that can fail: a fictitious seed moving the antiperiodic weight as well breaks the invariance of $2W$, as it must. Figure 16 is the same statement drawn. Read down the third block and the bound does not vanish on either branch — it changes the coset that carries it, and with it the rung and the number. *That is a more interesting result than the one we set out to report, and it is only visible because there are two columns.* A single-column paper cannot tell you which of its theorems were resting on an inherited number.

Every route in Parts I–VII enters the causal chain at the coefficient end and runs backwards. What would close it is not another consistency check but a calculation in the other direction: a Coleman–Weinberg sum over the six-dimensional Kaluza–Klein tower with these orbifold parities, returning the weight of a $P_6 = -1$ pair from its antiperiodic moding rather than reading it off. **And that calculation appears not to exist in published form**, which turns a private weakness into something worth doing on its own account. [17] give the general one-loop Wilson-line potential for five-dimensional $SU(N)$ on $S^1/\mathbb{Z}_2$, valid for arbitrary representations and boundary conditions — it is what §11 runs. We could find no six-dimensional counterpart: the recent $T^2/\mathbb{Z}_4$ study of [49] writes model-specific potentials rather than a general formula and does not treat $T^2/\mathbb{Z}_2$ with two independent parities at all. So the missing object is the six-dimensional analogue of [17], and deriving it would both close the one inferred link in this paper's chain and supply a formula the field does not have. Until then the strongest statement in §10 — (57), which reads the comb and therefore the seed, and not (56), which does not — has one determination and no second opinion.

- **The antiperiodic sector is unanchored, and that is not an accident.** Every published check a curvature can be anchored against sits at $B = 0$, and on that locus $\Delta_k = \mathcal{A}_k = S_k$: the parity-weighted moment and the parity-blind one are the *same function* there. So no published check can separate A from B, and in particular none of them tests the $-\frac{3}{4}$ of $D = A_2 - \frac{3}{4}B_2$ — the one coefficient every theory-side scale in this paper passes through. That is the blindness, and it is structural rather than a gap in the reading: each of the three things the surrounding literature does with a parity stays on the locus. [7] build the twisted trace itself — their (4.42)–(4.43) carry $\mathrm{Tr}(\lambda_R T_R^{A'} T_R^{A})$, the parity inserted in the trace, as the candidate brane term — and then prove it *vanishes* on $S^1/\mathbb{Z}_2$, their abstract concluding that all brane effects for the Higgs mass do. [8] split the sum by parity, but at $p = 5$ and comparing values at two orbifold points; the words *curvature*, *second derivative*, *Dynkin* and *trace* do not occur in their forty-five pages. (That comparison at two points is not idle for us — (34) is it, and it is what removes eight hundred GeV from the ceiling. It is blind to the split all the same: it reads the *value* of the potential, which is the $p = 5$ rung, and the anchor question is about the $p = 3$ one.) And [16], studying the same Wilson-line potential non-perturbatively on $\mathbb{R}^4 \times S^1$, records that increasing the number of fermions is equivalent to adding a parity pair — which is exactly true where they say it and exactly false here, and the two moments say why in one line: on our lattice a parity pair has $\Delta = 0$ for every one of the four multiplets but $B \neq 0$ for all four, so a pair moves along the *unanchored* axis while leaving the parity-weighted moment untouched, and a multiplicity does the reverse. Identifying them is identifying a direction the literature measures with one it does not. **Nobody has had a reason to leave the corner, because inside it the three operations coincide.** That is why an anchor for $B \neq 0$ has to be computed rather than found, and it is the sharpest thing this paper is missing.

  Two measurements say how sharp, and they are in `anchor_delta.py`. First, the residual is *not* a function of the parity-blind moments: rows (1) and (2) of [2]'s table have identical S_0, S_1, S_2 to the last digit — 87.5, 179.5, 707.5 — and their residuals differ by twenty-five per cent, which no ordering over five collinear points could have established. Second, the split is a near-cancellation: $D = A_2 - \frac{3}{4}B_2$ is $78.000 - 76.125$ on row (1), a 2.4% residue, and the leverage that gives is measured rather than estimated: rows (1) and (2) differ by one per cent in the split, $78.000 - 76.125$ against $79.000 - 75.375$, and their D differ by a factor of 1.93 — so $\alpha_{\min} \sim \sqrt{D}$ moves by 39%, the size of the residual itself. Rescaling the antiperiodic sector by a single λ does not repair the five rows, and it is $\lambda - 1$ that has to be judged: the required perturbation runs from 0.2% to 1.4%, a factor of seven, so this joins the one-parameter repairs already excluded in Part VI. But it is of one sign and of *that* order on every row, which changes what the caveat asserts. It is not that the potential is wrong by a factor of two; it is that one trace may be wrong in its second digit, in the one place where D has no room.

- **The other two congruences are read, and one of them is already in this paper twice.** (30) gives $\mathbb{Z}^5/L \cong \mathbb{Z}_2 \times \mathbb{Z}_{72} \times \mathbb{Z}_{2592}$. As Smith prints them the last two are unreadable

— five-digit coefficients modulo 72 and 2592 — but $72 = 8\cdot 9$ and $2592 = 32\cdot 81$ are coprime factorisations, so each is really *two*. In lowest terms, and *on the affine coset based at the gauge seed [2] print* — the residues below are the coset's, not the matter sublattice's, which has residue zero in every one of them — every bulk content satisfies

$$\begin{aligned} 8D + 2U + 2W &\equiv 1 \pmod 2, \quad 8D \equiv 2U + 4(2W) && \pmod 8, \\ A_4 + 8D + 3(2U) &\equiv V \pmod 9, \quad 2W \equiv -\big(2A_4 + 12(8D) + 3(2U)\big) && \pmod{32}, \end{aligned} \tag{64}$$

together with $V \equiv 0 \pmod{81}$, which is automatic because V is 81 times a multiplicity. The five moduli multiply to $2\cdot 8\cdot 9\cdot 32\cdot 81 = 373248$, the index, so this is the *complete* congruence description of the coset and not a sample of it. Checked in `congruences.py` on the eight generators, the gauge sector, the witness of Figure 10 and [2]'s five rows, and against Smith's own form on a sweep of 114 244 vectors, so that the readable statements are the same *set* and not a weakening.

Two things come out. **The parity character is not one of the factors**: it is the joint mod-two shadow of three of them, since the mod 8 statement gives $8D \equiv 2U$, the mod 32 one gives $2U \equiv 2W$, and the mod 2 one then forces all three odd. So the oddness of $2U$, which the previous edition of this list recorded as a gap, is *equivalent* to the oddness of $2W$ — by a congruence that never mentions the $-\frac{7}{2}$. And **the** mod 9 **statement is this paper's own divisibility**: reduced modulo three it reads $A_4 + 8D \equiv 0$, which is at once the step-three filter §8 searches by and the remark in §7 that [2]'s rows give $8D + A_4 = 273, 300, 198, 234, 270$, all divisible by three. Those are one fact, and carrying $2U$ buys a further power of three.

What they do *not* do is move the ceiling, and that is structural rather than a matter of checking cases: the projection of the content lattice to the $(A_4, 8D)$ plane has index exactly six — the gcd of its 2×2 minors — so the mod-six rule is already the complete statement there, and nothing involving $2U$, V or $2W$ can cut the plane further. Whatever these forbid, they forbid inside a fibre. The one escape would be a vertex whose fibre they empty, and the three this paper argues about, $A_4 = 104, 212, 215$ on $8D = 1$, each merely pin $2U$ to one class modulo three, which the eight multiplets realise. What an odd $2U$ *means* is therefore still open, but it is no longer a separate gap from the $2W$ theorem.

And the four-coordinate projection comes with it. The previous edition of the honest-limits table recorded that the conditions behind $\mathbb{Z}^4/L \cong \mathbb{Z}_{18}\times\mathbb{Z}_{648}$ of (29) were computed but not written out; eliminating $2W$ from (64) writes them. **And the two levels must not be run together here of all places.** The projected *matter* lattice is the kernel of the four congruence maps — $8D \equiv 0 \pmod 2$, $8D - 2U \equiv 0 \pmod 8$, $A_4 + 8D + 3(2U) - V \equiv 0 \pmod 9$ and $V \equiv 0 \pmod{81}$ — with residue *zero* in every one of them, as a subgroup must have. Adding the gauge base point [2] print translates that subgroup to the affine coset, and it is *there* that a content reads $1, 4, 0, 0$. The two readings genuinely differ in the second, where $4(2W)$ survives only because $2W$ is odd. Index $2\cdot 8\cdot 9\cdot 81 = 11664 = 18\cdot 648$, confirmed by closing the subgroup of $(\mathbb{Z}/32)^4$ and $(\mathbb{Z}/81)^4$ by brute force rather than by trusting the elimination. Two further questions survive. Is the lifted semigroup *normal*? In the projection it is: no point of the cone-and-lattice is unreachable, measured to $A_4 = 420$; if the lift has holes they are contents every visible law permits and no bulk can build. And what is the vector partition function $N(A_4, 8D, 2U, V)$, which should be quasi-polynomial on the chambers of the cone and would turn the degeneracies of Figure 5 from a measurement into a formula.

- **Where does the third minimum bind?** The previous edition of this list asked whether $W > 0$ is sufficient as well as necessary, and proposed the semi-infinite programme as the way to find out. §8 now runs it, and the answer has two halves. On the rung that decides the ceiling the programme generates exactly one cut, the one at $y = 1$, so there $W > 0$ is the whole of the true-vacuum condition and (36) needs no screen. But a content invisible to both symmetric points does exist — (41) demands a second cut on $8D = 3$, and §8 places it at the exact rational holonomy $y = 2/3$, where the depth is the closed-form functional $-\frac{121}{81}\zeta(5)[C_1 + C_2 + C_4]$ — so the phenomenon is real, it is exact, and it merely fails to reach the deciding rung. What is now open is smaller and sharper: is there a rung on which a cut other than $y = 1$ moves the *ceiling* rather than an interior vertex? And where the *other* cuts sit is now a measured negative rather than an untried question: across 6560 contents the 2956 interior minimisers stand only 1.05 times closer to the rational skeleton $\{1/3, 1/2, 2/3\}$ than chance does. The y at which a given content's third minimum sits still ought to be readable off its charges the way $\alpha_{\min}$ is, and is not — and the nearest rational holonomy is now known not to be the way to read it.

- **A remainder bound uniform in the rung.** (39) is proved over the first seventy odd rungs and not past them, and the reason is measurable: the bound's relative bite $\delta/\hat{x}$ climbs from 2.5×10^{-4} at $8D = 1$ to 9.0×10^{-1} at $8D = 1201$, so it loosens faster than the truncated ceiling falls and eventually certifies nothing. That is a defect of the *bound*, not of the ceiling — (25) proves $M(k) \leq 6268$ GeV on every rung $k \geq 3$ with no scan at all. What is missing is a bound on $|R'_j(\hat{x}(k))|$ decaying at least as fast as the curvature $\propto k$, which would close (39) for all k in one line. The per-multiplet remainders are explicit Li_5 tails, so this looks like an exercise in bounding $\sum_{n>N} \sin(nc\hat{x})/n^4$ uniformly in $\hat{x} \to \hat{x}_\infty$ rather than an open problem; we have not done it.

- **The 215-versus-212 margin is smaller than the expansion that produced it.** One part in 10^4, against a closed form good to 0.13–0.62 %. The remedy is not more digits but the corrected identities: redoing the elimination *without* discarding the remainder R gives $x^2(6\mu + A_4) = 12\zeta(3)D - 18R'/x + 6R''$ and a matching correction to (I), and §5 has just established that R is a convergent tail with a rigorous bound. Running the integer program over certified intervals for x rather than over the truncated surface would turn 10.01 TeV from a statement about the algebraic surface into one about the polylogarithmic potential. That is the single most valuable thing left in this paper.

- **What would turn the recast sensitivity into an exclusion. One thing now, and the public-data programme is exhausted.** Three closures have been run and none of them is the missing step. *The observed limits*: aimed at [39]'s published colour-singlet limit on their own selection, the machinery returns 17.5 against 17 TeV and 29.5 against 37. *The expected ones*: an Asimov Standard-Model dataset through our own χ^2 returns 1.08 and 0.85 against [39]'s expected 18^{+20}_{-2} and 34^{+8}_{-5} TeV, close enough to the observed 1.03 and 0.80 that the gap is the method and not this dataset. *And the template*, which was the sharpest of the three worries: the closure benchmark ought to be the vector-like VV operator and not only LL: the tower is $(\bar{q}\gamma^\mu T^a q)^2$, which sees both chiralities, while the 17 and 37 TeV are left-handed, and the published VV limits behave differently enough — the constructive one reaching some 50 TeV, the destructive one developing disconnected regions where the interference cancels against $1/\Lambda^4$ — that agreeing with one says little about the other. The machinery for it is in place — [52]'s operator basis is the chiral one of its eq. (2), so VV is $\lambda_1 = \lambda_3 = \lambda_5$ where the LL closure is λ_1 alone. **That run has now been done, and it settles the worry rather than confirming it.** Paired selection against selection, on the constructive side where [39] publish a limit for both operators, the closure ratio moves from 0.80 to 0.78 on their own contact-interaction selection, and from 0.69 to 0.66 over all seven mass bins. Two selections, the same direction, and **two to three points either time**. *So the chirality of the template is not what the closure deficit is made of* — which is worth knowing, because it was the most plausible explanation available and it is now the least. What the pairing does *not* do is give the deficit a single size: it is 0.78 on the matched contact-interaction selection and 0.66 over all seven bins, so the discrepancy is selection-dependent and only the LL-to-VV step is stable. On the destructive side there is nothing to pair: [39] publish no VV limit there. And the deficit stays what it was, **closure accuracy and not a transferable multiplicative systematic**; measuring it twice does not turn it into a band. **What remains is one step, and it is the one that costs something**: the limit should be set the way [39] set theirs, with detector-level Poisson likelihoods, profiled nuisances and CL_s. Until then (56) is a sensitivity and is written as one. **And a second systematic sits beside it, of a different kind and unmeasured**: the baseline is [39]'s next-to-next-to-leading-order QCD while the tower's deformation is validated against [52] at leading order, so the two enter (55) at different orders. The 0.2 % closure is an implementation check of the leading-order interference and not a statement about the missing K-factor; how far an NLO deformation would move (56) is not measured here, and it is the cheapest of the open collider questions to answer.

- **The thermal holonomy literature has not been swept.** §3 makes the two problems one family; that literature has $V''(0) \propto T(R)$ as textbook and $\mathbb{Z}_2$-projected traces as natural bookkeeping, so it is the highest-prior remaining place the bridge could already exist. It is named here as an open gate, not as a null.

- **Two loops.** The map computed here is a one-loop map. At two loops the potential sees more of the content — a weight $W(w, w') = \sum_a |\langle w|T^a_R|w'\rangle|^2$ rather than a multiplicity — and what that second rung is blind to is decidable without a loop integral.

- **Gauge coupling unification, which is their own open question.** [2] close by asking for it: $SU(3)_C$ and $SU(4)$ sit inside one $SU(7)$, so the couplings must meet where the symmetry is restored, and they leave the one-loop analysis to future work. Half of it is settled here and costs nothing: below $1/R_5$ the theory is the Standard Model, so $\alpha_2^{-1} - \alpha_3^{-1}$ is 19.2 at 2 TeV and 18.2 at the certified ceiling — it barely moves across the whole allowed range, because four-dimensional running has under five e-folds to work with and closes the gap at 0.61 per e-fold. **The entire burden therefore falls on the Kaluza-Klein tower**, and the ceiling of §8 is not a bystander: it fixes where that tower may begin. The other half — whether the $SU(7)$ tower, whose levels the orbifold projection leaves incomplete, can close 18.2 before perturbativity ends — is a computation, not an opinion, and it is the natural next paper.

- **Does the theorem's dimension have a general statement?** §7 shows the protection is absent from the five-dimensional class [17]'s formula covers — $SU(N)$ on $S^1/\mathbb{Z}_2$ with a single Wilson phase, where every rung is even and $D = 0$ is reached by three multiplets — while here the rungs below the fourth moment are odd. The mechanism is the parity character (63) of the gauge base point — not a single half-integer — and a second independent parity is what supplies the extra affine character that *can* make a rung odd — in this construction, which is as far as one model reaches. Whether *odd rung* $\Leftrightarrow$ *second parity* is a theorem — across $T^2/\mathbb{Z}_N$, and in seven dimensions — is not settled by one model.

- **Does the periodic/antiperiodic split organise the counterterms?** It does at the pole: (18) sends the antiperiodic half to zero at $p = 1$, so the branch point is fed by the periodic moment alone. Whether that coincides with the bulk-localised divergent structure, with the complementary parity sector feeding the boundary terms, is an interpretation and not a consequence of the rung: it needs an explicit counterterm calculation. Whether the same split works away from the pole — one half feeding the higher-derivative bulk operator, the other the boundary-localised one — is what the Hurwitz correspondence of §12 opens and does not answer.

- **The lattice, counted.** The multiplet lattice with its congruence is a rational cone intersected with an affine sublattice, so the number of contents at fixed $(A_4, 8D)$ is a vector partition function, and Theorem 2 supplies one residue-class constraint underlying its chamberwise quasi-polynomial structure. *Not* that six is the minimal quasi-period: nothing here shows that, and a congruence on the lattice is a weaker statement than a period. That is machinery this paper uses only in its polyhedral half.

The ledger

Established, and how	Where
$\alpha_{\min}$ in closed form: median 0.13 % over 272 contents, 0.209 % over the five published rows, against our own numerical minimisation of the same potential	§5; the no-logarithm control fails by +104 to +174 %
$F''(\alpha_{\min}) = \pi^2[2\zeta(3)D - A_4x^2/6]$: the logarithm and G cancel at the stationary point, so both columns of their Table 1 follow from (D, A_4) with no minimisation — *our* reconstruction of the two, not their published $\alpha_{\min}$	§6
The two logarithm-free identities, verified to 6×10^{-14}	§8, eq. (22)
$8D$ is an odd integer, so $D \neq 0$ and *the symmetric point* is never marginal — **conditional on [2]'s gauge seed**; the matter half of the argument is not	§7, §13; 4096 parity assignments, no counterexample
$8D \equiv 2A_4 + 3 \pmod 6$, from $c^4 \equiv c^2 \pmod 3$ — **holds on either gauge seed**	§7; the five published rows and all eight generators
$\lvert F(0)\rvert \geq \zeta(5)/32$, the bound attained — **conditional on [2]'s gauge seed**: it reads (63) exactly as the odd eighths do, and on the candidate seed $32F(0)/\zeta(5)$ is even	§7, §13
The ladder is geometric with ratio $c^2/4$, charge two its fixed point; and $O(5) - O(3) = 4[O(3) - O(1)]$ whenever no third mode opens	§4; 4096 assignments, and a prediction confirmed on all eight multiplets
The ladder continues past its own pole to $p = -1, -3, \ldots$, where ζ is rational: exactly one logarithm exists, at order x^4, and the remainder is a convergent tail for $\lvert\alpha\rvert < 1/c_{\max}$	§5; term ratios $(c\alpha)^2$ and $(c\alpha/2)^2$, with controls past both radii
$A_4 = 0$ forces $8D \equiv 3 \pmod 6$, so the analytic branch cannot reach the quantum $8D = 1$; and on that branch no content has both $D > 0$ and $W > 0$ — **stated on the seed [2] print**, the candidate coset having $A_4 \in \mathbb{Z} + \frac{1}{2}$, where $A_4 = 0$ does not occur at all	§7, eq. (21); eleven base solutions, one free multiplet, exact arithmetic, and it misses by one step
$2W$ is an odd integer, so $\lvert F(1) - F(0)\rvert \geq \frac{31}{32}\zeta(5)$: in *this* model the two symmetric points are never degenerate, hence never equivalent descriptions of one orbifold — unlike [8]'s, where they are	§8, eq. (35); **not** the same half-integer as Theorem 1, which an earlier version of this row said it was: $2W$ reads the *difference* of the two charge-one weights, so it holds on *either* gauge seed where $8D$ odd does not. Checked on 3003 contents
M_{KK}^2 sits on an arithmetic comb of spacing $8\pi^2 m_W^2/(\zeta(3)\, 8D)$, necessary and not sufficient	§9, eqs. (46)–(47)
The charge-two parity flip is the only log-free one, with signature $(-7, -16, -100/3)$	§4, eq. (9)
$1/R_5 \leq 10.03$ TeV for arbitrary bulk content, by an exact rational dual — **on the seed [2] print**	§8; dominates all 1097 window contents found to $N = 14$
$G = \frac{25}{12}A_4 - U\ln 2 - V\ln 3$ lifts the content to a *four*-dimensional lattice with $\mathbb{Z}^4/L \cong \mathbb{Z}_{18} \times \mathbb{Z}_{648}$, where the relaxation's vertex $(215, 1)$ is empty and the bound sharpens to 10.01 TeV *attained*	§8, eqs. (29), (32); rank and invariant factors in `lattice_rank.py` and independently in Sage, `lattice_rank.sage`; the maximum over a finite feasible set at fifty digits, the witness rebuilt through `moments()`
The remainder past three rungs is bounded, not estimated, and is itself *linear* in the multiplicities, so it rides inside the certificate's own dual: $1/R_5 \leq 10.037$ TeV on the untruncated potential over the first 70 odd rungs, 0.025 % above the truncated certificate; beyond them the remainder bound loosens and certifies nothing	§8, eqs. (38), (39); 32 contents × four α, zero violations, and the bound refuses at $u \geq 1$

The ledger, continued	**Where**
[8]'s stability criterion, summed over a content, is a fourth *linear* functional, so it fits inside the same integer program; imposing it gives $1/R_5 \leq 9.22$ TeV on the three-rung surface, attained, whose witness on the exact potential gives 9157 GeV. **The criterion is theirs; only its use here is new**	§8, eqs. (34)–(36); closed against exact to machine precision on six contents. Untruncated it is not yet a single number and not an interval either: the witness gives 9157 GeV, the first 70 odd rungs are capped at 10.037 TeV, and a bound over *all* rungs is open — §13
Part VI's escape costs 10/8 of D exactly, so affording it forces $8D \geq 11$ and lowers the ceiling to 3.97 TeV — the content with the largest hierarchy is the one that cannot pay	§8, eq. (45); same dual, and 857 of the 1097 can pay
Our machinery reproduces [7]'s criterion, its three critical flavour numbers, its $SU(2)$ minimum and the whole group-theory bracket of its curvature formula	§11; residue $2/\pi^2$ to eight digits, a pure convention
A nominal recast sensitivity of $1/R_5 > 10.86$ TeV from the spacelike form factor alone — no width, no $\mathrm{BR}(G_n \to jj)$ — putting the 3.97 TeV escape branch a factor of three below it	§10, eq. (56); against [39]'s 77×77 covariance with the scale profiled, on a machinery that returns their own contact-interaction limit as 17.5 against 17 and 29.5 against 37 TeV, and their *expected* one as 1.08 and 0.85 — so the gap to their CL_s is the method and not a fluctuation of these data. Putting the s channel back, with the total width profiled and its worst case at $\Gamma_{\text{exotic}} = 0$, costs 1.03 %
Read at the teeth of the per-rung ceiling comb rather than at an arbitrary mass, the recast says the allowed region is not partly disfavoured but wholly so: the highest per-rung ceiling any bulk content can reach on the seed [2] print, 10.03 TeV at $8D = 1$, sits at $\Delta\chi^2 = 12.0$ against a threshold of 3.84, and the next tooth at $\Delta\chi^2 = 244$	§10, eq. (57), Fig. 13; nominal, from the same Gaussian recast, and stated as such
Corrected here, and left visible	**Why**
"$1/R_5 \leq 2.18$ TeV with the Higgs window"	that is a bound on the *size* of the content ($\leq$ 6 multiplets), not on the model. Eight multiplets reach 9.14 TeV. §8
"at moment zero marginality is reachable"	true with pure component counts, false here — but not for the reason this row used to give. What forbids it is the parity character (63) of the gauge base point, not a single half-integer and not the sixth dimension on its own: the candidate seed is equally six-dimensional and has $\chi_{\text{gauge}} = 0$. §7, §13
Part VI §7's 1.94 and 1.20 quoted as a range	they are group means ($n = 2$ and $n = 3$); the row-wise range is 1.03–2.08, and §9 uses the range
"the ceiling is attained at $(A_4, 8D) = (215, 1)$"	the dual *certifies* it there and the lattice *reaches* it, but no content of any size can pay for it in G — short by 0.067 in a budget of 681. The vertex below can: 10.01 TeV at $(212, 1)$, attained, with a witness. §8, eq. (32)
"sixteen contents with $A_4 = 0$"	seventeen. The enumeration capped multiplicity at four per multiplet while the sentence said six; the missing one has $D < 0$, so the "five with $D > 0$" stands. §7

The ledger, continued	**Where**
"the expansion is asymptotic"	it is convergent for $\|\alpha\| < 1/c_{\max} = 1/3$, and every α used here is inside that by a factor of at least 2.7. §5
"$A_4 = 0 \Rightarrow m_h \leq 29.6$ GeV, an exact margin"	that number came from evaluating the closed form at $\alpha \simeq 0.89$, a factor 2.7 *outside* its own radius. Withdrawn and replaced by (21), which uses no expansion at all. §7
"the 9.16 of the enumeration against the 10.03 of the certificate is the gap between a relaxation and its lattice"	it is not: it is the true-vacuum condition, which the enumeration imposed silently through the global minimiser and the certificate did not impose at all. Named, it becomes 9.22 TeV attained. §8
"restoring the widths lowers the tower's contact coefficient by 2.55%, so Λ_8 is low by 1.3%"	there is no width to restore. In the spacelike channels every mode is below its two-parton threshold, so the propagator is real; the giveaway is that the shift does not switch off as $t \to 0$. Withdrawn, and the tower summed exactly instead, (52). §10
"3.97 TeV is already inside the excluded region"	the exclusions quoted are for a *narrow* resonance and this object has $\Gamma/M \simeq 0.16$; the s-channel rate also needs a branching ratio the model does not fix. Sensitivity, not exclusion — and the angular recast, now done, does not change that word: it is a Gaussian χ^2 over unfolded data and not [39]'s detector-level CL_s. §9, §10
Withdrawn, and left visible	**Why**
"the bracket of the gauge table is the number of colours, so the parity theorem and the death of the obstruction are one product"	true at their assignment, false on 3872 of 4096. A coincidence at one point. §12
"the hierarchy is a quantised near-cancellation between bulk and brane"	measured before it was written: a 10 % effect, not 90 %. What survives is a percentile statement about their five rows. §12
"the parity-split mechanism is ours"	it is published at $p = 5$ [8]. Only the $p = 3$ rung and its consequences are claimed. §4, §12
"the true-vacuum criterion of §8 is ours"	it is not: [8] §2.2 states it, proves its identities and draws its sign rule. Only its use inside the integer program is claimed. §8, §12
Open, and stated as open	**Why**
Their Table 1 $\alpha_{\min}$ column is not reproduced by our assembly	Part VI §7; §11 narrows the locus to the model dictionary but does not close it
The antiperiodic sector is unanchored	no independent published *curvature* anchor tests $B \neq 0$; the parity-split results that exist stop at the value rung $p = 5$. At $B = 0$, Δ and S coincide. §11
[7]'s $SU(3)$ adjoint minimum 0.29 against our exact 1/3	the centre control favours ours; not proved, so not called an erratum. §11
What an odd $2U$ *means* physically	the congruence is now written, (64), and ties $2U$ to $2W$; the interpretation is still missing. §13

The ledger, continued	Where
Is the lifted semigroup normal?	the projected one is, to $A_4 = 420$. §8
Where does the third minimum bind?	settled where the bound is decided: on $8D = 1$ the semi-infinite programme generates only the $y = 1$ cut. On $8D = 3$ it demands one, and that one is exact at $y = 2/3$, so the phenomenon is real and does not reach. Where the *other* cuts sit is a measured negative: the interior minimisers do not crowd the rational skeleton. §8
The thermal holonomy literature is not swept	§3 makes it the same family, so it is the highest-prior place the bridge could exist. §13
Is the recast sensitivity an exclusion?	10.86 TeV, and 13.75 on their own contact-interaction mass range, both clear 9.09 and 10.03. But a Gaussian χ^2 over unfolded data is not their detector-level CL_s, and the two closure benchmarks differ by interference physics rather than by a calibration. Observed and expected closures diagnose a discrepancy of order ten to twenty per cent in Λ on the matched contact-interaction selection, and larger over all seven bins; none of them defines a transferable uncertainty. Closing the likelihood is what would settle the ceiling. §10
Every theory-side scale carries the anchor residual	§2; the collider recast does not, since it never passes through $\alpha_{\min}$. The ceiling is quoted with it in §9
The anchor residual is a *trend*, not scatter, and both ceilings sit outside the range that measures it	sorted by D the five ratios are monotone, 2 of 120 pairings, $p = 0.017$; but A_4 orders them equally well and five rows cannot separate the two moments. The ceiling is certified at $D = 1/8$ against an anchored $[1.125, 3.625]$, the escape ceiling at $A_4 = 727$ against $[189, 271]$. §2, §9

Data availability

The paper in both languages, every script, every archived run and the figure sources are at github.com/karlesmarin/su7-compactification-bound, which is where corrections to this paper will appear; the other six parts of the series have a repository each, linked from that one. Every displayed number in this paper regenerates from the ancillary scripts, whose standard output is archived beside them in `outputs/`; the ledger above cites the script for each entry rather than the number alone, so a reader who disagrees with a claim can re-run the thing that made it. The principal artifacts are `bernoulli_clausen.py` (§3), `ladder_closed_form.py`, `moment_ladder.py`, `moment_split.py` and `mechanism.py` (§4), `amin_closed_form.py` (§5), `mh_closed_form.py` (§6), `parity_mod4.py` and `ladder_obstruction.py` (§7), `ceiling_ilp.py` with its independent Sage lattice computation `cone_moments.sage`, the semi-infinite programme `semi_infinite.py` and the untruncated certificate `rigorous_ceiling.py` (§8), `congruences.py` (§13), `prediction.py` and `hy_predictions.py` (§9), `vgiq_anchor.py` with `vgiq_charges.sage` (§11), moduli_space.py (§12), and inverse_map.py, which reproduces Part VI's §7 by a route that shares no arithmetic with it. formula_map.py states the formulas of Parts VI and VII in minimal form and checks the identities that link them, one to the next; the figures are drawn by make_figures_vii.py from the same archive. An interactive instrument, tools/hierarchy_explorer.html, ships with the paper and runs in a browser with no install and nothing leaving the machine: it computes the three rungs, the closed form and the ceiling for a matter content the reader types in, so the claims of §7 and §8 can be tried on a content this paper never saw.

And the collider layer, which needs three things that are not in this archive. The measurement is [39]'s, on HEPData under DOI 10.17182/hepdata.167852 and released CC0; fetch_hepdata.py fetches it, and the nineteen tables travel with the ancillary files because that licence permits it. The parton densities are CT10nlo [47] through LHAPDF [46], with NNPDF3.1 [50] and CT14nlo [51] for the set dependence of (56); all three come from the LHAPDF set repository at lhapdfsets.web.cern.ch, are freely available, and none is ours to ship. pdf_provenance.py reads each one's own .info and checks what has to hold before it is used: the physical (x, Q) our quadrature asks for lie inside the grid, the three share $\alpha_s(m_Z) = 0.118$, and — because the product grid over mass and boost has corners with $x > 1$ that $x_1 x_2 = M^2/s$ forbids — that all three return *exactly zero* there rather than extrapolating. They do; the unphysical corners carry no weight and the integral is the physical one. That was never checked before a referee asked which set we used. And the contact-operator coefficients come from CIJET [52], whose licence permits academic use but forbids redistribution in any form: the program is therefore *not* included and neither is its output. What travels instead is the three input cards — ours, and checked against [39]'s selection — together with the command and the runtime needed to regenerate the coefficient file, about nine hours on a desktop, written out in the archive's tools/cijet/README. A reader who wants the collider numbers can obtain every ingredient from its own source; what we may not do is hand any of them on.

Acknowledgements

This paper stands on [2]. Y. Komori and N. Maru wrote their model out fully enough that its vacuum can be recomputed from their own equations rather than assumed, which is what Parts VI and VII both do, and they were explicit about where they had stopped. N. Maru replied promptly to an earlier set of questions about the construction, for which we are grateful. He has not reviewed this paper: the reading of their equations here, and any error in it, is entirely ours.

The mechanism expanded here is not new. G. Cacciapaglia, A. S. Cornell, A. Deandrea, W. Isnard, R. Pasechnik, A. Preda and Z.-W. Wang [8] published the parity-split sum, its weight $\eta(5)/\zeta(5)$ and the stability criterion that follows from its sign in September 2024. Their §2 *is* the rung $k = 0$ of §4. Without that paper, what appears here as one rung of a ladder would have been written up as a discovery. What is ours is the rung above it — the curvature — its packaging as two standard traces, and the arithmetic that follows; §12 says which is which, item by item, and that section is the part of this paper we would most like a reader to check.

G. von Gersdorff, N. Irges and M. Quirós [7] wrote their five-dimensional computation out in enough detail — conventions included, down to the footnotes — that it can be used as an independent anchor by someone who was not there; §11 does exactly that, and it is their care with those conventions that lets the agreement mean anything. D. J. Gross, R. D. Pisarski and L. G. Yaffe [6] supply the polynomial at the even index, which is what makes the statement at the odd one sharp rather than merely negative. N. Haba and T. Yamashita [17] wrote the general formula the expansion of §4 starts from. The bulk-versus-brane split that our two traces turn out to be was worked out in [19, 20, 21, 22], and A. Hurwitz's functional equation [23, 24] is what closes §3; none of that is ours, and the paper is shorter for it.

Software, and what it was actually used for. The lattice computations of §7 and §8 are checked independently in SageMath [40] — `cone_moments.sage`, `lattice_lift.sage`, `lattice_rank.sage`, `dimension_five.sage` and `vgiq_charges.sage` — so that every rank, Smith normal form and invariant factor quoted here comes out of two implementations that share no code. The exact algebra of (43) and the formula ledger use SymPy [41]; the fifty-digit arithmetic behind the certificates and the tail bounds is mpmath [45]; and the numerical layer throughout rests on NumPy [42] and SciPy [43], with the figures drawn in Matplotlib [44]. The collider recast of §10 adds several that are not ours in any part: parton densities from CT10nlo [47], NNPDF3.1 [50] and CT14nlo [51], all through LHAPDF [46]; and the contact-operator coefficients from CIJET [52], whose authors' own conventions the matching of §10 is stated in. Naming them is not a courtesy: a reader who wants to re-run the archive needs to know which of these results rests on which tool.

A What is machine-checked, and what is not

The arithmetic spine of §7 is checked in Lean 4 [1]. The source, `OddEighths.lean`, lives in the repository named in the data-availability note; this appendix says what it certifies, and — at greater length, because it matters more — what it does not.

What is in it. Contents are non-negative integer combinations of eight bulk multiplets, so every statement is about a lattice with a base point. The lattice is written in the integer coordinates $(2A_4, 8D, 2W)$: on the candidate seed of §13 A_4 is half-integral, so A_4 is not an integer coordinate and $2A_4$ is. Three statements are proved for arbitrary content.

- `mod6_law`: for any gauge base point with $8D - 2A_4 = 9$ — which by `key_published` and `key_candidate` is *both* seeds — and any content, $8D - 2A_4 \equiv 3 \pmod 6$. This is Theorem 2.
- `twoW_odd`: with $2W_{\text{gauge}} = -3$, $2W$ is odd for every content.
- `oddEighths`: if the gauge contribution to $8D$ is odd then $8D$ is odd for every content. This is Theorem 1, and the hypothesis sits in the signature, where it cannot be quietly dropped.

Two kinds of step close them. The finite half is `decide`: each of the eight generators is checked one at a time, for $8D - 2A_4 \equiv 0 \pmod 6$ and for $2W$ even and $8D$ even. The infinite half is a short induction, `dvd_combo`, carrying divisibility from the generators to any combination with natural multiplicities. Nothing else is needed, which is itself worth reporting: the content of these theorems really is that small.

The certificate. `#print axioms` gives `[propext, Quot.sound]` for the three above and for `oddEighths_published`, and *no axioms at all* for `key_seed_independent`, the statement that the two seeds give the same 9. There is no `sorryAx` anywhere, which is the real proof that no step was skipped — a stronger check than searching the source for `sorry`, and `Classical.choice` is not used either. **And the run that says so is archived**, in `outputs/OddEighths_lean.txt`, on the same terms as every other number in this paper: a brick that compiles on the machine of whoever wrote it is not a brick that has been checked.

What is deliberately not in it, which is the longer list. *The split.* Lean cannot adjudicate [2]'s eq. (68): that is a physical input, not a logical step, and formalising $(\frac{3}{2}, \frac{1}{2})$ would be adjudicating someone else's equation with a proof assistant. It is not there and should not be. *The analysis.* The Lambert closed form, the polylogarithm ladder and the two identities are the analytically hard part of this paper and none of it is formalised; the cost is high and the return low, and claiming a machine-checked paper on the strength of a lattice argument would be false coverage. What is certified is the arithmetic, and it is labelled as the arithmetic.

And machine-checking says nothing about novelty. It is worth saying plainly, because the two are easy to conflate and this group has conflated them before: a Lean proof establishes that an argument is correct, not that it is new. We screened six papers for these statements ([10], [17], [48], [12]'s $T^2/\mathbb{Z}_3$ companion, [49], and one GUT-inspired effective-potential study), four at full text. The result is not "these are new". It is that the $\frac{31}{32}\zeta(5)$ structure is prior work and is cited as such in §7, and that *we found no precedent* for the quantisation or for the congruence. Five of the six are from one school, so the null is a null about that lineage rather than about the field.

A frontier we are declaring rather than discovering later. If any of this is ever extended to something that *does* adjudicate [2]'s equation — the single-orbit determinant itself, not our arithmetic — that goes to those authors before it goes anywhere else.

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[11] N. Haba, Y. Hosotani, Y. Kawamura, *Classification and dynamics of equivalence classes in SU(N) gauge theory on the orbifold $S^1/\mathbb{Z}_2$*, Prog. Theor. Phys. **111** (2004) 265, arXiv:hep-ph/0309088 — eqs. (4.13)–(4.15) reduce the content-dependence to two integer functionals and state the resulting degeneracy.

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